PROGRAMMING NONLINEAR MAGNETIC RESPONSE WITH TWELVE MECHANICAL MODES

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Abstract. We study prescribed finite-time responses to the unobserved conserved magnetic mean in the three-dimensional viscous, resistive magnetohydrodynamic equations. The control acts only in the momentum equation, through twelve fixed fundamental Fourier shear fields. A deterministic compiler programs an even polynomial velocity response and an odd polynomial magnetic fluctuation of that mean. Its single open-loop force is uniform over a bounded unknown preparation, independent positive spatially constant diffusivity histories with supplied Lipschitz bounds, sufficiently small absolute disturbances, and every energy-admissible weak realization. The degree, coefficient lists and Fourier supports are variable input; physical and binary costs have universal polynomial bounds in the numerical parameters and complete description length. The construction first evaluates odd magnetic registers, loads one collinear packet batch, writes a small even velocity, and amplifies it while preserving the magnetic output. A dyadic return word controls amplification on the whole error space; an inverse-carrier estimate controls the complete unwanted packet output. The result is an endpoint theorem for globally supported periodic forcing. It supplies no hardware, affordability or autonomous computation claim.

1. Introduction

Suppose a conducting fluid contains an unknown magnetic flux. A force can be chosen in advance, but neither the flux nor the transport coefficients can be measured during the experiment. Can this one force make the final velocity and magnetic field depend on that flux in a prescribed nonlinear way? The problem is different from steering one known preparation to one known state: the same physical experiment must implement an entire response map.

The input is the conserved magnetic mean H. Its normalized value $h = H/M$ belongs to the unit ball by the preparation bound. The requested endpoint is

$$(u, B) = \big(U(x, h),\ H + Z(x, h)\big), \qquad h = H/M, \tag{1.1}$$

up to a supplied absolute error. The coefficient fields of U, Z are solenoidal and mean zero. Magnetic sign symmetry requires U to be even and Z odd. Their polynomial degree and complete rational Fourier lists are variable input. No invertibility, monotonicity, separability or small nonlinear-part assumption is imposed.

1.1. Contribution and its relation to earlier control results.

The new operation is the simultaneous nonlinear joint response (1.1), with a universal resource exponent as the degree grows, under unknown preparation and independent time-dependent diffusivities. Its foundations are a fixed-depth linear magnetic assembly and memory-preserving mechanical operations. Its additional steps are actual magnetic register arithmetic, an amplifier bounded on every error mode, and one collinear readout whose complete higher output decreases with its carrier frequency. These steps are proved separately before they are composed. The linear construction appears explicitly in Sections 4–7; it is recovered with its original Sobolev input class in Corollary 11.1.

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Finite-mode control, saturation, return methods and reversible arithmetic are established tools. The twelve shear directions used here already form a saturating space in Nersesyan's work [5, Theorem 4.1 and Proposition 4.3]. The forcing-direction method of Agrachev–Sarychev [3] and its three-dimensional developments [4, 5, 6] provide the geometric background. Qualitative saturation does not by itself give a coefficient-blind force, a stable variable-depth register machine, or the numerical resource bounds required here. The explicit common-control construction in [1] is the quantitative control foundation.

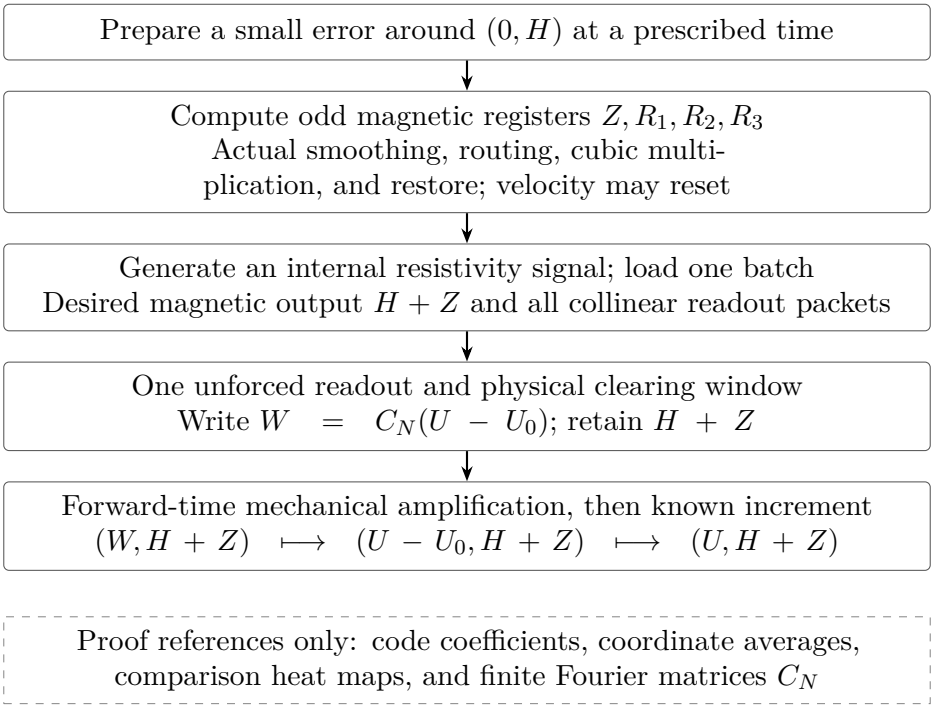
Earlier MHD results also have distinct and substantial scopes. Galan [7, Theorem 2.1] treats approximate torus controllability with controls in both equations. Rissel [8, Theorem 1.2] treats known smooth two-dimensional ideal data with two interior controls and a magnetic cohomology condition. Rissel–Wang [9, Theorems 1.2 and 1.5] establish global approximate boundary control under specified slip conditions; their general theorem allows additional induction terms. Momentum-only control is itself an established possibility: Tao–Yao–Yin [10] prove local trajectory controllability in a bounded two-dimensional domain, under their boundary and reference-trajectory hypotheses. None of these statements is used as an unknown-ensemble response or polynomial-resource theorem. Appendix C records the exact imports actually needed.

The separate seed operation in [2] starts with a prepared nonzero magnetic seed and produces a specified field. Programming a response to an unknown conserved mean is a different task. In particular, zero mean does not mean that the initial magnetic field vanishes. The endpoint here requests zero magnetic output when $H = 0$; it does not assert that mechanical control cannot act on a nonzero mean-zero seed.

1.2. **Conceptual overview: compute first, read once.** A magnetic register is a finite group of oscillatory Fourier coefficients whose amplitudes hold unknown odd functions of h. A force can route and multiply those amplitudes without measuring them. Logical arithmetic alone does not make a register physical: unwanted modes, diffusion, induced velocity and historical errors must be controlled in the actual equations.

The even velocity target has the exact decomposition $U = U_0 + \sum_j h_j R_j$ with odd R_j and no division by h_j. Thus all nonlinear arithmetic can be performed magnetically while velocity is still disposable. The computation returns to the same finite code space after every assignment, using genuine smoothing and an actual mechanical restore. Nonexpansive whole-register assignments keep old coefficient errors from accumulating exponentially with the polynomial degree.

The final conversion to velocity is made once. All packet carriers point in one common direction. Solenoidality then removes the large carrier factor from the first nonlinear interaction. After physical high-frequency clearing, the entire higher output is inverse-carrier small. A bounded dyadic return word first defines a small preconditioned target and then amplifies the written velocity. The inverse word has polynomial gain on the *whole* mean-zero error space, including modes absent from the target. It returns ideal magnetic transport to the identity. Finally a known velocity increment adds U_0 while retaining both unknown fields.



Solid boxes are executed by momentum forcing and unforced evolution. The dashed box lists mathematical descriptions used to analyze them. In particular, neither a coordinate average nor an inverse heat operator is applied to the physical state. The preconditioner acts on the *known target coefficients* in the compiler. Its reverse effect on the fluid is implemented by actual mechanical words.

1.3. **How to read the proof.** Theorem 2.2 states the complete input and realization quantifiers. Section 3 isolates the stable finite arithmetic. Sections 4–7 develop the reusable linear, geometric, memory and heat foundation. Sections 8, 9 and 10 supply the new physical operations. Section 11 fixes their noncircular parameter order and resource ledger. Long estimates are placed in the appendices.

Notation	Meaning
H, $h = H/M$	Conserved unknown mean and normalized logical input
U, Z, R_j	Even velocity target, odd magnetic fluctuation, odd cofactors
D, N, F	Polynomial degree, retained output band, logical code frequency bound
$\mathscr{C}$, $T_i(\phi)$	Normalized curl and ideal differential magnetic gate
μ	$\min(\nu_-, \rho_-)$, a supplied positive lower diffusivity bound
Q, σ	Final readout carrier and its critical packet budget
g, G_a	Preconditioning gain and whole-space amplification bound
P	Computably generated positive-coefficient polynomial envelope

The symbols h and H denote flux variables. Local durations are written d_c, d, ℓ, T or as explicitly subscripted clocks. The resource polynomial can have very large fixed constants and powers. This is a finite-time PDE control result, not a cost advantage for a physical processor.

2. The model and the response theorems

Let $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$ have volume one and put $\kappa = 2\pi$. For a vector field, $\widehat{v}(k) = \int_{\mathbb{T}^3} v(x) e^{-i\kappa k\cdot x}\, dx$ and

$$\|v\|_{H^s}^2 = \sum_{k\in\mathbb{Z}^3} (1 + \kappa^2|k|^2)^s |\widehat{v}(k)|^2.$$

Matrix norms sum these squared norms over the entries. The joint norm $\|(u,B)\|_{\mathcal{H}^s}^2 = \|u\|_{H^s}^2 + \|B\|_{H^s}^2$ is inhomogeneous: constant magnetic fields are included. Let $\mathbb{P}_k = I - k\otimes k/|k|^2$ for $k \neq 0$ and $\mathbb{P}_0 = I$. On solenoidal fields set

$$\mathcal{B}(v,z) = \mathbb{P}[(v\cdot\nabla)z], \quad L_v z = \mathcal{B}(v,z) + \mathcal{B}(z,v), \quad \mathcal{C}_v b = (b\cdot\nabla)v - (v\cdot\nabla)b.$$

The sign convention for $\mathcal{C}$ makes the ideal induction map a vector pushforward. It is the negative of the convention for C in [1]. The physical equations are

$$\begin{aligned} u_t &= \nu(t)\Delta u - \mathcal{B}(u,u) + \mathcal{B}(B,B) + f + h_u, \\ B_t &= \rho(t)\Delta B + \mathcal{C}_u B + h_B, \\ \operatorname{div} u &= \operatorname{div} B = 0, \qquad \langle u\rangle = 0. \end{aligned} \tag{2.1}$$

Only the momentum force is chosen, and it always belongs to

$$\mathcal{E} = \operatorname{span}_{\mathbb{R}}\{e_i\cos(\kappa x_j),\ e_i\sin(\kappa x_j) : i \neq j\}. \tag{2.2}$$

The disturbance pair (h_u, h_B) is measurable, solenoidal and mean zero. Thus $H = \langle B_0\rangle$ is conserved. Both coefficients are spatially constant.

Definition 2.1 (Admitted realization)**.** An admitted realization is a distributional solution of (2.1), weakly continuous into $\mathcal{H}^0$ on $[0,\tau]$, in $L^\infty(0,\tau;\mathcal{H}^0)\cap L^2(0,\tau;\mathcal{H}^1)$, with the given initial data, and with total-energy inequality

$$\begin{aligned} &\tfrac12\|(u,B)(t)\|_2^2 + \int_s^t \big(\nu(r)\|\nabla u(r)\|_2^2 + \rho(r)\|\nabla B(r)\|_2^2\big)\,dr \\ &\qquad \le \tfrac12\|(u,B)(s)\|_2^2 + \int_s^t \big[\langle f + h_u, u\rangle + \langle h_B, B\rangle\big]\,dr \end{aligned} \tag{2.3}$$

for $s = 0$ and almost every $s > 0$, and every subsequent energy trace. At the terminal time use the weak left trace, whose norm is at most the lower limit of the preceding total energies. This is the one-sided total-energy endpoint convention of [1, Definition 2.1]. On the strong controlled part constructed below it agrees with the ordinary strong trace.

2.1. **The variable-degree response theorem.** For complete rational Fourier-polynomial input lists let

$$U(x,h) = \sum_\alpha U_\alpha(x)h^\alpha, \qquad Z(x,h) = \sum_\alpha Z_\alpha(x)h^\alpha. \tag{2.4}$$

Use the constant and half-lattice convention in Section 3. Every coefficient is real, mean zero and solenoidal. Only even total degrees occur in U, and only odd total degrees in Z. Define $D = \max(1, \max|\alpha|)$ over occurring terms. Exponents may be binary encoded, but the numerical value of D is charged below. The rational input promise is

$$\sum_{\alpha,k\in\mathcal{K}_+} (1+7|k|_1)\big(|U^c_{\alpha,k}|_1 + |U^s_{\alpha,k}|_1 + |Z^c_{\alpha,k}|_1 + |Z^s_{\alpha,k}|_1\big) \le K. \tag{2.5}$$

It is a bound on aggregated coefficients, not an oracle for a target norm.

Theorem 2.2 (Nonlinear flux-response programming)**.** *There are effectively computable universal integers $A, p \ge 1$ and a total deterministic compiler with the following property. It receives binary rationals*

$$\begin{gathered} M \ge 1, \quad K \ge 2, \quad 0 < \nu_- \le \nu_+ \le 1, \quad 0 < \rho_- \le \rho_+ \le 1, \\ 0 < \tau \le 1, \quad 0 < \epsilon < 1, \quad \Lambda \ge 1, \end{gathered}$$

and complete lists (2.4) satisfying the stated parity, spatial constraints and (2.5). Support, integer frequencies, monomial indices, rational coefficients and denominators are variable input. Set

$$R_* = 2 + M + K + D + \nu_-^{-1} + \rho_-^{-1} + \tau^{-1} + \epsilon^{-1} + \Lambda, \qquad L = \text{complete binary input length.} \tag{2.6}$$

The output is a positive rational η and an ordinary rational piecewise-polynomial force $f : [0,\tau] \to \mathcal{E}$ such that

$$\Big(\|u(\tau) - U(\cdot, H/M)\|_2^2 + \|B(\tau) - H - Z(\cdot, H/M)\|_2^2\Big)^{1/2} \le \epsilon, \qquad H = \langle B_0 \rangle, \tag{2.7}$$

simultaneously for every pair of independent real coefficient functions ν, ρ with the supplied ranges and individual Lipschitz constants at most Λ, every unknown real solenoidal initial pair of joint inhomogeneous H^1 norm at most M and zero velocity mean, every mean-zero solenoidal disturbance pair with $\operatorname{ess\,sup}_t \|(h_u, h_B)(t)\|_{\mathcal{H}^0} \le \eta$, and every admitted realization in Definition 2.1.

The compiler neither observes the fields or their means nor evaluates the actual coefficients, their derivatives, samples or descriptions. The same output covers irrational H, zero components and $H = 0$. Moreover,

$$\eta^{-1}, \quad \sup_t \|f(t)\|_{W^{1,\infty}}, \quad \int_0^\tau \|f(t)\|_{W^{1,\infty}}\, dt \le A R_*^p. \tag{2.8}$$

Compiler bit operations, memory bits, total output bits, interval count, time-polynomial degrees and individual rational bit lengths are at most $A(L + R_*)^p$. The exponent is independent of D, catalogue, coefficient rectangle and actual histories. All preparation, generated directions, precision and fixed precomputation are included. No PDE, norm, spectral, optimization, function-evaluation or exact-real oracle is used.

This is polynomial in the stated *numerical* parameters together with L, not polynomial time in binary length alone. In particular a binary exponent of enormous numerical degree incurs its numerical degree cost. We do not assert a fixed time degree for the new compiler; the bound above is sufficient. The older construction below has a separately proved stronger degree-four guarantee. Set $\mu = \min(\nu_-, \rho_-)$ throughout; then $\mu^{-1} \le R_*$.

2.2. **The retained linear foundation.** The following statement fixes the exact older input promise. Its proof is included in Sections 4–7 and Appendix D; the nonlinear construction uses its operations rather than treating its endpoint as an arbitrary memory-preserving service.

Theorem 2.3 (Linear response with Sobolev input)**.** *There are effectively computable universal integers $A, p \ge 1$ and a total deterministic compiler with the following property. Its input consists of binary rationals*

$$M \ge 1, \quad K \ge 2, \quad 0 < \nu_- \le \nu_+ \le 1, \quad 0 < \rho_- \le \rho_+ \le 1,$$
$$0 < \tau \le 1, \quad 0 < \epsilon < 1, \quad \Lambda \ge 1,$$

and complete rational Fourier lists for a real vector field V and a real matrix field Π, promised to satisfy

$$\begin{aligned} &\langle V \rangle = 0, \quad \operatorname{div} V = 0, \quad \|V\|_{H^1} \le K, \\ &\langle \Pi \rangle = I, \quad \operatorname{div}(\Pi e_j) = 0 \ (j = 1,2,3), \quad \|\Pi\|_{H^1} \le K. \end{aligned} \tag{2.9}$$

Support, integer frequencies, rational polarizations, and coefficient numerators and denominators are variable input. Put

$$R = 2 + M + K + \nu_-^{-1} + \rho_-^{-1} + \tau^{-1} + \epsilon^{-1} + \Lambda, \qquad L = \text{complete binary input length.} \tag{2.10}$$

The output is a positive rational η and an ordinary rational piecewise-polynomial force $f : [0,\tau] \to \mathcal{E}$, of degree at most four on each rational partition interval, such that

$$\big(\|u(\tau) - V\|_2^2 + \|B(\tau) - \Pi H\|_2^2\big)^{1/2} \le \epsilon, \qquad H = \langle B_0 \rangle, \tag{2.11}$$

simultaneously for every pair of real coefficient functions with the supplied ranges and individual Lipschitz constants at most Λ, every real solenoidal initial pair of joint H^1 norm at most M and zero velocity mean, every admitted disturbance with $\operatorname{ess\,sup}_t \|(h_u, h_B)(t)\|_{\mathcal{H}^0} \le \eta$, and every corresponding admitted realization.

The compiler neither observes the fields or their means nor evaluates the actual coefficients, their derivatives, samples or descriptions. It imposes no determinant, positivity, invertibility, commuting-column or smallness condition on Π. *All* H, *including irrational* H *and* $H = 0$, *are covered by the same output. Moreover,*

$$\eta^{-1}, \quad \sup_{t} \|f(t)\|_{W^{1,\infty}}, \quad \int_0^{\tau} \|f(t)\|_{W^{1,\infty}}\, dt \le AR^{p}. \tag{2.12}$$

Bit operations, memory bits, total output bits, interval count and individual rational bit lengths are at most $A(L + R)^{p}$. *The constants include all fixed preprocessing and are independent of the input and target catalogue.*

The resource variable R in (2.10) belongs to this linear statement; R_* in (2.6) belongs to the variable-degree theorem. Their promises must not be identified. The polynomial conversion proving linear recovery is given in Section 11.

2.3. **Exact earlier control import.** We use [1, Theorem 2.2] with target zero for common preparation near $(0, H)$, including its disturbance and all-realization scope. Its input radius is the supplied bound for the actual preparation and its time horizon is a prescribed positive rational. Time restriction and translation preserve both coefficient ranges and Lipschitz bounds. The final disturbance allowance is no larger than the one that theorem returns. These facts verify all its hypotheses at each use. Its Definition 2.1 supplies the energy-trace convention.

The explicit returning words, actual-viscosity correctors, fixed-depth spatial synthesis and actuator lowering of the same source are identified in Appendix C. Their additional memory and amplification contracts are established here from their actual equations. A velocity setter permitted to erase velocity is never used to preserve a computed unknown velocity. Coordinate averaging and Fourier multipliers occur as comparison objects until their physical realizations have been proved.

3. Stable arithmetic of an unobserved flux

The conserved mean supplies three real inputs, but it is neither measured nor inserted into the force. Arithmetic below describes which coefficients the magnetic field is to carry. Its physical implementation is a separate task, proved in Section 8.

3.1. **Canonical input and uniform truncation.** Let $\mathcal{K}_+$ contain the nonzero integer vectors whose first nonzero component is positive. A real coefficient field has the unique convention

$$F(x) = F_0 + \sum_{k \in \mathcal{K}_+} [F_k^{c} \cos(\kappa k \cdot x) + F_k^{s} \sin(\kappa k \cdot x)].$$

There is one constant cosine coefficient and no constant sine coefficient. The input promise makes $F_0 = 0$. A negative wavevector changes the sine sign and leaves the cosine coefficient unchanged. Duplicate entries are aggregated as rational numbers before the promise is tested. Put

$$|F|_{\mathrm{a}} = \sum_{k \in \mathcal{K}_+} (|F_k^{c}|_1 + |F_k^{s}|_1), \qquad |F|_{\mathcal{X}} = \sum_{k \in \mathcal{K}_+} (1 + 7|k|_1)(|F_k^{c}|_1 + |F_k^{s}|_1).$$

Here the vector norm $|\cdot|_1$ sums its three component moduli. Since $\kappa < 7$, Minkowski's inequality gives $\|F\|_{H^1} \le |F|_{\mathcal{X}}$. For a polynomial, coefficient mass means the sum over all its monomials.

Lemma 3.1 (Paid spatial truncation)**.** *Set* $N = \lceil 64K/\epsilon \rceil$. *Discarding listed frequencies with* $|k|_\infty > N$ *gives* U_N, Z_N *with*

$$\sup_{|h| \le 1} \big(\|U(h) - U_N(h)\|_2 + \|Z(h) - Z_N(h)\|_2\big) \le \frac{K}{1 + 7(N+1)} < \epsilon/64. \tag{3.1}$$

For fixed $r, s \geq 0$, the retained targets have spatial H^{r+1} and parameter derivative order s envelopes $C_{r,s}K(1+N)^r(1+D)^s$. Canonicalization and dense retained storage cost polynomially many bits and operations in $L + N + D$.

Proof. Every monomial has modulus at most one on $|h| \leq 1$. Each omitted coefficient has weight at least $1 + 7(N + 1)$, proving (3.1) by the triangle inequality. A spatial derivative multiplies a retained complex coefficient by at most $\kappa\sqrt{3}N$, and a fixed parameter derivative multiplies it by at most D^s. Summing the promised masses proves the envelopes. There are at most $\binom{D+3}{3}$ monomial keys and $C(N + 1)^3$ spatial keys. A scan compares each binary input frequency with N; it never enumerates the discarded large frequency. Exact rational aggregation has polynomial bit cost in the complete input length, and multiplication by multi-index ratios adds only polynomially many bits. □

3.2. **Reducing the velocity target to odd fields.** Magnetic sign reversal requires odd magnetic registers. It does not require division by a nonzero component of the input.

Lemma 3.2 (Odd factorization of an even response)**.** *Let $U_0 = U_N(\cdot, 0)$ and define*

$$R_j(x, h) = \sum_{\substack{|\alpha|>0 \text{ even} \\ \alpha_j>0}} \frac{\alpha_j}{|\alpha|}(U_N)_\alpha(x)h^{\alpha-e_j}. \tag{3.2}$$

Then $U_N = U_0 + \sum_{j=1}^3 h_j R_j$. Each R_j is odd, solenoidal and mean zero, and

$$\sum_j \sum_\beta |(R_j)_\beta|_{\mathcal{X}} \leq \sum_{\alpha\neq 0} |(U_N)_\alpha|_{\mathcal{X}}.$$

Proof. For each nonzero α, the coefficients $\alpha_j/|\alpha|$ are nonnegative and sum to one. Multiplying (3.2) by h_j and summing proves the identity, including when $h_j = 0$. Removing one exponent reverses total parity and leaves the spatial constraints unchanged. The same nonnegative weights prove the mass bound. □

3.3. **Eleven logical registers.** Every odd polynomial F has the unique parity decomposition

$$F(h) = \sum_{S\in\mathcal{S}} h_S P_S(t_1, t_2, t_3), \qquad t_i = h_i^2, \quad h_S = \prod_{i\in S} h_i, \quad \mathcal{S} = \{\{1\}, \{2\}, \{3\}, \{1, 2, 3\}\}. \tag{3.3}$$

Indeed write each exponent as $2\beta + \mathbf{1}_S$. This partitions rather than duplicates coefficients. Three nested Horner accumulators evaluate each P_S, retaining the factor h_S throughout. Their only variable multiplication is $r \mapsto h_i^2 r$.

The eleven registers are four outputs Z_N, R_1, R_2, R_3, three nested work registers, and four permanent sources $h_1, h_2, h_3, h_1h_2h_3$. The last source is formed once by the cubic operation in Lemma 8.5; its normalization uses the known M^{-3}. A register can hold an entire spatial coefficient array. Its physical code therefore grows with the retained Fourier list, while the number of logical registers stays fixed.

To make the schedule explicit, for each output and each S, run the outer t_1 index downward. For each outer index, run the middle t_2 index downward, evaluating its inner t_3 row by downward Horner. Inject the known coefficient array times the permanent source h_S at the appropriate inner step. Transfer the finished inner accumulator into the middle one while clearing it; transfer the middle result into the outer one while clearing it. Finally accumulate the outer result into its output and clear the outer register. Empty rows contribute zero. Dense degree ranges of length at most $D + 1$ give at most $C(D + 1)^3$ assignments over the sixteen output/parity sectors.

Proposition 3.3 (Nonexpansive arithmetic)**.** *In the sum of the coefficient norms of work and output registers, the intended live state has mass at most K. Permanent sources have bounded*

additional mass. The schedule uses at most $S_a = C(D+1)^3$ whole-register assignments. With source error e_s and fresh absolute error at most δ per assignment,

$$E_{\text{final}} \le E_{\text{initial}} + Ke_s + S_a\delta. \tag{3.4}$$

If a square multiplier has uniform error at most δ_h, the right side may instead be multiplied by $\exp(CS_a\delta_h)$, with an additional term $CKS_a\delta_h$ inside. Thus inverse-polynomial local accuracy suffices.

Proof. A destructive transfer is the single map

$$(d, s) \longmapsto (td + s, 0), \qquad |t| \le 1,$$

so $|td + s|_{\mathcal{X}} \le |d|_{\mathcal{X}} + |s|_{\mathcal{X}}$. An accumulate-and-clear operation $(o, x) \mapsto (o + x, 0)$ obeys the same inequality. Applying them separately and charging a copy before clearing would lose this property. Known injections read a permanent source without clearing it; the sum of their gains is bounded by the injected coefficient mass K. Every coefficient is injected once. Induction proves the live mass bound and (3.4). A perturbed multiplier increases one step's error by at most δ_h times its live mass and changes the step norm to at most $1 + C\delta_h$. Multiplying those scalar bounds gives the stated estimate. Choosing $S_a\delta_h \le 1$ bounds its exponential by an absolute constant. No statement here implements a physical copy, projection or multiplication. □

Example 3.4 (A genuinely nonlinear response)**.** With $s_j = \sin(\kappa x_j)$ and $c_j = \cos(\kappa x_j)$, take

$$U(x, h) = \frac{h_1^2 + h_2h_3}{32} e_1s_2(x), \qquad Z(x, h) = \frac{h_1h_2h_3}{32} e_2c_3(x).$$

The coefficient fields have zero mean and divergence. Their joint weighted mass is $3(1+7)/32 = 3/4$, so $K \ge 2$ suffices. The degrees are two and three. Here $U_0 = 0$ and

$$R_1 = \tfrac{1}{32}h_1e_1s_2, \quad R_2 = \tfrac{1}{64}h_3e_1s_2, \quad R_3 = \tfrac{1}{64}h_2e_1s_2.$$

The construction requests zero magnetic output at $H = 0$, conserves the mean at every other flux, and never needs to test whether any component vanishes. These identities check the target algebra; they are not a numerical simulation of the compiler.

4. Five gates on one unknown field

The construction in this section proves the retained linear foundation. Here V, Π and its local cutoff N have the Sobolev promise of Theorem 2.3; nonlinear input truncation was treated separately in Lemma 3.1. The outer pilot frequencies Q_1, Q_2, Q_3 and the microscopic gate carrier Q_{g} below are distinct from the single final readout carrier Q of Section 10. Write $D_j = \kappa^{-1}\partial_j$ and $\mathscr{C} = \kappa^{-1}\,\mathrm{curl}$. For a smooth scalar ϕ, define the differential gate

$$T_i(\phi)B = B + \mathscr{C}(e_i\phi B_i). \tag{4.1}$$

Its added field is solenoidal, has zero mean and has zero ith component. Consequently $(T_i - I)^2 = 0$ on smooth fields. This elementary nilpotence does not make the gate a bounded operator on H^1; it differentiates B_i. Section 6 constructs a finite physical approximation on a smooth reference and separately bounds its action on a rough discrepancy.

Example 4.1 (Why a reference is necessary)**.** For $\phi = 1$ and $b_n = n^{-1}e_1\sin(\kappa n x_2)$, the fields b_n are solenoidal and bounded in H^1, but

$$(T_1(1) - I)b_n = -e_3\cos(\kappa n x_2)$$

has H^1 norm growing linearly with n. Thus the differential gate is not bounded $H^1 \to H^1$. Our finite implementation approximates its action on the supplied smooth reference; it propagates the actual rough error with its own, finite polynomial gain.

4.1. **Canonical targets and an explicit curl potential.**

Lemma 4.2. *The complete input can be canonicalized and truncated at $N = \lceil 128MK/\epsilon \rceil$ in polynomial bit work in $L + R$ so that*

$$\|V - V_N\|_2 + \sup_{|H|\le M} \|(\Pi - \Pi_N)H\|_2 \le \epsilon/32. \tag{4.2}$$

The truncated fields retain (2.9)'s mean and divergence constraints. There is an explicitly computable real rational Fourier matrix Γ, of radius at most N, such that $\mathscr{C}\Gamma = \Pi_N - I$ columnwise. For each fixed m, all required mth-order Fourier and supremum envelopes of V_N, Π_N, Γ are bounded by a fixed polynomial in K, N.

Proof. Normalize the signs of wavevectors and coalesce duplicate sine and cosine coefficients, including their rational polarizations, by exact arithmetic. Do this before using a norm promise: a redundant list may contain arbitrarily large cancelling terms. Discard a mode by comparing the binary integer $|k|^2$ with N^2; no enumeration up to the original largest frequency occurs. For $|k| > N$, $(1 + \kappa^2|k|^2)^{-1/2} \le (\kappa N)^{-1}$. Parseval and the matrix Cauchy–Schwarz inequality give tail bounds $K/(\kappa N)$ and $MK/(\kappa N)$, respectively, proving (4.2). Orthogonal Fourier truncation preserves the remaining constraints.

For $k \neq 0$ and column j set

$$\widehat{\Gamma^{(j)}}(k) = \frac{i\,k \times \widehat{(\Pi_N - I)^{(j)}}(k)}{|k|^2}, \qquad \widehat{\Gamma}(0) = 0. \tag{4.3}$$

The symbol of $\mathscr{C}$ is $ik\times$, so the divergence constraint proves the desired identity. Conjugate symmetry proves reality; the corresponding real sine/cosine coefficients are rational. No inverse of Π_N is taken. There are at most $(2N+1)^3$ retained frequencies. Cauchy–Schwarz bounds their coefficient sum by $C(N+1)^{3/2}K$; the multiplier in (4.3) has size at most one. Replacing κ by seven gives, for instance, the rational envelope

$$\mathcal{F}_m(F) = 1 + \sum_k (|F_k^c|_1 + |F_k^s|_1)(1 + 7|k|_1)^m \le C_m(1+K)(1+N)^{m+3}. \tag{4.4}$$

It bounds all needed fixed-order norms by the triangle inequality. Exact coalescing and rational arithmetic have polynomial bit cost; long input denominators affect that cost, rather than these numerical bounds. □

Write Γ_i for row i. Then

$$\Pi_N H = H + \sum_{i=1}^{3} \mathscr{C}(e_i \Gamma_i H). \tag{4.5}$$

The difficulty is that each gate acts on the current field, so its action on constants cannot simply be added to the effects of preceding gates. Temporary oscillations solve this problem.

4.2. **Encoding, reading and removing the pilots.** Choose integers satisfying

$$Q_3 \ge 16(N+1), \quad Q_2 \ge 16(Q_3 + N + 1), \quad Q_1 \ge 16(Q_2 + N + 1). \tag{4.6}$$

Set $w(\theta) = (\sin\theta, \cos\theta, \sin 2\theta)$. Phase averaging gives

$$\langle w_j \rangle_\theta = 0, \qquad \langle w_j w_l \rangle_\theta = \tfrac{1}{2}\delta_{jl}. \tag{4.7}$$

These identities are used as coefficient-extraction identities for Fourier polynomials; phase averaging is not an additional experiment.

Proposition 4.3 (Fixed-depth row writing)**.** *Five gates (4.1), separated by three idealized carrier selections, take the same constant field H to $\Pi_N H$ for every $H \in \mathbb{R}^3$. Their scalar coefficients depend on Π_N and the three integers, but not on H. At selection j, the input has a decomposition*

$$L_j H + W_j(Q_j)H, \tag{4.8}$$

where L_j is independent of Q_j, the unwanted term has frequencies of modulus at least $Q_j/2$, and

$$\sup_{|H|\le M} \|W_j(Q_j)H\|_{H^1} \le C_j(1+Q_j)^2. \tag{4.9}$$

Here C_j has a computable polynomial bound in the target envelopes and the already specified lower pilot frequencies. All intermediate reference matrices are finite Fourier fields with effective fixed-order envelopes.

Proof. Starting from H, use the two encoding gates

$$T_2\left(\frac{\cos(\kappa Q_1 x_3)}{Q_1}\right), \qquad T_3\left(-\frac{\cos(\kappa Q_1 x_2)}{Q_1}\right).$$

Since $\mathscr{C}(e_2\psi) = (-D_3\psi, 0, D_1\psi)$ and $\mathscr{C}(e_3\psi) = (D_2\psi, -D_1\psi, 0)$, their exact output is

$$B_1 = H_1 + \sin(\kappa Q_1 x_3)H_2 + \sin(\kappa Q_1 x_2)H_3, \qquad B_2 = H_2, \quad B_3 = H_3. \tag{4.10}$$

No second magnetic field has been prepared or added.

Let the zero-mean row

$$\chi_2(x_3) = \left(-\frac{\cos(\kappa Q_2 x_3)}{Q_2}, \frac{\sin(\kappa Q_2 x_3)}{Q_2}, -\frac{\cos(2\kappa Q_2 x_3)}{2Q_2}\right)$$

satisfy $D_3\chi_2 = w(\kappa Q_2 x_3)$. Put $A_1 = \Gamma_1 + \chi_2$ and

$$\phi_1 = A_{11} + 2A_{12}\sin(\kappa Q_1 x_3) + 2A_{13}\sin(\kappa Q_1 x_2). \tag{4.11}$$

In the product $\phi_1 B_1$, the part with no nonzero Q_1 carrier is $A_{11}H_1 + A_{12}H_2 + A_{13}H_3 = A_1 H$. The cross product of the two different pilots still has a nonzero carrier. Applying $T_1(\phi_1)$ and selecting away those carriers leaves

$$\Sigma_1 H = H + \mathscr{C}(e_1\Gamma_1 H) + e_2 w(\kappa Q_2 x_3)H. \tag{4.12}$$

The original pilots in B_1 are selected away as well; they do not cancel.

Define $\chi_3(x_1)$ by the same formula with Q_3, x_1 so $D_1\chi_3 = w(\kappa Q_3 x_1)$, and put

$$A_2 = \Gamma_2 + \chi_3, \qquad \phi_2 = 2\sum_{j=1}^{3} A_{2j} w_j(\kappa Q_2 x_3). \tag{4.13}$$

The second component of (4.12) is $(e_2^T + D_3\Gamma_1)H + w(\kappa Q_2 x_3)H$. Equation (4.7) therefore makes the carrier-free part of $\phi_2 B_2$ exactly $A_2 H$. The product with its base row has a nonzero carrier. Applying $T_2(\phi_2)$ and selecting at Q_2 gives

$$\Sigma_2 H = H + \mathscr{C}(e_1\Gamma_1 H) + \mathscr{C}(e_2\Gamma_2 H) + e_3 w(\kappa Q_3 x_1)H. \tag{4.14}$$

Finally set

$$\phi_3 = 2\sum_{j=1}^{3} \Gamma_{3j} w_j(\kappa Q_3 x_1). \tag{4.15}$$

The third component in (4.14) is a base row of radius at most N plus $w(\kappa Q_3 x_1)H$. Its product with ϕ_3 has carrier-free part $\Gamma_3 H$. Gate $T_3(\phi_3)$ and selection at Q_3 yield (4.5).

Here selection can be defined precisely as Fourier cutoff at radius $Q_j/2$. In the first step the unwanted carrier vectors are nonzero combinations of $Q_1 e_2, Q_1 e_3$ with offsets of radius at most $N + 2Q_2$. In the second step their nonzero carrier indices are at most four, with offsets bounded by $2N + 2Q_3$; the third has offsets at most $2N$. The separations (4.6) leave a gap at $Q_j/2$ in every case. The retained matrices are respectively $L_1 = \Sigma_1$, $L_2 = \Sigma_2$ and $L_3 = \Pi_N$, each independent of the frequency currently removed. All coefficient sums follow by finite convolution. At each step only one curl and the single derivative in the H^1 norm cost powers of the current carrier. Scalar amplitudes and all lower-frequency factors have bounds independent of that carrier. This proves (4.9) and gives the claimed computable envelopes. □

Example 4.4 (A singular response with noncommuting columns)**.** Take

$$\Pi = I + 2\cos(\kappa x_2)e_1e_1^T, \qquad \Gamma_3 = (2\sin(\kappa x_2), 0, 0), \quad \Gamma_1 = \Gamma_2 = 0.$$

It has solenoidal columns and mean I. Its determinant $1 + 2\cos(\kappa x_2)$ vanishes on two tori, and $[\Pi e_1, \Pi e_2] = 2\kappa\sin(\kappa x_2)e_1$ is nonzero. After the first two row-writing stages the field is $H + e_3 w(\kappa Q_3 x_1)H$. The last scalar is $\phi_3 = 4\sin(\kappa x_2)\sin(\kappa Q_3 x_1)$. Its product with B_3 has low part $2\sin(\kappa x_2)H_1$, because $\langle \sin^2\theta\rangle = 1/2$, whereas its products with $H_3, \cos\theta H_2, \sin 2\theta H_3$ have zero low part. Taking $\mathscr{C}(e_3\,\cdot)$ writes $2\cos(\kappa x_2)H_1e_1$ as required. The construction has used neither H_1 nor an inverse of Π. The exact-arithmetic code in the source archive reproduces this example and an example using all nine potential entries. It checks finite Fourier identities, not the physical compiler or an infinite-dimensional estimate.

4.3. **From coefficient selection to actual operations.** The three selectors in Proposition 4.3 are not applied controls. A positive heat interval damps their high carriers but also changes the retained field. The duration will be chosen from that retained field *before* its discarded carrier frequency is chosen. The physical heat operation proved in Section 7 is valid for that already chosen duration, even when the complete intermediate field subsequently has a much larger smooth envelope.

This is also why the reference-plus-discrepancy formulation matters. The reference at the next stage is $\Sigma_j H$ with a known envelope. The small remaining high modes and physical errors form an arbitrary H^1 discrepancy; the next differential gate is not applied to it formally. Instead its finite physical implementation has a paid polynomial error propagation bound.

5. Velocity operations preserving smooth magnetic memory

The magnetic field produced by an earlier operation is retained during all later mechanical operations. This requires more than an endpoint theorem for velocity. We use the actual returning words and physical writing leaves of [1], and prove their smooth-memory extensions.

5.1. **A fixed magnetic metric and its coercivity.** Write $J = I - \Delta$. If Φ is a volume-preserving periodic flow map, let

$$(\mathcal{M}_\Phi z)(x) = D\Phi(y)z(y), \qquad y = \Phi^{-1}(x).$$

It preserves solenoidality. It also preserves the mean of a solenoidal field: on a periodic lift $\Phi(y) = y + \xi(y)$, $\int D\Phi z = \int z - \int \xi \operatorname{div} z$. Naturality of the Lie bracket gives

$$\mathcal{M}_\Phi^{-1}\mathcal{C}_v\mathcal{M}_\Phi = \mathcal{C}_{\mathcal{M}_\Phi^{-1}v}. \tag{5.1}$$

A *fixed magnetic metric* associated with an accumulated pushforward G is the norm $\|G^{-1}b\|_{H^1}$. The word "fixed" refers to the metric at both endpoints of a mechanical operation. Its coefficients may depend on actual profiles in the proof, provided that supplied uniform geometric envelopes bound them. The force uses only those envelopes.

Lemma 5.1 (Geometric Gårding bound)**.** *Suppose Φ and Φ^{-1} have supplied bounds through a fixed sufficient derivative order. There are computable polynomial bounds for a_0^{-1}, ω_0 such that*

$$\operatorname{Re}\langle \mathcal{M}_\Phi^{-1}\Delta\mathcal{M}_\Phi z, Jz\rangle \le -a_0\|z\|_{H^2}^2 + \omega_0\|z\|_{H^1}^2. \tag{5.2}$$

No time derivative of the geometry is required.

Proof. In material coordinates put $A = D\Phi$, $g = A^{-1}A^{-T}$ and $L = \partial_i(g^{ij}\partial_j)$. The chain rule gives

$$\mathcal{M}_\Phi^{-1}\Delta\mathcal{M}_\Phi z = A^{-1}L(Az) = Lz + 2A^{-1}g^{ij}(\partial_i A)\partial_j z + A^{-1}(LA)z. \tag{5.3}$$

The second-order principal part is scalar and uniformly elliptic, with polynomial inverse ellipticity. Pair with Jz and integrate by parts. The principal Hessian term is negative; differentiating g, and the last two terms, cost at most $P\|z\|_{H^1}\|z\|_{H^2}$. Young's inequality

proves (5.2), with the constant Sobolev mass included in the lower-order term. All coefficients are obtained by finitely many derivatives, products and inverse-matrix formulas with supplied determinant and inverse bounds. □

For Euler propagators, bounded invertibility alone does not imply this coercivity. We use the stronger result [1, Lemmas 8.1–8.2] for fixed products of fundamental shears. Its proof writes the conjugated Laplacian as a scalar elliptic principal part plus an $H^1 \to L^2$ remainder. The remainder is controlled by multiplier commutators on both $H^{-1} \to L^2$ and $L^2 \to H^1$; those two scales permit closure under the fixed products. For a generic smooth field ξ used only at small impulse a, direct commutator integration instead gives

$$\|e^{aL_\xi}\Delta e^{-aL_\xi} - \Delta\|_{H^2\to L^2} \le P|a|. \tag{5.4}$$

Indeed $[\Delta, L_\xi] : H^2 \to L^2$ has bound $C\|\xi\|_{C^3}$: the leading third derivatives cancel and $\mathbb{P}$ commutes with Δ. The intervening propagators are bounded by $e^{C|a|\|\xi\|_{C^k}}$ on the finitely many needed orders. A small impulse makes this a fixed bound and preserves coercivity. We make no long-impulse claim for a generic Fourier field.

5.2. **A raw reset bound before choosing endpoint accuracy.** The important uniformity is in the velocity remainder and the raw endpoint, not in the transient large carrier velocity.

Lemma 5.2 (Uniform raw reset)**.** *Let $\mathsf{B} \ge 2$ bound the incoming joint physical H^1 norm. There is a common twelve-mode reset with an arbitrarily prescribed positive rational duration cap, raw endpoint velocity L^2 tolerance σ_0, and a polynomial bound*

$$\sup(\|u-\widehat{w}\|_{H^1} + \|B\|_{H^1}) \le P_0, \qquad \int \mu\mathcal{D}^2 \le P_0^2, \qquad \|(u,B)_{\mathrm{raw}}\|_{\mathcal{H}^1} \le P_0. \tag{5.5}$$

Here $\widehat{w}$ is its exact product induction carrier, $\mathcal{D}$ is the corresponding joint transformed H^2 norm, and P_0 depends polynomially on $\mathsf{B}, \mu^{-1}, 1+\Lambda$, independently of σ_0. The chosen force, inverse duration, counts and binary costs are polynomial in the supplied bounds, inverse cap and σ_0^{-1}. The carrier vanishes at the raw endpoint.

Proof. The precise structural imports are [1, Lemma 4.1, Proposition 4.5, Corollary 4.6, Lemma 4.7]. One fixed word of small impulse s has induction endpoint I and Euler endpoint $W(s)$, with

$$\|W(s) - e^{s^4 D}\|_{H^m\to H^m} \le C_m s^5, \quad \|e^{tD}\|_{H^m\to H^m} \le 1, \quad \|W(s)^j\|_{H^m\to H^m} \le e^{C_m j s^5}. \tag{5.6}$$

Its stable forward repetition contracts $H^1 \to L^2$ to a specified tolerance with polynomial s^{-1}, n, while ns^5 stays small. Emit w_t and define only in the proof

$$a_t + L_w a = \nu(t)\Delta w, \qquad a(0) = 0.$$

The actual-viscosity corrector satisfies, in each fixed required order,

$$\sup\|a\|_{H^m} \le C_m(s + ns^5 + \Lambda s\delta)e^{C_m n s^5}. \tag{5.7}$$

Its large time derivative cancels; the remaining momentum source is $\nu\Delta a - \mathcal{B}(a,a)$. No value of ν is emitted.

On one complete word restart $Q_t = -L_w Q$, $R_t = \mathcal{C}_w R$ at identity, and write $u = w + a + Qz$, $B = Rb$. After exact cancellation the equations are

$$\begin{aligned} z_t &= \nu Q^{-1}\Delta Q z - Q^{-1}L_a Q z - Q^{-1}\mathcal{B}(Qz,Qz) + Q^{-1}\mathcal{B}(Rb,Rb) \\ &\quad + Q^{-1}\{\nu\Delta a - \mathcal{B}(a,a) + h_u\}, \\ b_t &= \rho R^{-1}\Delta R b + R^{-1}\mathcal{C}_{a+Qz}Rb + R^{-1}h_B. \end{aligned} \tag{5.8}$$

Partial maps and inverses have fixed signed-Sobolev bounds. Their conjugated diffusions equal $\Delta + O(s)$ from H^2 to L^2. Pairing with Jz, Jb, the product estimate

$$\|\mathcal{B}(v,z)\|_2 + \|\mathcal{C}_v z\|_2 \le C\|(v,z)\|_{\mathcal{H}^1}^{3/2}\|(v,z)\|_{\mathcal{H}^2}^{1/2} \tag{5.9}$$

and Young's inequality yield, with $Y = \|(z,b)\|_{\mathcal{H}^1}^2$ and $\mathcal{D} = \|(z,b)\|_{\mathcal{H}^2}$,

$$Y' + c\mu\mathcal{D}^2 \le C\mu^{-3}(1+Y)^3. \tag{5.10}$$

For example the separate velocity Lorentz term is $C\|b\|_{H^1}^{3/2}\|b\|_{H^2}^{1/2}\|z\|_{H^2}$; its integrated dissipation, rather than a pointwise magnetic H^2 bound, pays this term. Smooth correctors are made small by (5.7); disturbances of norm at most one are included.

At complete words $z^+ = W(s)z^-$ and $b^+ = b^-$. Since $\|W(s)\|_{H^1\to H^1} \le 1 + Cs^5$, weighting word j by $(1+Cs^5)^{-2j}$ removes upward energy jumps. This weighting is uniformly equivalent to one when ns^5 is small. For $\delta \le c\mu^3(1+\mathsf{B})^{-4}$, first-exit integration of (5.10) gives

$$\sup(\|u-w\|_{H^1} + \|B\|_{H^1}) \le C\mathsf{B}, \qquad \int \mu\mathcal{D}^2 \le C\mathsf{B}^2. \tag{5.11}$$

These constants do not depend on the demanded contraction. Stable forward products, never an inverse long contracting product, give by variation of constants

$$\|u(\delta) - W(s)^n u_{\mathrm{in}}\|_2 \le C\{\mathsf{B}\sqrt{\delta} + \mathsf{B}^2\mu^{-1/4}\delta^{3/4} + \delta\} + \|a(\delta)\|_2. \tag{5.12}$$

The Lorentz integral follows from Hölder and $\int\|B\|_{H^2}^2 \le C\mu^{-1}\mathsf{B}^2$. Choose the contraction below $\sigma_0/(4C\mathsf{B})$, the smooth allowance below $\sigma_0/(4C)$, and then a polynomial duration cap making the right side of (5.12) below $\sigma_0/2$. The raw bound (5.11) is unaffected.

We now check actual lowering, using the concrete construction of [1, Section 8.6]. On a slot of length d_{c} put

$$\psi(z) = z(1-z)(1-2z), \quad q = a\psi(t/d_{\mathrm{c}}), \quad Q_0 = ad_{\mathrm{c}}\,\tfrac{1}{2}(t/d_{\mathrm{c}})^2(1-t/d_{\mathrm{c}})^2.$$

Then q and Q_0 have zero endpoint values, $\int\psi = 0$, $\int\psi^2 = 1/210$, and $\sup|Q_0| = |a|d_{\mathrm{c}}/32$. For $E = e^{-Q_0 L_F}$, actual force $q'F + f_{\mathrm{low}}$ and $u = qF + Ey$, substitution gives

$$\begin{aligned} y_t = {}& \nu E^{-1}\Delta Ey - E^{-1}\mathcal{B}(Ey,Ey) - q^2E^{-1}\mathcal{B}(F,F) \\ & + \nu qE^{-1}\Delta F + E^{-1}\{f_{\mathrm{low}} + \mathcal{B}(B,B) + h_u\}. \end{aligned} \tag{5.13}$$

The mean quadratic force is $-a^2\mathcal{B}(F,F)/210$. Its oscillatory primitive is $O(a^2d_{\mathrm{c}})$ and its conjugation mean error is $O(\sup|Q_0|a^2)$. The viscosity term has the exact primitive identity

$$qE^{-1}\Delta F = \frac{d}{dt}\left(\int_0^{Q_0(t)} e^{sL_F}\Delta F\,ds\right). \tag{5.14}$$

Integration against Lipschitz ν gives a zero-endpoint proof primitive of size $C_F\sup|Q_0|(1+\Lambda d_{\mathrm{c}})$ and a mean residue $C_F\Lambda\sup|Q_0|$. The primitive contains only smooth quadratic, viscosity and old-force discrepancies. The rough Lorentz term remains in (5.13), with no high-derivative assumption.

The continuous smooth correction b_ℓ of the lowering construction has the exact additive residual identity

$$r_{\mathrm{new}} = r_{\mathrm{old}} + \nu\Delta b_\ell - L_{b_{\mathrm{old}}+\zeta}b_\ell - \mathcal{B}(b_\ell,b_\ell). \tag{5.15}$$

If H_* bounds the already fixed old spatial norms, first time derivatives and clocks, and A_* bounds new amplitudes, zero slot integrals and $\sum d_j^2 \le Td_{\max}$ give

$$\sup\|b_\ell\|_{H^m} \le P(H_*,A_*)[\delta_{\mathrm{amp}} + d_{\max}(1+\Lambda T)]. \tag{5.16}$$

Choose absolute amplitude precision and then mesh so this remains small after multiplication by every already fixed amplitude in (5.15). This is a whole-operation bound, without a cell-count multiplier.

The magnetic coordinate is not E. With $M = e^{Q_0\mathcal{C}_F}$,

$$-L_F - \mathcal{C}_F = -2\mathbb{P}[(\cdot\nabla)F], \qquad \|(E-M)w_0\|_{H^1} \le C_F\sup|Q_0|\|w_0\|_{H^2}. \tag{5.17}$$

The exact new induction map is MR_0 and its generator is $qF + Mw_0$, whereas the momentum normal form has $qF + Ew_0$. For nested lowering the exact product generator obeys $\widehat{w}_j = \zeta_j + (\phi_j)_*\widehat{w}_{j-1}$. Its difference from the simultaneous carrier sum is bounded by the small

partial impulse times already fixed old spatial norms. The Euler product defect $L_A - \mathrm{Ad}_E L_A$ has $H^2 \to L^2$ norm at most $P \sup |Q_0| \|A\|_{C^k}$ by commutator integration. The induced diffusion defect has the paid $H^{r+2} \to H^r$ bounds for $-2 \le r \le 2$. Choose these defects below a fixed fraction of the coercivity margin, after fixing all old amplitudes. They are absorbed in (5.10); smooth residues are bounded by (5.16). The quadratic terms still satisfy (5.9) with fixed map bounds.

At any time there is only a fixed-depth product of partial lowering maps; completed excursions are identity. Its geometry therefore has uniform bounds independent of the number of cells. All retained old endpoints have the original Euler endpoint and exact induction return, so the restart calculation remains unchanged. This proves (5.5) for the twelve actual actuators. Refining the absolute defects below the divided σ_0 allowance preserves (5.12), without enlarging its raw state bound. Each next mesh is chosen after its amplitudes and old clocks; fixed lowering depth gives polynomial reciprocal clocks and costs. Coarse per-cell composition of state-envelope polynomials is neither used nor valid here. □

5.3. **The magnetic discrepancy and the final smoothing tail.** The following asymmetric product bound is particularly useful:

$$\|\mathcal{C}_v z\|_2 \le C\|v\|_{H^1}\|z\|_{H^1}^{1/2}\|z\|_{H^2}^{1/2}, \qquad \|\mathcal{C}_v \beta\|_2 \le C\|\beta\|_{W^{1,\infty}}\|v\|_{H^1}. \tag{5.18}$$

For $v \cdot \nabla z$ use $L^6 L^3$; for $z \cdot \nabla v$ use $L^\infty L^2$ and the three-dimensional H^1/H^2 Agmon inequality. Thus no H^2 velocity factor is needed in this magnetic estimate.

Lemma 5.3 (Memory-preserving reset)**.** *Supply an incoming joint state bound, a fixed prefix G, and fixed-order smooth and geometric envelopes for the family $G\beta$. For every positive rational duration cap and endpoint tolerances, a common reset makes velocity H^1-small and has, in the same G-metric,*

$$E_{\mathrm{out}} \le (1 + P\delta)E_{\mathrm{in}} + P\sqrt{\delta}(1 + \eta). \tag{5.19}$$

Here δ is the total reset time after a harmless fixed enlargement of P. In particular, on a supplied input radius it can ensure $E_{\mathrm{out}} \le E_{\mathrm{in}} + r$ for any prescribed $r > 0$. All costs and inverse tolerances are polynomial in the supplied parameters. Its force uses family envelopes, never the values of β, H or actual coefficient profiles.

Proof. Let $R(t)$ be the reset's exact product induction map and $N(t) = R(t)G$, so $N(0) = N(\delta) = G$. Put $u = \widehat{w} + a$, $z = N^{-1}B - \beta$, $r = N^{-1}a$. Lemma 5.2 bounds a in H^1 independently of the requested raw velocity accuracy; its small smooth corrections and lowering generator defect are included in a. Naturality gives the exact equation

$$z_t = \rho N^{-1}\Delta N(z + \beta) + \mathcal{C}_r(z + \beta) + N^{-1}h_B. \tag{5.20}$$

Let $E = \|z\|_{H^1}$ and $D = \|z\|_{H^2}$. Pair with Jz and use Lemmas 5.1–5.2 and (5.18). Young's inequality absorbs $C\|r\|_{H^1}E^{1/2}D^{3/2}$ and the smooth source times D, giving

$$(E^2)' + c\mu a_0 D^2 \le PE^2 + P(1 + \eta^2). \tag{5.21}$$

The diffusion of β uses its supplied smooth envelope. Integrate for a polynomially short time. Both endpoint maps are G, proving (5.19) before smoothing. There is no fixed norm conversion loss at each call.

Let the raw endpoint have $\|u\|_2 \le \sigma_0$ and joint H^1 bound P_0 independent of σ_0. Append an unforced tail of length l. The scalar nonautonomous Stokes propagator and the product bound into $H^{-3/4}$ used in [1, Lemma 5.1] give

$$\|u(l)\|_{H^1} \le \frac{C\sigma_0}{\sqrt{\mu l}} + C\mu^{-7/8}P_0^2 l^{1/8} + C\eta\sqrt{l/\mu}. \tag{5.22}$$

Use the ordinary short coupled bound on this tail. First choose l from the velocity and magnetic budgets, the raw bound and total duration cap; then choose σ_0 for the first term; then compress the raw reset further if needed. The independence of P_0 proved above makes

this ordering noncircular. Equation (5.21) applies to the tail with identity carrier and the same endpoint metric, paying its magnetic effect. Combining the two intervals proves the asserted total-time bound. All clocks are fixed fractions of explicit polynomial minima, so their reciprocals also have polynomial bounds. □

5.4. **Smooth-memory velocity writing.**

Lemma 5.4 (A smooth-memory velocity setter)**.** *Supply a rational finite Fourier velocity target Z, smooth-family and prefix envelopes for $G\beta$, a duration cap and positive endpoint budgets σ, ϑ. There is a common twelve-mode writer, of time degree at most two, such that for sufficiently small incoming velocity and magnetic error*

$$\|u_{\mathrm{out}} - Z\|_{H^1} \le \sigma, \qquad E_{\mathrm{out}} \le (1+\vartheta)E_{\mathrm{in}} + \vartheta. \tag{5.23}$$

Its admissible joint input radius may be chosen $r = \min(1,\sigma)/64$ in the velocity H^1 and fixed magnetic metric, independently of the target's amplitude; the required shorter clocks pay that amplitude. Numerical and binary costs are polynomial in the supplied envelopes and inverse budgets. The force uses only the target and family envelopes. An initial reset extends the velocity entrance condition to any supplied bounded state ball.

Proof. We extend the actual operations of [1, Sections 8.3–8.5, Proposition 8.3], not its constant-magnetic-target conclusion. On a closed translation leaf the physical carrier w and seed force are the published rational polynomials. With known intended velocity prefix Z_0, define only in the proof

$$v_t + L_w v = \nu\Delta w + f_{\mathrm{seed}}, \quad v(0) = Z_0, \qquad u_* = w + v, \quad B_* = R_w(t)G\beta.$$

The homogeneous Euler and induction endpoints are identity. Frozen viscosity affine maps form a group; comparing with the actual coefficient gives endpoint velocity bias $P\Lambda d_{\mathrm{c}}$. Fixed-order spatial bounds on v, B_* are polynomial in the known impulses, prefix, seed weights and family envelope, independently of compressed leaf length d_{c}. The right-minus-left residuals are exactly

$$R_u = \nu\Delta v - \mathcal{B}(v,v) + \mathcal{B}(B_*,B_*), \qquad R_B = \rho\Delta B_* + \mathcal{C}_v B_*. \tag{5.24}$$

They have polynomial smooth bounds. The new nonconstant reference derivatives and its Lorentz term have thus been paid without an H-dependent force.

Subtract the reference and use the separate Euler and induction error coordinates. Large carrier terms cancel. Smooth-linear terms cost P times the H^1 error; quadratic errors obey (5.9). Lemma 8.2 of [1] and Lemma 5.1 give, for squared joint error $\mathcal{Z}$ and joint H^2 error $\mathcal{D}$,

$$\mathcal{Z}' + P^{-1}\mathcal{D}^2 \le P(\mathcal{Z} + \mathcal{Z}^3 + 1 + \eta^2). \tag{5.25}$$

On a fixed error ball, a polynomial clock cap gives the leaf bound

$$e_+ \le (1 + Pd_{\mathrm{c}})e_- + P\sqrt{d_{\mathrm{c}}}(1+\eta) + P\Lambda d_{\mathrm{c}}. \tag{5.26}$$

Both homogeneous endpoints return, so no fixed condition number is paid. For magnetic error alone, repeat (5.21) using the bounded actual velocity remainder. It gives a factor $1 + Pd_{\mathrm{c}}$ on the incoming magnetic error and a small additive term, without adding the incoming velocity error as a fixed magnetic loss. That error is part of the uniformly bounded velocity remainder.

At a quadratic node install $\alpha\xi$, hold unforced for d_{c}, then remove it. The signed-square certificate chooses the sign and dose $\lambda \ge 0$; choose rational α with $d_{\mathrm{c}}\alpha^2$ within an absolute dose tolerance of λ. Choose d_{c} so $|\alpha| d_{\mathrm{c}} \|\xi\|_{C^k}$ is small. During the hold use

$$\begin{gathered} u_* = \alpha\xi + v, \qquad v_t + \alpha L_\xi v = \nu\alpha\Delta\xi - \alpha^2\mathcal{B}(\xi,\xi), \quad v(0) = Z_0, \\ B_*(t) = \mathcal{M}_\xi(\alpha t)G\beta. \end{gathered} \tag{5.27}$$

At fixed dose, only $|\alpha|d_{\mathrm{c}}$ and $\alpha^2 d_{\mathrm{c}}$ enter its smooth bounds. Small-impulse variation of constants gives

$$\|v(d_{\mathrm{c}}) - [Z_0 - \lambda\mathcal{B}(\xi,\xi)]\|_{H^1} + \|B_*(d_{\mathrm{c}}) - G\beta\|_{H^1} \le P\sqrt{d_{\mathrm{c}}} + P\delta_{\mathrm{dose}}. \tag{5.28}$$

In particular, displacement of the already written field during the hold is paid, not assumed absent. The homogeneous rough-error maps have norm at most $e^{P|\alpha|d_{\mathrm{c}}} \le 1 + P\sqrt{d_{\mathrm{c}}}$ in the same endpoint metric; we do not assert operator-norm closeness to identity on an H^1 ball. Equation (5.4) supplies velocity coercivity. After (5.27) the residuals are still (5.24), with bounded spatial H^1 norm, so their actual $L^1_t H^1_x$ norm is at most Pd_{c}. Thus the small integrated residual and the separate $P\sqrt{d_{\mathrm{c}}}$ endpoint mismatch give

$$e_+ \le (1 + P\sqrt{d_{\mathrm{c}}})e_- + P\sqrt{d_{\mathrm{c}}}(1+\eta) + P\delta_{\mathrm{dose}}. \tag{5.29}$$

No smallness is inferred from an unrelated time norm. Hold amplitudes are fixed before selecting shorter child translation clocks.

There are at most two quadratic levels. Passive-plane quadrature has polynomial count and bounded total absolute seed weight. Consequently all intended prefixes are bounded by sums of prescribed increments. Allocate the flattened primitive budgets so

$$\sum_{\text{closed}} P_j d_j + \sum_{\text{holds}} P_j\sqrt{d_j} \le c, \quad \sum_j P_j\sqrt{d_j}(1+\eta) + \sum_{\text{closed}} P_j\Lambda d_j + \sum_j P_j\delta_{\mathrm{dose},j} \le c\min(\sigma,\vartheta),$$

with separately divided directional errors and a fixed sufficiently small c. Make the corresponding magnetic-only sum below $\vartheta/4$. The product of all factors is then below two in joint error and below $1+\vartheta$ in magnetic error. Equation (5.25) uses a fixed input error ball; shrinking the clocks, rather than that ball, pays large target amplitudes. Thus the radius $r = \min(1,\sigma)/64$ suffices for velocity accuracy uniformly over target amplitude. This avoids a circular input-radius condition when a setter target is selected by an earlier hold amplitude. All temporary envelopes are substituted only down the fixed-depth tree, not once per Fourier coefficient. The published rational dose and quadrature arithmetic then gives the asserted polynomial resources. Only family bounds occur in these choices. If needed, prepend Lemma 5.3 with divided magnetic and velocity budgets. □

6. A physical whole-field magnetic gate

The algebraic operation used in the response construction differentiates the field to which it is applied. It is therefore important to formulate its physical realization around a specified smooth reference. The finite physical map has a bounded action on a rough error, whereas its differential limit need not have such an action uniformly.

Write $D_j = \kappa^{-1}\partial_j$ and $\mathscr{C} = \kappa^{-1}\,\mathrm{curl}$, and put

$$T_i(\phi)b = b + \mathscr{C}(e_i\phi b_i). \tag{6.1}$$

All derivative orders in this section are fixed in advance. A polynomial envelope means a polynomial with nonnegative, effectively computable rational coefficients in the displayed numerical parameters. Different occurrences of P may be enlarged by a fixed power. Its construction uses finite product, chain-rule, interpolation and Young inequalities; it never involves an unevaluated solution norm.

Proposition 6.1 (Whole-field gate)**.** *Fix the spatial order required by the memory services of Section 5. Let ϕ and Σ be rational finite Fourier data of radii at most N and F, where Σ has solenoidal columns and mean I. Supply rational bounds $M, R_u, E \ge 2$, with $|H| \le M$, $\|u_{\mathrm{in}}\|_{H^1} \le R_u$, and with E bounding all fixed-order smooth envelopes used below for ϕ and Σ. Let μ be a rational positive lower bound for both diffusivities and let Λ bound their temporal Lipschitz constants.*

Given positive rational $S, \zeta, \sigma \le 1$, there is a deterministic compiler producing a rational duration $0 < D \le S$, a force $f \in \mathcal{E}$, and positive rational r, η, G, with $G \ge 1$, such that

$$\begin{aligned} B_{\mathrm{in}} = \Sigma H + e, \quad \langle e \rangle = 0, \quad \|e\|_{H^1} \le r \quad \Longrightarrow \quad & \|u_{\mathrm{out}}\|_{H^1} \le \sigma, \\ & \|B_{\mathrm{out}} - T_i(\phi)(\Sigma H)\|_{H^1} \le G\|e\|_{H^1} + \zeta. \end{aligned} \tag{6.2}$$

The same output works for every admitted coefficient pair, unknown H, incoming fields, disturbances bounded by η, and energy-admissible weak realization. Its force uses the reference family and envelopes, not their unknown values. For universal effectively computable C, d,

$$\begin{aligned} r^{-1} + \eta^{-1} + G + \sup_t \|f(t)\|_{W^{1,\infty}} + \int \|f(t)\|_{W^{1,\infty}}\, dt \le CX^d, \\ X = 2 + M + R_u + E + F + N + \mu^{-1} + \Lambda + S^{-1} + \zeta^{-1} + \sigma^{-1}. \end{aligned} \tag{6.3}$$

The same type of bound holds for the physical state envelope, interval count and, after adding the complete encoding length to X, all binary resources. The emitted time polynomials have degree at most two.

We prove the proposition through an explicit passive construction, a physical assigned-flow realization, and a short actual coupled filtering interval.

6.1. **The lifted wavepacket geometry.** It suffices to treat $i = 3$. Set

$$Q_{\mathrm{g}} = q^4, \quad \Omega = \kappa Q_{\mathrm{g}}, \qquad U = \operatorname{curl}\left(-e_2 \frac{\phi(x)\sin(\Omega x_3)}{\Omega}\right), \quad V = e_2 \sin(\Omega x_3). \tag{6.4}$$

In coordinates (x, y, z) this is

$$U = (\phi\cos(\Omega z) + \Omega^{-1}\phi_z \sin(\Omega z),\ 0, -\Omega^{-1}\phi_x \sin(\Omega z)).$$

Both fields are solenoidal and have mean zero if $Q_{\mathrm{g}} > N$. Their Fourier data are rational: $\Omega^{-1}\phi_j = Q_{\mathrm{g}}^{-1} D_j \phi$.

Let $\mathcal{M}_W(t)$ be induction pushforward by the time-t flow of W; its generator is $\mathcal{C}_W$. Introduce an independent periodic phase θ, restricted at the end to $\theta = \Omega z$, and abbreviate $c = \cos\theta$, $s = \sin\theta$. The lifted particle generators are

$$X_U = (\phi c + \Omega^{-1}\phi_z s)\partial_x - \Omega^{-1}\phi_x s \partial_z - \phi_x s \partial_\theta, \qquad X_V = s\partial_y. \tag{6.5}$$

The graph $\theta = \Omega z$ is invariant because $\theta' = \Omega z'$. Put $S_\Omega = \operatorname{diag}(1, 1, \Omega)$. For the rescaled field $\widetilde{b} = S_\Omega b$, the lifted induction generators are $\widetilde{L}_U = A_U - X_U$ and $\widetilde{L}_V = A_V - X_V$, where

$$A_U = \begin{pmatrix} \phi_x c + \Omega^{-1}\phi_{xz} s & \phi_y c + \Omega^{-1}\phi_{yz} s & -\phi s + 2\Omega^{-1}\phi_z c + \Omega^{-2}\phi_{zz} s \\ 0 & 0 & 0 \\ -\phi_{xx} s & -\phi_{xy} s & -\phi_x c - \Omega^{-1}\phi_{xz} s \end{pmatrix}, \qquad (A_V)_{23} = c, \tag{6.6}$$

and the other entries of A_V vanish. These formulas follow by conjugating the physical stretching matrices by S_Ω.

Lemma 6.2 (Polynomial full-prefix geometry)**.** *For each fixed m, a rational envelope E_m is computable from finitely many derivatives of ϕ such that, whenever $(|a| + |b|)E_m \le c_m$, the particle flow, inverse flow and every prefix of $\mathcal{M}_V(b)\mathcal{M}_U(a)$ have fixed-order geometric bounds polynomial in Ω. The same holds for the induction operators on each fixed signed Sobolev space. In particular one may use*

$$\|\Phi^{\pm 1}\|_{C^j} \le C_j(1+\Omega)^j, \qquad \|\mathcal{M}_\Phi^{\pm 1}\|_{H^j \to H^j} \le C_j(1+\Omega)^{j^2+j+1} \quad (j \ge 0), \tag{6.7}$$

after increasing the envelope and imposing the indicated smallness condition. The geometric norm is that of the periodic displacement, with the identity part included separately.

Proof. Every fixed-order coefficient norm in (6.5)–(6.6) is bounded independently of $\Omega \geq 1$ by a rational derivative envelope. For an ODE flow F_t, its first derivative solves $J' = DX(F_t)J$. If the integral of the common coefficient envelope is at most one, then $\|J\| \leq e < 3$. The jth derivative satisfies

$$\partial_t D^j F = DX(F)D^j F + \sum_{\substack{\pi \text{ partition of } \{1,\dots,j\} \\ |\pi|\geq 2}} D^{|\pi|}X(F) \prod_{A\in\pi} D^{|A|}F.$$

Thus constants $B_1 = 3$ and $B_j = 3(1 + \sum_{\pi,|\pi|\geq 2} \prod_{A\in\pi} B_{|A|})$ bound all these derivatives by induction. Negative time gives the same inverse bounds. Apply this argument to the two consecutive lifted vector fields, with their total impulse. The matrix ODE for rescaled induction and its differentiated equations have the same finite recurrence.

Restriction to $\theta = \Omega z$ costs at most $(1 + \Omega)^j$ for j physical derivatives. This proves the first bound in (6.7). The formula

$$\mathcal{M}_\Phi b = (D\Phi\, b) \circ \Phi^{-1}$$

and volume preservation bound its H^j norm by a polynomial in the C^{j+1} norm of Φ and the C^j norm of its inverse. Counting at most j chain-rule factors gives the deliberately generous second exponent. The same argument applies to the inverse; duality and the corresponding adjoint composition formula give fixed negative orders. All the recurrences are finite. In particular the proof does not estimate the physical flow by the exponential of $|a|\|\nabla U\|_\infty$. □

6.2. **The low jet and its normalization.** For the normalized gate (6.1), choose

$$a = q^{-2}, \qquad b_* = \frac{2}{\kappa^2 q^2}. \tag{6.8}$$

The extra factor κ^{-1} in b_* is required: the older unnormalized operation $b + \operatorname{curl}(e_3\phi b_3)$ would instead use $2/(\kappa q^2)$.

Lemma 6.3 (Finite jet)**.** *Let b be a smooth finite Fourier reference of radius F, and let ϕ have radius N. For q above a computable polynomial in fixed-order smooth envelopes, and $|b_0| \leq 2q^{-2}$, write*

$$\mathcal{M}_V(b_0)\mathcal{M}_U(a)b = J_4 + R_5, \quad J_4 = \sum_{r+s\leq 4} \frac{b_0^r a^s}{r!s!} \mathcal{C}_V^r \mathcal{C}_U^s b. \tag{6.9}$$

Then

$$\|R_5\|_2 \leq Pq^{-6}, \qquad \|R_5\|_{H^1} \leq Pq^{-2}. \tag{6.10}$$

Every degree-j term of J_4 is a sum of $A_n(x)e^{in\Omega z}$ with $|n| \leq j$ and slow support in radius $F + jN$. Writing $J_4 = J_L + J_H$ for its zero and nonzero fast harmonics, at $b_0 = b_*$ one has

$$\|J_L - T_3(\phi)b\|_{H^1} \leq Pq^{-4}. \tag{6.11}$$

For each fixed r, the Fourier Wiener norms satisfy

$$\|J_4\|_{X^r} \leq P_r q^{2+4r}, \qquad \|J_L\|_{X^r} \leq P_r, \quad \|v\|_{X^r} = \sum_k (1 + |k|)^r |\widehat{v}(k)|. \tag{6.12}$$

If $Q_{\mathrm{g}} \geq 2(F + 4N + 1)$, J_H is supported at physical frequencies of modulus at least $Q_{\mathrm{g}}/2$.

Proof. Taylor's formula can be applied to the lifted system in supremum derivative norms. The initial rescaled field $S_\Omega b$ has size at most Ω times its smooth envelope. Repeated application of the frequency-independent lifted coefficients, together with Lemma 6.2, bounds the lifted C^1 remainder by $P\Omega(|a| + |b_0|)^5$. Returning from S_Ω costs no factor; restriction costs at most one further Ω for one physical derivative. Thus the L^2 and H^1 remainders are bounded respectively by $P\Omega(|a| + |b_0|)^5$ and $P\Omega^2(|a| + |b_0|)^5$, proving (6.10). This uses a supremum-norm restriction estimate, not an L^2 trace from a four-dimensional torus.

For completeness we compute the quadratic low term. Let $\langle\cdot\rangle_\theta$ denote average in the independent phase. First-order terms have zero average and direct differentiation gives

$$\langle \mathcal{C}_V^2 b\rangle_\theta = \tfrac{1}{2}\partial_{yy} b. \tag{6.13}$$

For the U self term, only $\Omega b_3 e_3$ in the initial rescaled vector could create a physical horizontal contribution of order Ω. Let $f = b_3$. At $\Omega = \infty$, the leading lifted operator obeys

$$\widetilde{L}_{U,0}(f e_3) = (-\phi f \sin\theta, 0, -\partial_x(\phi f)\cos\theta).$$

Its first component after another application is $2\phi^2 f_x \sin\theta\cos\theta$, of zero phase average, and its second component is zero. The third component is divided by Ω on returning to physical variables; every omitted coefficient has an extra Ω^{-1}. Consequently

$$\|\langle \mathcal{C}_U^2 b\rangle_\theta\|_{H^r_{\mathrm{slow}}} \le P_r. \tag{6.14}$$

No solenoidal cancellation stronger than the displayed calculation is being assumed here.

For the mixed term the leading physical components are

$$(\mathcal{C}_U b)_1 = -\Omega\phi b_3 \sin\theta + O(P), \qquad (\mathcal{C}_U b)_3 = -\partial_x(\phi b_3)\cos\theta + O(P/\Omega),$$

and $\mathcal{C}_V c = \Omega c_3 \cos\theta\, e_2 - \sin\theta \partial_y c$. Therefore

$$\langle \mathcal{C}_V \mathcal{C}_U b\rangle_\theta = \frac{\Omega}{2}(\partial_y(\phi b_3), -\partial_x(\phi b_3), 0) + O(P) = \frac{\Omega}{2}\operatorname{curl}(e_3 \phi b_3) + O(P). \tag{6.15}$$

Since $a b_* \Omega/2 = \kappa^{-1}$, this is exactly the desired normalized coefficient. The self terms, multiplied by $a^2/2$ and $b_*^2/2$, cost Pq^{-4}. Every generator changes fast Fourier parity, so the cubic zero-phase term vanishes. The quartic low term is at most $P\Omega(|a| + |b_*|)^4 \le Pq^{-4}$ in fixed slow norms. This proves (6.11).

Each generator adds a fast harmonic $+1$ or -1; multiplication by a coefficient of U adds at most N to slow support, and differentiation does not enlarge it. This proves the support assertion. The same lifted calculation in Fourier coefficient sums bounds a degree-j physical coefficient by $P_j\Omega$. After multiplication by its impulses the largest term is the degree-one term, of size Pq^2. Each fixed physical derivative costs at most $P_r Q_{\mathrm{g}}^r$, yielding (6.12). The separation assertion is the triangle inequality. □

6.3. **From memory services to a physical assigned flow.** We spell out the physical bridge, since a prescribed passive velocity is not an available actuator. The reset and setter of Section 5 have the following quantitative features. Given one common incoming state bound, a smooth magnetic family, and a fixed accumulated induction metric, they can be compiled on an arbitrary rational duration cap with a prescribed endpoint velocity and with

$$E_+ \le (1+\vartheta)E_- + \vartheta. \tag{6.16}$$

Their force depends only on family envelopes. They have a raw remainder bound independent of the requested final velocity accuracy, and polynomial costs in all envelopes and inverse tolerances. Their actual rational clocks are known after compilation. In the magnetic discrepancy estimate the two endpoint norms use the same fixed metric. The arbitrary smooth reference and the metric are allowed to depend on actual coefficients only in the proof. These assertions come from the actual returning words, viscosity correctors and lowered closed leaves, not from the earlier endpoint theorem.

Lemma 6.4 (Assigned-flow lift)**.** *Let W be a known mean-zero rational Fourier solenoidal field and let $0 < \alpha \le 1$ be a rational impulse. Supply fixed-order smooth bounds for a reference family β, the admitted incoming velocity, and the full assigned particle flow Φ_s of W, including every prefix and inverse for $0 \le s \le \alpha$. Suppose all these bounds and their required inverse ellipticity constants are polynomial envelopes. Given a duration cap S and endpoint tolerances $\xi, \sigma > 0$, there is a twelve-mode physical programme of duration at most S such that*

$$\|u_{\mathrm{out}}\|_{H^1} \le \sigma, \qquad \|B_{\mathrm{out}} - \mathcal{M}_{\Phi_\alpha}\beta\|_{H^1} \le G_0 \|B_{\mathrm{in}} - \beta\|_{H^1} + \xi. \tag{6.17}$$

Here G_0 and the reciprocal admissible incoming radius have polynomial bounds, as do all physical and binary costs in the supplied envelopes, $\alpha^{-1}, S^{-1}, \xi^{-1}, \sigma^{-1}, \mu^{-1}, \Lambda$. The force needs only the reference-family envelopes. It has time degree at most two.

Proof. All constants in this proof are computed from the supplied envelopes before the final physical holds and inserted controller clocks are selected. Choose an impulse subdivision $d = \alpha/J$, a physical hold length $d_{\mathrm{c}} \le d$, and $A = d/d_{\mathrm{c}} \ge 1$. At every hold start, the setter puts the velocity within a fixed small H^1 radius σ_0 of AW; a reset follows the hold. During a hold the chosen force is zero.

The correct smooth reference is not the frozen velocity AW. Let u_j^r solve the actual-viscosity unforced Navier–Stokes equation from AW on that hold. In Euler time $s = A(t - t_j)$ write $u_j^r = Av_j$. Then

$$\partial_s v_j = \frac{\nu(t_j + s/A)}{A}\Delta v_j - \mathcal{B}(v_j, v_j), \qquad v_j(0) = W. \tag{6.18}$$

High-order Sobolev energy, discarding the favorable diffusion, gives for $d \le c/P$

$$\begin{aligned} &\sup_{s\le d}\|v_j(s)\|_{H^{m+6}} \le 2\|W\|_{H^{m+6}}, \\ &\sup_{s\le d}\|v_j(s) - W\|_{C^{m+2}} \le P(d + d_{\mathrm{c}}), \qquad \int_{\mathrm{hold}} \|u_j^r\|_{W^{2,\infty}}\, dt \le Pd. \end{aligned} \tag{6.19}$$

The middle estimate follows by integrating (6.18) two derivatives below its energy level; the viscous part has integral $d/A = d_{\mathrm{c}}$. These estimates require neither actual coefficient values nor their derivatives to be computed.

Concatenate these reference flows over Euler time $[0, \alpha]$, omitting the inserted controller durations, and call the resulting particle flow Ψ_s. Conjugate by the full assigned flow: $Y_s = \Phi_{-s} \circ \Psi_s$. Its generator is the pullback of $v_j - W$ by Φ_s, so its fixed-order norm has total integral at most

$$P\alpha(d + d_{\mathrm{c}}). \tag{6.20}$$

If this quantity is small, finite differentiated flow equations give polynomial bounds for the whole Ψ path and inverse, and

$$\|\mathcal{M}_{\Psi_\alpha}\beta - \mathcal{M}_{\Phi_\alpha}\beta\|_{H^1} \le P\alpha(d + d_{\mathrm{c}}). \tag{6.21}$$

The large accumulated strain is not exponentiated: only the small generator of Y is. This step uses the full assigned-flow geometry in the hypothesis.

Let $\mathcal{M}_t$ be the accumulated induction reference map, extended constantly over inserted services. During a hold put

$$w = u - u_j^r, \quad B = \mathcal{M}_t(\beta + z), \quad X = \|w\|_{H^1}^2, \quad Y = \|z\|_{H^1}^2, \quad V_2 = \|w\|_{H^2}, \quad Z_2 = \|z\|_{H^2}.$$

Lie-bracket covariance gives exactly

$$z_t = \rho\mathcal{M}_t^{-1}\Delta\mathcal{M}_t(z + \beta) + \mathcal{C}_{\mathcal{M}_t^{-1}w}(z + \beta) + \mathcal{M}_t^{-1}h_B. \tag{6.22}$$

The conjugated diffusion has a scalar elliptic principal part, so direct integration by parts gives

$$\mathrm{Re}\langle \mathcal{M}_t^{-1}\Delta\mathcal{M}_t z, (I - \Delta)z\rangle \le -P^{-1}\|z\|_{H^2}^2 + P\|z\|_{H^1}^2. \tag{6.23}$$

Coefficient and inverse-metric bounds follow polynomially from the already controlled particle geometry. No temporal derivative of $\mathcal{M}_t$ occurs in this inequality. A bound on the map and its inverse alone, without this scalar-principal representation, would not establish it.

We give the hold-energy calculation explicitly. Under a bootstrap $X, Y \le 1$, geometry and reference bounds imply $\|B\|_{H^1} \le P$ and $\|B\|_{H^2} \le P(1 + Z_2)$. Subtracting (6.18) in physical time and pairing the velocity error with $(I - \Delta)w$ yields

$$X' + c\mu V_2^2 \le \ell(t)X + PX^3 + P(1 + Z_2), \qquad \int_{\mathrm{hold}} \ell(t)\, dt \le Pd. \tag{6.24}$$

Indeed the self term is bounded by $C\mu^{-3}X^3$ after Young, the reference linear terms cost $C\|u_j^r\|_{W^{2,\infty}}X$, and

$$\|\mathcal{B}(B,B)\|_2^2 \le C\|B\|_{H^1}^3\|B\|_{H^2} \le P(1+Z_2).$$

For $\mathcal{J} = \sup_{\mathrm{hold}} X + \mu\int_{\mathrm{hold}} V_2^2$ and $\mathcal{D} = \int_{\mathrm{hold}} Z_2^2$, integration, when Pd is small, gives the valid estimate

$$\mathcal{J} \le CX_{\mathrm{in}} + Pd_{\mathrm{c}}(1+\mathcal{J}^3) + P\sqrt{d_{\mathrm{c}}}\sqrt{\mathcal{D}}. \tag{6.25}$$

The last term is the Cauchy–Schwarz bound for $\int Z_2$. A bound by $Pd_{\mathrm{c}}\mathcal{D}$ alone is not valid.

For induction use the elementary product estimate

$$\|\mathcal{C}_r z\|_2 \le C\|r\|_{H^1}\|z\|_{H^1}^{1/2}\|z\|_{H^2}^{1/2}. \tag{6.26}$$

For $r\cdot\nabla z$ use $L^6\cdot L^3$; for $z\cdot\nabla r$ use $L^\infty\cdot L^2$, with $\|z\|_\infty + \|\nabla z\|_3 \le C\|z\|_{H^1}^{1/2}\|z\|_{H^2}^{1/2}$. Also $\|\mathcal{C}_r\beta\|_2 \le P\|r\|_{H^1}$. Pairing (6.22) with $(I-\Delta)z$ and using (6.23) gives

$$Y' + P^{-1}Z_2^2 \le P(1+Y+X+X^2Y). \tag{6.27}$$

This sharper product estimate avoids any term involving V_2Y. In particular one need not make the erroneous estimate $\int\sqrt{X}V_2 \le Pd_{\mathrm{c}}\mathcal{J}$; direct Cauchy–Schwarz would give only $\mu^{-1/2}\sqrt{d_{\mathrm{c}}}\,\mathcal{J}$.

For $X\le 1$, (6.27) gives

$$\sup Y + P^{-1}\mathcal{D} \le (1+Pd_{\mathrm{c}})Y_{\mathrm{in}} + Pd_{\mathrm{c}}.$$

Substitute this in (6.25). If $Y_{\mathrm{in}} \le 1/4$, σ_0 is a sufficiently small fixed constant and $d_{\mathrm{c}} \le P^{-c}$, the bootstrap improves to $\mathcal{J} \le C\sigma_0^2 + P\sqrt{d_{\mathrm{c}}} < 1/2$ and $\sup Y < 1/2$. The endpoint magnetic estimate is therefore

$$\|z_{\mathrm{out}}\|_{H^1} \le (1+Pd_{\mathrm{c}})\|z_{\mathrm{in}}\|_{H^1} + P\sqrt{d_{\mathrm{c}}}. \tag{6.28}$$

All constants are independent of A. In particular no constraint $d_{\mathrm{c}} \lesssim A^{-4}$ has been used.

We now specify a noncircular clock and error allocation. Enlarge P once to bound the final metric conversion and all fixed recurrence constants. Fix the hold-start velocity tolerance σ_0 from the preceding bootstrap, before choosing an amplitude, and put

$$r_* = \frac{\min(1,\sigma_0)}{64}, \qquad 0<\gamma\le\min\left\{\frac{1}{64}, \frac{\xi}{64P}, \frac{r_*}{2^{12}P}\right\}.$$

The radius r_* is admissible for every setter with endpoint tolerance σ_0 by Lemma 5.4, independently of its target amplitude. Take

$$J = \left\lceil\frac{CP(1+\alpha)^2}{\gamma}\right\rceil, \qquad d = \frac{\alpha}{J},$$

$$d_{\mathrm{c}} = \min\left\{d, \frac{S}{16J}, P^{-c}, \left(\frac{\gamma}{64PJ}\right)^4\right\}, \qquad A = \frac{d}{d_{\mathrm{c}}}. \tag{6.29}$$

These choices fix A before any inserted controller duration. The geometric error (6.20), $PJ\sqrt{d_{\mathrm{c}}}$ and PJd_{c} are within their allocated fractions of γ. Every minimum is a minimum of explicit positive rationals, so its reciprocal has a polynomial bound.

Choose the reset velocity tolerance at most $r_*/4$, and at most σ for the last reset; setters request endpoint tolerance σ_0. Each hold begins with error at most σ_0 about AW and ends with a raw velocity in one common ball of radius $P(1+A\|W\|_{H^1})$. The magnetic field is in one common ball of radius P, independently of A at the service endpoints. All inserted calls can consequently be compiled for these same incoming bounds; their larger internal carriers are not recursively used as the next call's state radius. Choose their duration caps last, and request (6.16) with $\vartheta = \gamma/[64(J+1)]$. Request final velocity at most σ and cap each inserted duration by $S/(16J+16)$. Include an initial reset if needed. There are at most $2J+1$ inserted services. Their reciprocal clocks, costs and disturbance allowances remain

polynomial because A is already fixed and all common envelopes and inverse tolerances are polynomial.

At every inserted endpoint the accumulated induction metric is unchanged. Thus all applications of (6.28) and (6.16) concern the same discrepancy variable. The product of their multiplicative factors is at most

$$\exp\{PJd_{\mathrm{c}} + (2J+1)\vartheta\} \le 1 + C\gamma,$$

and their summed additive errors are at most $P\gamma$ after the fixed enlargement. Choose the initial magnetic discrepancy below $r_*/16$ and the divided common service radii. The choices above keep the discrepancy below $r_*/4$ at every inserted endpoint, not just below the coarser hold bootstrap radius. Together with the reset velocity bound this verifies every setter's entrance condition. Its radius is fixed before J, d_{c}, A; only the shorter internal clocks depend on A. All reciprocal entrance radii remain polynomial. Convert to physical norm only once, at the last endpoint, and add (6.21). This proves (6.17). Holds use at most $S/16$ and inserted calls less than $S/8$, so the duration cap has ample room. The local raw energy estimates give controlled strong continuation on every actual segment. Identical-data weak–strong uniqueness gives the same conclusion for all admitted weak realizations.

The emitted controls are those of the memory services, translated to known rational clock positions, and the zero force on holds. Reset carriers $6z(1-z)$ have linear derivative pulses; seeds are quadratic; lowering uses the derivative of a cubic pulse. Hence every actual coefficient has degree at most two. The proof-only u_j^r, Ψ, ν, ρ and smooth reference values are never computed. □

6.4. **Coupled filtering, precision and the gate clock.** The finite packet estimate in Appendix B is an actual unforced MHD estimate. For $T = \mu^{-1}q^{-7}$ it gives, on the jet of Lemma 6.3,

$$P\{q^{-4} + \mu^{-1}q^{-7} + q^{6}e^{-cq}\} \tag{6.30}$$

in joint H^1 from $(0, J_L)$. For incoming joint L^2, H^1 errors e_0, e_1 its additional error is

$$P\{e_1 + q^{3}e_0 + \eta\sqrt{T}\}. \tag{6.31}$$

These bounds keep the constant-flux Alfvén coupling in the exact linear part; magnetic heat is not substituted as an available control.

Proof of Proposition 6.1. Set $t_* = \min(\zeta, \sigma)$. First choose an integer q such that

$$q^4 \ge 32(F + 4N + 1), \qquad (q^{-2} + 2q^{-2})E_m \le c_m,$$
$$P\{q^{-2} + \mu^{-1}q^{-7} + q^{6}e^{-cq}\} \le t_*/64, \qquad \mu^{-1}q^{-7} \le S/8. \tag{6.32}$$

Here the polynomial P is obtained from the finite passive and packet estimates, prior to physical lifting. This is an effective polynomial choice: $e^{-cq} \le 12!c^{-12}q^{-12}$, and one may choose $q = \lceil (CX)^d \rceil$ with a fixed integer d larger than all the degrees in these finitely many inequalities.

Only the impulse b_* is irrational. Compute a rational $b_0 > 0$ satisfying

$$|b_0 - b_*| \le \min\{q^{-2}/100, t_*/[CP(1+q)^{c_m}]\}. \tag{6.33}$$

An elementary convergent series gives a paid enclosure of π. The H^1 derivative of the smooth-reference map with respect to b_0 is $\mathcal{C}_V\mathcal{M}_V(b_0)\mathcal{M}_U(a)(\Sigma H)$ and has a polynomial bound by Lemma 6.2. Thus (6.33) pays the reference error. No operator-norm continuity assertion on a rough H^1 ball is used.

Apply Lemma 6.4 to U for impulse a and then to V for impulse b_0, with duration caps $S/4$. The second reference is the smooth family $\mathcal{M}_U(a)(\Sigma H)$; its known envelope follows

from the lifted calculation, so no flow values or unknown H need be evaluated. Choose the two endpoint errors backward to ensure joint H^1 discrepancy at most

$$\delta_{\mathrm{packet}} = \frac{t_*}{CP(1+q)^3}$$

about the exact two-flow image with zero velocity, apart from a term $G_1\|e\|_{H^1}$ with polynomial G_1. All their input radii and clocks are now fixed by the lemma. The reciprocal impulses are polynomial in q and the paid rounding precision, so the lift's resource bound applies.

Append the actual unforced interval $T = \mu^{-1}q^{-7}$. The nominal finite jet has zero velocity, while the exact ideal magnetic image differs from it by the remainder in (6.10). Its contribution in (6.31) is at most $P(q^{-2} + q^3q^{-6})$. Add (6.11), (6.30) and the rounded-impulse error. With the assigned safety fractions this gives both conclusions of (6.2), with a polynomial G.

Choose r as the minimum of the derived positive input radii divided by their already computed polynomial propagation factors and of $t_*/(64G)$. In particular, the packet's velocity error arising from the incoming magnetic discrepancy is also below its assigned fraction of σ; velocity accuracy is not independent of that discrepancy. Choose η as a fixed fraction of the minimum of all actual subcall disturbance budgets and the packet budget. These choices have polynomial reciprocals; none is inferred from a large uniqueness exponent. Every discrepancy retains zero mean because the references and actual evolution have the same conserved flux. All initial state and reference bounds supplied to calls are the common bounds proved above.

The two assigned-flow calls and the final interval have total rational duration less than S. Each service's actual duration is known when compiled. No terminal unprescribed waiting interval is inserted. The force always lies in the twelve allowed directions, and the degree-two statement follows from the actual emissions in the lift proof.

Finally, finite Fourier convolution and derivative recurrences, polynomial subdivision counts, rational bisection and absolute coefficient precision have polynomial bit cost in the complete encoding length and X. The number of nested spatial generation/lowering levels is fixed. Repeated calls use common envelopes, so their count multiplies one polynomial rather than iterating it. Symbolic substitution through this finite dependency graph defines C, d and proves (6.3). □

Remark 6.5 (Why the passive statement is insufficient). The passive map prescribes U and V in the induction equation, while the physical input acts only in momentum. In particular U need not solve steady Euler, and its Lorentz reaction is not absent. The assigned-flow proof above pays those two effects with actual-viscosity Navier–Stokes references and joint energy estimates. Its resets retain the already written smooth magnetic family. An endpoint velocity theorem alone would not supply any of these properties. Likewise, the final packet interval is a coupled evolution, not a chosen magnetic heat operator.

Remark 6.6 (The rough error is not differentiated in the limiting map). For $\phi = 1$ and $b_n = n^{-1}e_3\sin(\kappa n x_1)$, the solenoidal fields b_n have bounded H^1 norm, while $\mathscr{C}(e_3(b_n)_3) = -e_2\cos(\kappa n x_1)$ has H^1 norm growing linearly in n. Thus $T_3(1)$ is not bounded $H^1 \to H^1$. Proposition 6.1 compares the physical output with $T_i(\phi)$ applied only to ΣH; the rough discrepancy is propagated by a bounded finite physical map. These are different statements.

7. Positive heat for full intermediate fields

The short packet filter in the magnetic gate has a special amplitude and time scale. The three outer selectors need a different operation: the total heat time is chosen from the retained field before the full carrier-dependent smooth envelope is known. Repeated rapid velocity resets supply this operation, with one magnetic energy estimate over all holds.

Lemma 7.1 (Arbitrary prescribed positive-heat time)**.** *Supply rational $P \geq 2$, $0 < \zeta \leq 1$, $0 < T \leq 1$, $\mu > 0$ and Λ. Assume $\mu \leq \nu(t), \rho(t) \leq 1$ and that their individual temporal*

Lipschitz constants are at most Λ. Let P bound the incoming velocity H^1 norm and the fixed required smooth norms of a solenoidal reference family β_0. Suppose

$$\|B_0-\beta_0\|_{H^1}\le \zeta/16, \qquad \langle B_0-\beta_0\rangle = 0. \tag{7.1}$$

There is a common rational twelve-mode programme of exactly duration T and a positive inverse-polynomial disturbance allowance such that

$$\|B(T)-e^{D_\rho\Delta}\beta_0\|_{H^1}\le\zeta, \qquad \|u(T)\|_{H^1}\le U, \qquad \mu T/2\le D_\rho\le T, \tag{7.2}$$

where $U>0$ is the explicit rational quantity in (7.3). The force uses only the family envelope, not reference values or actual coefficient profiles. All physical and complete binary resources are polynomial in $P,\zeta^{-1},\mu^{-1},T^{-1},1+\Lambda$ and the supplied encoding. The total time T is arbitrary; no condition $TP^c\ll 1$ is imposed. The same proof gives an input-error gain at most two under these safety choices, with the small additive budget divided further if requested.

Proof. Choose a fixed computable $C\ge 100$ dominating the constants below, enlarged once to cover replacing P by $P+2$. Set

$$U=\frac{\zeta\mu^2}{2^{12}CP^2}, \qquad \eta_{\mathrm{heat}}=\frac{\zeta\mu}{2^{12}C}, \qquad d_0=\frac{\zeta^8\mu^{24}}{2^{128}C^{16}P^{32}},$$
$$J=\lceil T/d_0\rceil, \qquad r=\frac{\zeta}{2^{12}J}. \tag{7.3}$$

Compile one reset using Lemma 5.3, with the identity endpoint metric, endpoint velocity at most $U/4$, and magnetic endpoint increment at most r whenever the discrepancy is at most ζ. It is valid for the whole common input ball of velocity radius P and magnetic radius $P+1$, and for every reference in the smooth envelope P. Give it duration cap $T/(2J)$. Its actual rational duration ℓ is known after compilation. Put

$$d=T/J-\ell, \qquad T/(2J)\le d\le d_0. \tag{7.4}$$

Repeat this same reset and an unforced hold of length d, J times. This fills exactly T. During resets freeze the proof reference; during holds evolve it by $\beta_t=\rho(t)\Delta\beta$. Scalar heat does not increase its fixed smooth norms. Its actual values remain unknown and are never used by the force.

The raw interior reset bound is computed once from the joint input radius $2(P+1)$, giving a polynomial $\overline{P}$. It covers both the first entrance, where velocity can have norm P, and every later entrance. It is the same bound for all resets. At endpoints the reference and error restore the magnetic radius $P+1$, rather than feeding $\overline{P}$ as a new input radius at the next cell. To realize the increment r, the reset clocks are chosen from the explicit caps in the preceding section; for example $P_r\ell\zeta+2P_r\sqrt{\ell}\le r$ suffices for its memory estimate. Its smoothing tail and contraction clocks are also subject to (7.4). All reciprocal clocks have polynomial bounds. Restrict the global disturbance below this template's direct allowance and below η_{heat}.

We close velocity and magnetic bootstraps on every hold. At its start $\|u\|_{H^1}\le U/4$. While $\|B-\beta\|_{H^1}\le\zeta$, the magnetic norm is at most $P+1$. The scalar nonautonomous Stokes formula and the product estimate into $H^{-3/4}$ give, while $\|u\|_{H^1}\le U$,

$$\sup_{\mathrm{hold}}\|u\|_{H^1}\le U/4+C\mu^{-7/8}(P+2)^2d^{1/8}+C\eta\sqrt{d/\mu}\le U. \tag{7.5}$$

The kernel integrates as $\int_0^d(\mu t)^{-7/8}dt=8\mu^{-7/8}d^{1/8}$. The choice of d_0 makes its nonlinear increment a small multiple of U: $d_0^{1/8}$ is proportional to $\zeta\mu^3P^{-4}$ and $\mu\le 1$. The disturbance increment is smaller still. The same choice lies below the polynomial local continuation threshold for radius $P+2$, so the bootstrap is applied to the actual local strong solution and gives a strict margin.

For the magnetic error $e = B - \beta$ write $E = \|e\|_{H^1}$ and $D = \|e\|_{H^2}$. Its exact hold equation is

$$e_t = \rho \Delta e + \mathcal{C}_u e + \mathcal{C}_u \beta + h_B. \tag{7.6}$$

Its mean remains zero by (7.1), the conserved physical mean, and heat and reset-reference mean preservation. Thus $-\langle \Delta e, Je \rangle$ controls the full inhomogeneous H^2 norm by Poincaré. Pair (7.6) with Je and use (5.18). Young's inequality gives

$$\begin{aligned} (E^2)' + c\mu D^2 &\le CUE^{1/2}D^{3/2} + C(PU + \eta)D \\ &\le C\mu^{-3}U^4E^2 + C\mu^{-1}(P^2U^2 + \eta^2). \end{aligned} \tag{7.7}$$

Set $a = C\mu^{-3}U^4$ and $b = C\mu^{-1}(P^2U^2 + \eta^2)$. At a reset endpoint, while $E \le \zeta$, its compiled increment gives

$$E_+^2 - E_-^2 \le 2\zeta r + r^2. \tag{7.8}$$

Concatenating the hold inequalities and these jumps gives one estimate over the whole programme:

$$E(t)^2 \le e^{aT} \left[E(0)^2 + bT + J(2\zeta r + r^2) \right] \tag{7.9}$$

at all complete holds, and with the corresponding partial-hold bound. The initial square is at most $\zeta^2/256$; $J(2\zeta r + r^2) \le \zeta^2/2048 + \zeta^2/2^{24}$. Equations (7.3) make aT universally small and bT a sufficiently small fixed multiple of ζ^2. After the fixed safety enlargement of C, the right side is below $\zeta^2/4$. This strictly closes the magnetic bootstrap and hence (7.5) at every hold. Inside resets use their common raw bound; there is no assertion that their carrier velocities stay small. The force and reference inputs at the following endpoints are restored to the same common bounds.

All reference heat factors commute. Their final product is

$$\beta(T) = \exp\left(\Delta \sum_{\text{holds}} \int \rho(t)\, dt \right) \beta_0.$$

The total hold time Jd is in $[T/2, T]$, proving the interval for D_ρ in (7.2). It is a proof quantity and is not evaluated.

Finally,

$$J \le 1 + 2^{128} C^{16} P^{32} \zeta^{-8} \mu^{-24}. \tag{7.10}$$

Every reset uses the same state envelope and has polynomial cost in these parameters and T^{-1}, Λ. Thus J multiplies a polynomial cost; a state-envelope polynomial is not composed J times. Repeating one rational template at rational translated cell times pays polynomial output and bit heights and preserves time degree. The disturbance allowance is the minimum of the hold allowance and this template's direct allowance, so its reciprocal is polynomial. Equation (7.9) also shows an initial-error gain $e^{aT/2} < 2$. Its additive terms can be reduced by dividing the corresponding budgets. This proves the stated stability form. □

The global inequality (7.9) is essential. Summing a local $d^{1/8}$, $d^{1/4}$ or $\sqrt{d}$ magnetic error over T/d cells would not prove the result. A larger full smooth envelope increases the number of cells in (7.10) and shortens their local clocks; it does not constrain the prescribed total T. Therefore a filter duration may be chosen from its retained low reference before choosing the unwanted carrier frequency.

The zero-mean condition in (7.1) has a precise analytic role. Without it a nonzero constant error has no diffusion, so the full-H^2 dissipation in (7.7) would require a lower-order mass term. All response references have the physical mean H, and thus satisfy exactly the stated condition.

8. Actual magnetic registers and a fixed-depth decoder

The arithmetic of Section 3 is useful only if a completed assignment returns the fluid to a common code format. A small error in a physical field cannot be projected away in a proof. We first construct an actual restoration operation, and then use its actual output coefficients as the reference for the next assignment. Velocity is disposable throughout this section.

8.1. **Code coefficients and physical norms.** Choose distinct real sine/cosine quadratures w_a at nonzero integer frequencies n_a, with

$$\int_0^1 w_a w_b = \tfrac{1}{2}\delta_{ab}, \qquad 1 \le |n_a| \le F.$$

The index a specifies a logical register and a retained real spatial coefficient. To use solenoidal basis fields at spatial frequency k, one may store the three scalar coefficients of $\mathbb{P}_k e_j$; for an originally solenoidal vector v, $\sum_j v_j \mathbb{P}_k e_j = v$. All inverses concern the known integer $|k|^2$. A bank has the form

$$B = H + \mathrm{Enc}(r), \qquad \mathrm{Enc}(r) = e_1 \sum_a r_a w_a(x_2). \tag{8.1}$$

The finitely many cyclic orientations of this format are also allowed. Changing the format label is bookkeeping; no coordinate transformation of the actual state is assumed to be an available control.

Use the sum of the coefficient norms $|\cdot|_{\mathcal{X}}$ of the work and output arrays as the logical norm $|r|_{\log}$. Permanent sources have a separate error budget. This norm is not the physical H^1 norm. If J_c is the number of code slots, elementary orthogonality gives explicit polynomials $P_{\mathrm{enc}}, P_{\mathrm{ext}}$ such that

$$\|\mathrm{Enc}(r)\|_{H^m} \le C_m (1+F)^m |r|_{\log}, \qquad |\operatorname{Ext}(e)|_{\log} \le P_{\mathrm{ext}}(N, J_c)\|e\|_{H^1}. \tag{8.2}$$

Here Ext takes twice the appropriate real Fourier inner products. For example $P_{\mathrm{ext}} = 2(1+21N)J_c$ is a safe bound. We can take $J_c \le C(N+1)^3$ and $F \le CJ_c$. Thus a common canonical state bound

$$P_c = C(1 + M + (1+F)(K+5)) \tag{8.3}$$

is polynomial in the numerical inputs. Completed assignments return to this same bound, rather than retaining their large internal packet norm.

8.2. **Variations, directional diffusion and restoration.** The memory inequality alone is insufficient for the next operation. We need its genuine variation estimates. Appendix G proves the following strengthened form directly from the actual transformed equations, including the lowering and final Stokes tail.

Lemma 8.1 (Normalized reset and transition variations)**.** *On a supplied joint H^1 ball, a reset can have arbitrarily small velocity-output derivative, while its normalized magnetic derivative C satisfies*

$$\|C\|_{H^1 \to H^1} \le 1 + \delta, \qquad \|(C - I)v\|_{H^1} \le \varepsilon \|v\|_{H^3}. \tag{8.4}$$

Its second variation on a common continuation ball is bounded by $Pd^{1/4}$ from $H^1 \times H^1$ to H^1, where d includes the recovery tail. One may choose the strong estimate in (8.4) with exponent $1/8$ in time. All requested small bounds have polynomial clocks and costs. The same conclusions hold for a fundamental-shear transition, normalized by its intended induction map, followed by a reset. The force depends only on the incoming bounds and fixed geometry. A previously accumulated prefix is allowed: both normalized endpoint norms use its same material coordinate, without a physical norm conversion at every call.

For clarity, the shear pulse used here is physical. On an interval of length d_{c} emit $q'F$, where

$$q(t) = \frac{a}{d_{\mathrm{c}}} 6(t/d_{\mathrm{c}})(1 - t/d_{\mathrm{c}}), \qquad F = e_i \sin(\kappa x_j), \quad i \neq j.$$

Only in the proof solve $x' + \kappa^2 \nu(t) x = q'$, $x(0) = 0$. Then xF is the exact actual-viscosity carrier, and

$$\left| \int_0^{d_{\mathrm{c}}} x - a \right| \leq C|a| d_{\mathrm{c}}, \qquad |x(d_{\mathrm{c}})| \leq C|a|. \tag{8.5}$$

The pulse need not have zero final velocity. Its following reset removes this unknown viscous tail while preserving magnetic memory. In particular no actual viscosity is inserted into the force.

Four shear phases supply a directional diffusion, not an applied coordinate average. If $s = \sin(\kappa x_j)$ or $\cos(\kappa x_j)$ and $E = e_i \otimes e_j$, direct differentiation gives

$$\begin{aligned} \mathcal{M}_a^{-1} \Delta \mathcal{M}_a = \Delta + a[-2s'\partial_{ij} - s''\partial_i + 2s''E\partial_j + s'''E] \\ + a^2[(s')^2 \partial_{ii} - 2s's''E\partial_i]. \end{aligned} \tag{8.6}$$

The average over $\sin, \cos, -\sin, -\cos$ is exactly

$$\Delta + \frac{\kappa^2 a^2}{2} \partial_{ii}. \tag{8.7}$$

The vector correction cancels as well as the scalar transport correction; this statement includes constant fields.

Lemma 8.2 (Actual directional filter)**.** *Supply a joint incoming H^1 bound P, rational $0 < T \leq 1$, rational $a \geq 1$ and an error budget $\delta > 0$. There is an ordinary twelve-mode programme of exactly duration T, with velocity returned small, whose magnetic output is uniformly within δ in H^1 of*

$$S_i B_{\mathrm{in}}, \qquad S_i = \exp[D_i(\Delta + \chi \partial_{ii})], \quad \chi = \kappa^2 a^2/2, \quad \mu T/2 \leq D_i \leq T. \tag{8.8}$$

The unknown D_i is an analysis quantity. The force uses no coefficient or state values. All costs are polynomial in $P, a, T^{-1}, \delta^{-1}, \mu^{-1}, 1 + \Lambda$. This is a uniform comparison on a supplied ball; the actual MHD endpoint map is nonlinear.

The proof in Appendix G includes three points that are essential here. Actual jumps are normalized with their genuine derivatives. An initial rough jump is paid by an adjoint smoothing estimate, rather than by an invalid small H^1 operator difference. Finally actual coupled holds are compared with their diffusion using one macro energy estimate, and complete macros are telescoped against contractive enhanced heat maps. There is no factor obtained by raising a raw Gårding bound to the number of cells.

Lemma 8.3 (Physical finite-rank restore)**.** *Let C_{la} be a known rational matrix mapping selected input quadratures $w_a(y)$ to output quadratures $w_l(z)$, and put*

$$D_z \chi_l = w_l, \qquad \Phi(y, z) = 2 \sum_{l,a} C_{la} \chi_l(z) w_a(y). \tag{8.9}$$

An actual twelve-mode operation can restore and route the code, with arbitrarily prescribed positive error and duration caps and polynomial resources. Its comparison is

$$\widehat{\mathcal{R}}_{D,C} = e^{D_y \Delta} P_y T_1(\Phi) e^{D_x \Delta} P_x e^{D_z \Delta} P_z, \tag{8.10}$$

where P_i denotes coordinate averaging only in this comparison formula. Every D_i has known positive lower and upper bounds. It preserves the unknown mean and has finite-rank nonconstant output. A selected coefficient is multiplied by C_{la} and by

$$e^{-\kappa^2 n_a^2 (D_x + D_z)} e^{-\kappa^2 n_l^2 D_y}. \tag{8.11}$$

These factors have modulus at most one and are not inverted. Thus every nonexpansive logical matrix remains nonexpansive after attenuation. Its bias is at most $PF^2(D_x+D_y+D_z)(K+1)$ *on the logical ball.*

Proof. First consider the exact linear comparison $\mathcal{R}_C = P_y T_1(\Phi) P_x P_z$. If B is solenoidal of mean H, then $P_x P_z B = (b_1(y), H_2, b_3(y))$ and

$$\mathcal{R}_C B = H + e_2 \sum_l w_l(z) \sum_a C_{la} \left(2 \int_0^1 w_a(y) b_1(y)\, dy \right). \tag{8.12}$$

This follows from $\mathscr{C}(e_1 f) = (0, D_z f, -D_y f)$. The last component averages to zero; the constant part of b_1 makes no code contribution. For equal code frequencies and $C = I$, this operator is an H^1 contraction. General routes have a known polynomial physical condition number; nonexpansivity for them concerns their logical coefficient matrix.

Replace the three averages by Lemma 8.2 and write the ideal diffusive composition as $\mathcal{R}_{D,C} = S_y T_1(\Phi) S_x S_z$. Positive baseline heat makes this regularized composition bounded on all of H^1. A three-term telescoping expansion gives

$$\|\mathcal{R}_{D,C} - \widehat{\mathcal{R}}_{D,C}\|_{H^1 \to H^1} \le P(D_{\min}^{-1}, \Phi, F, C) e^{-c \min_i \chi_i D_i}. \tag{8.13}$$

Indeed each $S_i - e^{D_i \Delta} P_i$ is exponentially small on its nonzero coordinate modes. For the first telescoping term use the positive $S_z : H^1 \to H^2$ before T_1; for the second use $(S_x - e^{D_x \Delta} P_x) S_z : H^1 \to H^2$; for the third use $S_z - e^{D_z \Delta} P_z : H^1 \to H^2$. In every case the one derivative in T_1 costs only a polynomial inverse baseline time. This proves the full regularized estimate, without asserting that T_1 is bounded on rough H^1 data.

Choose the three duration upper bounds from the attenuation budget first. The family $S_x S_z B$ has a known fixed-order smooth envelope $P(D_{\min}^{-1}) P$, independent of enhancement, even for an unknown incoming H^1 field. Lemma G.1 extends the physical whole-field gate to this unknown smooth family. Its proof pays a Fourier tail as well as the existing finite jet; neither a finite-support nor a linearity-in-H hypothesis is silently deleted. Choose that gate's error, gain and incoming allowance next, then choose the enhancements and actual directional-filter errors backward from (8.13). Only the physical implementation error before the gate is multiplied by its gain. Neither the input field nor its coordinate averages is evaluated. Formula (8.10) gives (8.11) by Fourier orthogonality, proving the remaining assertions. All choices are finite polynomial choices. □

8.3. **A decoder whose depth does not grow with the catalogue.** The initial code bank already stores every coefficient. Three row gates therefore suffice after routing it to a suitable high scale.

Lemma 8.4 (Fixed-depth physical decoder)**.** *Given known rational solenoidal mean-zero Fourier fields* F_l*, and bounds for unknown real coefficients* r_l*, an actual fixed-depth twelve-mode programme takes an encoded bank to*

$$H + \sum_l r_l F_l \tag{8.14}$$

within any prescribed positive H^1 *tolerance, with velocity returned small. Its force depends only on the known fields and coefficient envelopes. It has polynomial costs in the catalogue, envelopes, inverse time and inverse tolerance, with an exponent independent of the number of coefficients. The actual computed* r_l *need not be evaluated.*

Proof. Compute normalized curl potentials $\mathscr{C} A_l = F_l$ using the rational Fourier formula (4.3). Write A_{il} for component i. Use Lemma 8.3 to route the input to $H + e_1 \sum_l r_l w_l (S_1 x_2)$, writing the formulas for that output format. For another known format use the corresponding cyclic row order and cyclic indices in the emitted fields, rather than applying a coordinate transformation to the physical state.

The first router is an actual restore on the full incoming H^1 ball. An old off-code field with zero extraction is annihilated by its finite-rank comparison, including the preceding diagonal heat. Choose this router's fresh physical error after the destination scales are fixed, and then extract its actual routed coefficients for the next proof reference. Unknown attenuation and this fresh coefficient error have separately allotted logical budgets. Only newly introduced physical errors are multiplied by later high-frequency gains; historical coefficient errors pass through the known routing matrix. In particular the final carrier need not have been known during earlier arithmetic.

Choose lower scales S_2, S_3 and primitives $D_3\chi_{l,2} = w_l(S_2x_3)$, $D_1\chi_{l,3} = w_l(S_3x_1)$. Use successively

$$\phi_1 = 2\sum_l (A_{1l} + \chi_{l,2})w_l(S_1x_2),$$
$$\phi_2 = 2\sum_l (A_{2l} + \chi_{l,3})w_l(S_2x_3), \qquad \phi_3 = 2\sum_l A_{3l}w_l(S_3x_1). \tag{8.15}$$

After gate T_1 and removal of nonzero S_1 carriers, the reference is

$$H + \mathscr{C}\left(e_1\sum_l A_{1l}r_l\right) + e_2\sum_l r_l w_l(S_2x_3).$$

After gate T_2 and selection at S_2, it is

$$H + \mathscr{C}\left(e_1\sum_l A_{1l}r_l\right) + \mathscr{C}\left(e_2\sum_l A_{2l}r_l\right) + e_3\sum_l r_l w_l(S_3x_1).$$

The third gate and selection give (8.14). At every step $\langle w_l w_a\rangle = \delta_{la}/2$ extracts the desired row; cross products, the mean and the earlier low field times a pilot retain a nonzero current carrier. If J bounds base code indices and N_o the output band, safe separations are

$$S_3 \ge C(N_o + 1), \quad S_2 \ge C(N_o + JS_3 + 1), \quad S_1 \ge C(N_o + JS_2 + 1).$$

Each selector is replaced by the physical positive filtering argument of Lemma D.1; its retained reference is independent of its discarded carrier. Choose output tolerances and scales backward through this fixed list. The general heat service accepts the full resulting smooth envelope. Thus one router, three row gates and three filters prove the assertion with fixed depth. Copies and idle banks are included by adding their known basis fields to F_l. □

8.4. **Cubic multiplication and destructive assignments.**

Lemma 8.5 (Physical whole-register multiplication)**.** *On the common code ball, a twelve-mode operation implements*

$$(d, s) \longmapsto (h_i^2 d + s, 0) \tag{8.16}$$

and preserves the other live registers, with any prescribed fresh logical and off-code errors, polynomial clocks and universal polynomial resource exponents. It does not use the values of h_i or of the registers. Known injections and accumulate-and-clear operations are included. The three different permanent flux sources similarly give $h_1h_2h_3$ once, with the known normalization M^{-3}.

Proof. The short-triad estimate proved in Appendix G uses carrier $Q_c = q^4$, amplitude $A_c = q^6$, and pulse time $t_c = q^{-11}$, so $A_c^3 Q_c t_c^2 = 1$. For a real scalar register $r(z)$ put $g = (2\kappa)^{-1}\partial_z^{-1}r$, inverting only its known nonzero code frequencies. Prepare the three real packets

$$2A_c h_i e_2 \cos(\kappa Q_c x_1), \qquad 2A_c h_i e_3 \cos(\kappa Q_c x_2), \qquad 2A_c g(x_3)(e_1 - e_2)\sin(\kappa Q_c(x_1 + x_2)). \tag{8.17}$$

The two opposite horizontal triads give the slow magnetic coefficient $h_i^2 r(z)(e_1 - e_2)$. The sine quadrature is essential. Their exact coefficient and the convergent remainder estimate are derived in the appendix; fixed constants involving κ use paid rational enclosures.

Lemma 8.4 prepares all these packets and the desired idle bank together, consuming the old bank. For (8.16), include s directly at the destination code in the idle bank and omit the old destination and source codes. Include every other live register there unchanged. Permanent sources can occur in both active packets and idle memory. Thus overwrite and clearing are physical features of the prepared field, not a projection performed later in the proof.

On the actual unforced pulse, the leading slow field is the intended idle bank plus the desired cubic code. In the convergent expansion every mixed tree with a bounded idle leaf gains $A_c^{-1} = q^{-6}$ compared with three high leaves. Its slow remainder is below the common Pq^{-2} budget; all-low evolution is Pt_c. The expansion and arbitrary-mode perturbation estimate are proved in the appendix. They include both Lorentz and induction feedback and the exact constant-flux propagator.

Apply the actual general heat service for time q^{-7}. Every unwanted horizontal carrier has modulus at least Q_c, at every homogeneous degree, so its complete physical output is damped. The retained code incurs a small baseline heat bias, and velocity is reset. Finally apply an actual finite-rank restore to the canonical code. A slow cubic remainder projected into selected coefficients is a fresh logical error of size Pq^{-2}; the physical off-code error can independently be made as small as requested. All fresh preparation errors receive the smaller tolerance required by the pulse's polynomial incoming gain.

The operations are of fixed depth and return to the same code band. Known coefficient injections are linear decoder calls. Taking different first two flux sources in (8.17), and the third permanent source, initializes $h_1 h_2 h_3$. The initial h_i banks are obtained from the linear construction with the known factor M^{-1}. No unknown flux component is divided out. □

8.5. **The actual state is restored before its coefficients are used.**

Lemma 8.6 (Stable actual register machine)**.** *The logical programme of Proposition 3.3 has an actual twelve-mode realization. It uses at most $C(D+1)^3$ whole-register assignments and returns at every boundary to the common code band and state bound (8.3). For any requested arithmetic error $\delta_a > 0$, all costs and reciprocal disturbance and continuation allowances have fixed polynomial bounds in the numerical inputs and δ_a^{-1}. The exponent is independent of D.*

Proof. At a boundary decompose the actual field, only for the proof, as

$$B_n = H + \mathrm{Enc}(c_n) + e_n^{\mathrm{off}}, \qquad \mathrm{Ext}(e_n^{\mathrm{off}}) = 0, \quad \|e_n^{\mathrm{off}}\|_{H^1} \le \theta, \quad \|u_n\|_{H^1} \le \theta_u. \tag{8.18}$$

The vector c_n is the actual extracted coefficient vector, never a measured control input. Suppose its logical error is E_n and its live mass is at most $K+1$.

First use genuine unforced smoothing from this actual incoming state. The zero-disturbance comparison starts at that state, retaining every historical error. On a short rational time s the H1 Duhamel estimate in the appendix gives

$$\|B^0(s) - e^{\Delta \int_0^s \rho} B_n\|_{H^1} \le P_c^C s^{1/8}, \tag{8.19}$$

and a polynomial positive-time smooth envelope. Actual disturbances have a separately paid $P_c^C \eta \sqrt{s/\mu}$ effect. Choose s first from the local logical tolerance, then compute the smooth envelope. This is not an assertion that a rough disturbed solution becomes arbitrarily smooth.

Perform an actual identity restore next. Its ideal comparison attenuates the old code diagonally by factors of modulus at most one. Equations (8.2) and (8.19) pay the old off-code contribution, nonlinear smoothing, baseline attenuation and physical restore error. Only after this physical restore extract its actual coefficients c_n^{pre}. With fresh divided budgets,

$$|c_n^{\mathrm{pre}} - c_n^{\mathrm{ideal}}|_{\log} \le E_n + \delta_{\mathrm{pre}}, \qquad \|e_n^{\mathrm{off,pre}}\|_{H^1} \le \theta_{\mathrm{pre}}. \tag{8.20}$$

All higher derivatives of the new finite-band reference $H + \mathrm{Enc}(c_n^{\mathrm{pre}})$ depend only on $F, K+1, M$, not on the preceding restore's microscopic parameters.

Compile the next assignment from that common code envelope and its desired output budget. This fixes its carrier and polynomial fresh-error gain. Then choose θ_{pre} and velocity preparation tolerance below the divided fresh budget over this gain. The physical pre-restore can be compiled to that accuracy without changing the new reference envelope. This order is noncircular. Historical coefficient errors belong to the actual reference c_n^{pre}; they are not multiplied by a microscopic incoming-field gain.

Lemma 8.5 and its actual final restore give

$$c_{n+1} = A_h(c_n^{\mathrm{pre}}) + \text{fresh coefficient error}, \qquad \|e_{n+1}^{\mathrm{off}}\|_{H^1} \le \theta, \quad \|u_{n+1}\|_{H^1} \le \theta_u.$$

The exact A_h is one nonexpansive transfer/accumulate-and-clear map. Errors of permanent sources, including a multiplier slightly exceeding one in modulus, are paid within the fresh logical allowance using the bounded live mass. Thus

$$E_{n+1} \le E_n + \delta_{\log}. \tag{8.21}$$

Choose $\delta_{\log} \le \delta_a/[C(S_a+1)]$. Choose a multiplier allowance δ_h with both $CS_a\delta_h \le 1/4$ and $CKS_a\delta_h \le \delta_a/C$. Choose the permanent-source allowance $e_s \le \min(\delta_h/C, \delta_a/[CK])$, and its fresh drift at every assignment below $e_s/[C(S_a+1)]$. The square is Lipschitz by a fixed constant on the bounded source ball, so these choices pay its multiplier error. Choose the off-code tolerance below the same budget over the known extraction and fresh-error gains. Induction keeps the live error below one and proves the state bound at every boundary.

Use one common smoothing envelope, code band and polynomial upper bound for each finite assignment template. The number S_a multiplies its cost; a state-envelope exponent is not iterated S_a times. Each assignment has a fixed rational duration cap obtained by dividing the arithmetic horizon by its known count. All actual subcall durations are known after compilation. Direct strong continuation and disturbance estimates precede identical-data weak–strong transfer. The emitted derivative pulses, seeds and lowering pieces have degree at most two; all holds emit zero. This completes the physical realization. □

9. A mechanical amplifier with a whole-space bound

The final velocity readout will be small. We prepare its target by a known contracting Fourier multiplier, and recover its size by a mechanical word. The word reverses an inviscid Euler operation; physical time and both diffusivities remain positive. The important bound concerns every Fourier mode of an incoming error, including modes outside the target band.

9.1. The full matrix symbol and dyadic coverage.

Let $d(\xi)$ be the continuous transverse matrix in [1, Section 4], and let

$$S(\xi) = \sum_a \omega_a R_a d(R_a^T \xi) R_a^T$$

be the fixed rational rotational design of its Proposition 4.4. All constants describing this finite design are universal constants.

Lemma 9.1 (Two-sided size information)**.** *There are effectively computable fixed $c_d, C_d > 0$ such that, for $\xi \neq 0$ and $p(r) = \min(r^2, r^{-2})$,*

$$\operatorname{Sym} S(\xi) \le -c_d p(|\xi|)\mathbb{P}_\xi, \qquad \|S(\xi)\| \le C_d p(|\xi|). \tag{9.1}$$

These are full matrix statements on the transverse plane.

Proof. The first assertion is [1, Proposition 4.4]. For the second use its actual full-matrix construction, rather than deducing a norm from the symmetric part. Its moment matching gives leading matrices $-11r^2\mathbb{P}_\xi/10$ and $-3r^{-2}\mathbb{P}_\xi/35$ and remainder bounds $2^{110}r^3$ and $2^{110}r^{-3}$, respectively. Hence the norm is at most $2r^2$ for $r \le 2^{-120}$ and at most r^{-2} for $r \ge 2^{120}$, after a fixed enlargement. On the intervening fixed annulus, its Lemma 4.2 and $\sum|\omega_a| \le 2$ give $\|S\| \le 4608(2^{120}+2)^4$. Multiplying this bound by 2^{240} and enlarging by two gives a valid C_d. □

Let N_0 be the fixed frequency denominator clearer in [1], and let q_0 be its positive common weight denominator. Put $\gamma_0 = q_0\kappa^4$. For a Euclidean band radius $N \geq 1$, define

$$J_N = \lceil \log_2 N \rceil, \quad \lambda_j = N_0\, 2^j, \quad D_j(k) = \gamma_0 S(k/\lambda_j), \quad D_N(k) = \sum_{j=0}^{J_N} D_j(k). \tag{9.2}$$

Symbols are zero at $k = 0$ and on the longitudinal direction. The logarithmic integer is computed by binary comparisons.

Lemma 9.2 (Uniform norm and band gap)**.** *Fixed computable $c_0, C_0 > 0$ satisfy*

$$\sup_{k\neq 0} \sum_j \|D_j(k)\| \leq C_0, \qquad \operatorname{Sym} D_N(k) \leq -c_0 I_{k^\perp} \quad (1 \leq |k| \leq N). \tag{9.3}$$

Choose a fixed integer $b_0 \geq c_0^{-1}$, let $T_a = b_0\lceil \log_2 g \rceil$ for $g \geq 2$, and set

$$C_N = e^{T_a D_N}, \qquad A_N = e^{-T_a D_N}. \tag{9.4}$$

There is a fixed computable integer $\beta \geq 1$ such that, on every fixed Sobolev order,

$$\|C_N \mathbb{P}_{\leq N}\|_{H^s_{\sigma,0}\to H^s_{\sigma,0}} \leq g^{-1}, \qquad \|A_N\|_{H^s_{\sigma,0}\to H^s_{\sigma,0}} \leq C g^\beta. \tag{9.5}$$

Here $\mathbb{P}_{\leq N}$ denotes Fourier restriction to $0 < |k| \leq N$. The second bound is on the whole mean-zero space.

Proof. Split the dyadic arguments at a value in $[1, 2]$. Both tails of $p(r/2^j)$ have ratio $1/4$, so their finite sum is at most four. This proves the first inequality with $C_0 = 4\gamma_0 C_d$. If $|k| \geq N_0$, one chosen scale has argument in $[1, 2]$. If $1 \leq |k| < N_0$, scale zero gives $p \geq N_0^{-2}$. Thus $c_0 = \gamma_0 c_d \min(1/4, N_0^{-2})$ is sufficient. Each matrix is dissipative on $k^\perp$. Differentiating the squared Euclidean norm of its exponential gives contraction on the band. On all modes $\|e^{-tD_N(k)}\| \leq e^{C_0 t}$. Since $T_a \leq b_0(1 + \log_2 g)$, an integer $\beta \geq 2C_0 b_0 + 2$ suffices. Scalar Sobolev weights commute with these matrices, and Parseval proves every asserted operator bound. Rational lower and upper enclosures for the fixed constants determine b_0, β without evaluating an unknown quantity. □

9.2. **Actual words, their prefixes and noncommuting scales.** At scale λ_j, use the actual rotated periodic shears $R_a e_i\{\cos, \sin\}(\kappa\lambda_j R_a e_l \cdot x)$ from [1, Section 4]. Their frequencies are integer and their polarizations rational. Run the fixed word with parameter s/λ_j. Its fourth-order term is $s^4 D_j$. The division by λ_j cancels the fourth power of spatial dilation.

Lemma 9.3 (A finite reverse word)**.** *For a prescribed finite list of signed Sobolev orders, a computable $B_N \leq C(1 + N)^b$, with fixed b, bounds the dilated endpoint and affine fifth-order remainders. There are rational $s > 0$ and an integer $n = \lceil T_a/s^4 \rceil$, with polynomial reciprocal and count in N, g, ω^{-1}, such that the repeated reverse superword has Euler endpoint $\mathcal{Q}_{\mathrm{end}}$ satisfying*

$$\|\mathcal{Q}_{\mathrm{end}} - A_N\|_{H^m\to H^m} \leq \omega. \tag{9.6}$$

At every complete scale-word boundary its exact Euler prefix is ω-close to a known Fourier multiplier $\mathcal{A}$ with

$$\|\mathcal{A}\|_{H^m\to H^m} \leq C g^\beta, \qquad \|\mathcal{A}^{-1}\|_{H^m\to H^m} \leq 1. \tag{9.7}$$

Each complete induction word returns exactly to identity. Partial-word maps and inverses are bounded by two on the required orders after a further polynomial reduction of s.

Proof. Apply the proof of [1, Proposition 4.5] after spatial dilation. The endpoint transport still returns exactly, so its bounded interaction equation has five operator derivatives. Every fixed spatial or parameter derivative costs a fixed power of λ_j; the pressure inverse has a

fixed ellipticity neighbourhood after $s(1+N)^b$ is made small. The finite recursions therefore give a polynomial B_N. They give

$$\sum_j \|W_j(s/\lambda_j)^{-1} - e^{-s^4 D_j}\|_{H^m \to H^m} \le B_N s^5.$$

Here B_N includes the sum over the at most $1 + \lceil \log_2 N \rceil$ scales and the maximum over the prescribed orders m. The same proof and the affine inverse below give its affine counterpart. The partial norm bound follows from differentiated shear energy with small impulse times the required spatial envelope; it is not operator-norm closeness of transport to identity.

Different $D_j(k)$ need not commute. Taylor's integral remainder, applied pointwise to their product, nevertheless gives

$$\left\| \prod_j e^{-s^4 D_j(k)} - e^{-s^4 \sum_j D_j(k)} \right\| \le C s^8 \tag{9.8}$$

because $\sum_j \|D_j(k)\| \le C_0$. Every contiguous nominal scale subproduct has norm at most $e^{C_0 s^4}$. The discrete variation-of-constants formula for inserting the non-multiplier word remainders gives a superword bound $\exp(C_0 s^4 + C B_N s^5)$. In particular it does not multiply the separately maximized scale norms once per scale. One telescoping comparison over all repetitions now gives

$$C(n B_N s^5 + s^4) \exp(C_0 T_a + C n B_N s^5). \tag{9.9}$$

The last s^4 pays $|n s^4 - T_a|$. Taking, among finitely many caps, $s \le c\omega/[B_N(1+T_a)g^\beta]$ makes this at most ω after enlarging the fixed exponent once. The same argument applies to every incomplete list of complete scale words. Their nominal inverse is a product of dissipative factors and has norm at most one. Taking $\omega < 1/4$ also bounds the actual prefix inverse by two. Exact induction return is [1, Lemma 4.1], including inverses. □

9.3. **Known increments in an accumulated Euler metric.** We need to install generated shears without resetting an unknown velocity already written. A velocity setter alone does not do this. The following extension uses the actual closed leaves.

Lemma 9.4 (Joint-memory known increments)**.** *Let (v_0, b_0) be an unknown smooth solenoidal pair in a supplied fixed-order envelope. A known rational finite Fourier increment Z can be added to velocity, leaving b_0 unchanged to prescribed accuracy, by the twelve momentum directions. In addition to physical H^1, the estimate may be taken in the fixed norm*

$$\|(v,b)\|_{\mathcal{K},G} = \left(\|\mathcal{K}^{-1}v\|_{H^1}^2 + \|G^{-1}b\|_{H^1}^2\right)^{1/2}, \tag{9.10}$$

where $\mathcal{K} = P_\circ(\mathcal{A} + E)$, $\mathcal{A}$ is a nominal prefix from Lemma 9.3, E is sufficiently small on the required signed scales, and $P_\circ$ is any partial word immediately following that boundary. The empty partial word is allowed. The magnetic map G is the corresponding supplied smooth induction prefix. For any positive duration cap and $\theta, \delta > 0$ it gives

$$e_{\mathrm{out}} \le (1+\theta) e_{\mathrm{in}} + \delta \tag{9.11}$$

on a paid continuation ball. All costs, reciprocal clocks and reciprocal disturbance allowances are polynomial in the supplied envelopes, N, g, μ^{-1}, Λ and inverse budgets. The force uses Z and envelopes, never v_0, b_0 or actual profiles.

Proof. First consider a closed translation leaf from [1, Section 8.5]. Retain its known carrier w and seed force, but initialize its proof reference at the unknown v_0:

$$v_t + L_w v = \nu(t)\Delta w + f_{\mathrm{seed}}, \quad v(0) = v_0, \qquad u_* = w + v, \quad B_* = \mathcal{M}_w b_0.$$

The complete homogeneous Euler and induction maps are identity. Frozen-viscosity affine terms cancel, and actual Lipschitz variation costs a polynomial times Λd_{c} on a leaf of length d_{c}. The right-minus-left residuals are exactly

$$R_u = \nu\Delta v - \mathcal{B}(v,v) + \mathcal{B}(B_*, B_*), \qquad R_B = \rho \Delta B_* + \mathcal{C}_v B_*. \tag{9.12}$$

They have polynomial smooth bounds independent of compressed leaf length. The large seed derivative and carrier terms have cancelled.

For the metric in (9.10), use the multiplier commutator calculation of Appendix F: $[\phi, \mathcal{A}] : H^{s-1} \to H^s$ has polynomial bounds at the required orders. If L has scalar elliptic second-order principal part, then $\mathcal{A}^{-1} L \mathcal{A} = L + R_1$ with $R_1 : H^1 \to L^2$ polynomially bounded. Consequently the genuine Gårding argument of [1, Lemmas 8.1–8.2] persists through the fixed accumulated Euler prefix. Small actual-prefix errors have an absorbable $H^2 \to L^2$ form defect; see (9.17). Insertions inside a scale word also contain $P_\circ$. The scalar-principal commutator calculation in Appendix F pays its additional $P(N, g)s$ diffusion-form defect. It does not require $P_\circ - I$ to have small operator norm on H^1. The magnetic coordinate uses Lemma 5.1.

For completeness, subtract the reference before estimating errors. After removing the two carrier propagators, the homogeneous linear terms contain only the smooth remainder v, B_* and bounded coordinate maps. With $\mathcal{E}$ the squared joint H^1 error and $\mathcal{D}$ the joint H^2 error, the product estimates and Young's inequality give

$$\mathcal{E}' + a_* \mathcal{D}^2 \le P(\mathcal{E} + \mathcal{E}^3) + 2 r_1(t) \sqrt{\mathcal{E}} + P\eta^2, \qquad a_*^{-1} \le P. \tag{9.13}$$

Here r_1 bounds the smooth residual in the transformed H^1 norm. The only quadratic error source is the actual bilinear MHD error; its L^2 bound is $P\mathcal{E}^{3/4}\mathcal{D}^{1/2}$. This proves the displayed cubic energy term directly. There is no differentiation of an endpoint inequality with an additive error. On a fixed error ball, choose $\int P\,dt$ and $\int r_1\,dt + \eta(\int P\,dt)^{1/2}$ small. The regularization $\sqrt{\mathcal{E} + \varepsilon^2}$ and Gronwall give (9.11). Its homogeneous linear equation has no r_1 term, so it has the same near-unit factor.

To realize a signed quadratic node use known translations $+bK, -2bK, +bK$, separated by two unforced holds of length $d_c/2$, where $b^2 d_c$ approximates its known dose c. Before creating a hold, omit a dose whose absolute contribution is below its divided output tolerance. Every retained nonzero dose then has a polynomial reciprocal; no clock is selected by dividing by an arbitrarily small input coefficient. The total Euler and induction phases are zero. During a hold keep

$$v_t + qL_K v = \nu q \Delta K - q^2 \mathcal{B}(K, K), \quad q = b, -b, \qquad B_* = \mathcal{M}_K \left(\int q \right) b_0.$$

The quadratic term must be in this reference. At fixed dose its smooth bounds depend on bd_c and $b^2 d_c$, not separately on d_c^{-1}. Choose b polynomially large so $bd_c \|K\|_{C^m}$ is small. Variation of constants then gives the intended increment $-c\mathcal{B}(K, K)$ with mismatch $P(c/b + c/b^2)$, including the small displacement during each hold. The remaining residual (9.12) has actual $L^1_t H^1_x$ norm at most Pd_c. Frozen affine maps cancel on the phase return; the temporal residue is paid using the supplied Lipschitz bound. Select shorter child translation clocks after b and their envelopes have been fixed.

The passive-plane quadrature and at most two quadratic polarization stages of [1, Sections 8.3–8.4] now give every known Fourier increment. Its closed-leaf phase magnitudes have bounds depending on the spatial band, independently of the seed weights: amplitudes multiply the seed, not the conjugating geometry. Cap every generic quadratic phase by a fixed polynomial fraction before fixing its dose amplitude. Thus all leaf geometry and inverse ellipticity bounds can be fixed before the large carrier speed used below. The intended prefixes are bounded by sums of known increments. Divide θ, δ over the flattened primitive list; its product of endpoint factors is $\exp(O(\sum \theta_j))$, not a fixed loss per term. The same-coordinate energy proof above applies to actual twelve-direction lowering: preserve the old endpoints and choose each next mesh, after its inner amplitudes, so its continuous residual, Euler generator defect and signed diffusion defect lie below the divided fixed coercive margins. These are the actual operations of [1, Section 8.6]. All depths are fixed. Absolute rational dose and phase precision, including tiny coefficients, therefore has polynomial cost. □

9.4. **The physical amplification service.**

Lemma 9.5 (Whole-space mechanical amplification)**.** *Fix the spatial derivative order used above once. Supply rational $N \geq 1$, $g \geq 2$, $P \geq 2$, $0 < S, \delta \leq 1$ and the coefficient bounds. Let (v_*, b_*) be any unknown smooth solenoidal pair whose required fixed-order norms are bounded by P, with $\langle v_* \rangle = 0$. There is one ordinary rational twelve-mode force, of rational duration $0 < D \leq S$, and positive rational r, η, such that*

$$\begin{aligned} e_{\mathrm{in}} &:= \|(u_{\mathrm{in}} - v_*, B_{\mathrm{in}} - b_*)\|_{\mathcal{H}^1} \leq r \quad \Longrightarrow \\ &\|(u_{\mathrm{out}} - A_N v_*, B_{\mathrm{out}} - b_*)\|_{\mathcal{H}^1} \leq C g^{\beta} e_{\mathrm{in}} + \delta. \end{aligned} \tag{9.14}$$

The discrepancies are solenoidal and have zero mean, with no restriction on their Fourier support. The same force works for the complete unknown profile and reference family, all admitted disturbances of norm at most η, and every corresponding energy-admissible realization. The fixed β, C are independent of N, degree and target catalogue. For a universal polynomial in

$$X = 2 + N + g + P + \mu^{-1} + \Lambda + S^{-1} + \delta^{-1},$$

the reciprocal radii $r^{-1}, \eta^{-1}, D^{-1}$, force peak and impulse, counts and complete binary resources are bounded by that polynomial (with complete input length added for binary resources). The new force has time degree at most two.

Proof. Choose the rational word parameters of Lemma 9.3, simultaneously through the fixed orders needed for diffusion forms and smooth-reference sources. Their numerical tolerances are polynomial fractions of $\delta/(Pg^{\beta})$. Choose s also so every partial word and inverse is bounded by two and its Laplace commutator is small.

We first retain viscosity in the reference. For a shear F the one-letter affine action at a constant analysis value $\bar{\nu}$ is

$$z \longmapsto Q_F(a) z + \bar{\nu} a \Delta F, \qquad L_F \Delta F = 0.$$

Negative impulse is its exact affine inverse. Thus reversing a word $z \mapsto W_s z + \bar{\nu} b_s$ gives

$$z \longmapsto W_s^{-1} z - \bar{\nu} W_s^{-1} b_s. \tag{9.15}$$

Its fifth-order bias from [1, Lemma 4.7] survives reversal. For actual ν, define the continuous affine corrector only in the proof. Freezing within each complete word, with all its inserted physical intervals included in the clock, gives the coarse bound

$$\|a_{\mathrm{end}}\|_{H^m} \leq C g^{\beta} B_N \{ n s^5 + \Lambda D n s \}. \tag{9.16}$$

The estimate follows by variation of constants and $|\nu(t) - \nu(t_j)| \leq \Lambda D$; the total absolute affine impulse is at most $B_N n s$. Its partial spatial norms have the bound $P(N) g^{\beta}(s + n s^5 + \Lambda D n s)$. The defining equation cancels the large derivative of this corrector. No coefficient value is inserted into the force.

To implement a signed letter aF, use Lemma 9.4 to add the known carrier $\mathrm{sign}(a)\lambda F$, hold unforced for $|a|/\lambda$, and remove it by another known increment. The hold has exactly the required inviscid impulse. Its smooth velocity remainder is the Euler propagation of v_* plus the affine corrector; its norms depend on the fixed impulses, not the carrier speed λ. The magnetic reference follows the exact induction word. It returns to b_* at every complete scale word.

Here is a noncircular order of physical choices. First fix all word impulses, counts and their common smooth envelopes. Choose a positive rational hold-time budget d_{hold} from S, the second term of (9.16), and the smooth nonlinear, Lorentz and diffusive source budgets. All these hold envelopes are independent of λ. If A_{abs} is the known sum of absolute letter impulses, choose λ by a rational ceiling so $A_{\mathrm{abs}}/\lambda \leq d_{\mathrm{hold}}/4$. Only then fix the installation/removal envelopes and choose their shorter clocks and tolerances. Their sum is at most $d_{\mathrm{hold}}/4$, and their divided errors are smaller than the reserved budgets. Every reciprocal is polynomial: counts, A_{abs} and its reciprocal have polynomial bounds from the fixed word and its rational

s/λ_j. These rules also give a polynomial lower bound for the positive total duration. They do not impose $d_h P(\lambda)^c \ll 1$ after setting λ proportional to d_{hold}^{-1}.

We give the diffusion and rough-error checks, which are needed in addition to the inviscid endpoint. At a complete scale-word boundary write $\mathcal{Q} = \mathcal{A} + E$, where $\mathcal{A}$ is its known Fourier prefix and E is small on the required signed scales. Since $\mathcal{A}$ commutes with Δ,

$$\|[\Delta, \mathcal{Q}]\|_{H^{s+2}\to H^s} \le 2\|E\|_{H^s\to H^s} + 2\|E\|_{H^{s+2}\to H^{s+2}}.$$

Within a partial word, the additional commutator is at most $P(N)s$ times the preceding Cg^β bound. The inverses are bounded by four. Thus choosing s, E smaller by fixed polynomial factors gives an absorbable diffusion defect on every hold. During a generated-increment leaf, let $L = \mathcal{Q}_{\mathrm{leaf}}^{-1}\Delta\mathcal{Q}_{\mathrm{leaf}}$. Appendix F proves

$$\begin{aligned} \mathcal{A}^{-1}L\mathcal{A} &= L + R_1, \quad \|R_1\|_{H^1\to L^2} \le P, \\ \|\mathcal{Q}^{-1}L\mathcal{Q} - \mathcal{A}^{-1}L\mathcal{A}\|_{H^2\to L^2} &\le P(\|E\|_{L^2\to L^2} + \|E\|_{H^2\to H^2}), \end{aligned} \tag{9.17}$$

The right side is put below a fixed coercivity fraction. If the insertion is inside a scale word, its actual external prefix is $P_\circ\mathcal{Q}$. Write $L = L_0 + R_1$ with scalar principal part L_0. Conjugating L_0 by $P_\circ$ adds an $H^2 \to L^2$ defect bounded by $P(N)s$, and conjugating R_1 retains its $H^1 \to L^2$ bound. After $\mathcal{A}$ and E are included, the extra form defect is still $P(N, g)s$ and is absorbed by the earlier cap on s. Its constants use the amplitude-independent leaf geometry just established. The magnetic conjugation is differential, with the same fixed complete-word endpoint metric. This supplies coercivity rather than inferring it from bounded invertibility.

Subtract the smooth references and use these exact Euler/induction coordinates. The linear error terms involve the bounded smooth remainder and magnetic reference. Quadratic errors and disturbances give (9.13). On holds its coefficients and smooth residuals have common polynomial bounds independent of carrier speed. On increment calls their bounds may depend on λ, and their clocks were chosen afterwards. Arrange the sum of integrated linear coefficients to be below $1/16$, and the sum of smooth residual contributions and rough disturbance contributions to be below $\delta/(Cg^\beta)$. Choose the incoming continuation radius as a polynomial fraction of the smallest fixed coordinate bootstrap radius. First-exit integration of (9.13) then gives a uniform coordinate error at most twice its incoming norm plus the assigned additive budget. This is an energy proof for arbitrary rough modes, not a differentiation of an endpoint error estimate.

At the terminal time the magnetic coordinate is identity and the velocity coordinate has norm at most Cg^β. The word approximation and (9.16) pay its smooth reference bias, giving (9.14). No endpoint condition number has been paid once per letter. In particular, this proof does not assert that positive diffusion is uniformly close to identity on a rough H^1 ball.

All estimates apply to the actual disturbed local strong solution. Their strict bounds prevent its H^1 continuation exit through each finite segment. The identical-data weak–strong argument of [1, Section 10] transfers the endpoint to every admitted energy realization with its specified trace. Its uniqueness exponential is not used to select η.

Finally each known increment uses fixed-depth spatial certificates, ordinary rational seed pulses, and derivatives of cubic lowering pulses. Holds emit zero. Their time degree is at most two. Every refinement above is a positive rational minimum after its dependent amplitudes and envelopes are fixed. Repeated words reuse these common envelopes; counts multiply costs. The terminating majorant procedure in Appendix F therefore proves the stated numerical and binary bounds. □

9.5. **A finite known preconditioner.** Only C_N on the finite target band is computed. Each block is a known 3×3 matrix of norm at most one, and its exponent has norm at most $C_0T_a = O(\log g)$. Appendix F gives an absolute-error rational algorithm. It queries neither the physical state nor its coefficient functions. Applied to solenoidal target coefficients, it reduces their component-sum envelope to at most $\sqrt{3}/g$ times its initial value, up to the

explicitly allotted rounding error. After the physical amplifier, rounding is multiplied by at most Cg^{β} and is budgeted accordingly.

10. One collinear batch writes the even velocity

All unknown odd quantities are computed before this section begins. The final magnetic loader prepares the desired magnetic output and one complete packet batch, with velocity close to zero. There is no further packet loading after the unknown velocity has been written. The essential estimate below is stronger than a cubic error in the packet size: its complete unwanted physical output tends to zero with the carrier frequency. A subsequent whole-space amplifier can therefore have any fixed polynomial gain without forcing an incompatible packet-size choice.

10.1. **The exact propagator and the first interaction.** Write $B = H + b$ and $z = (u, b)$, with zero spatial means. For $k \neq 0$ put $\lambda_k = \kappa|k|$ and define the pair Fourier norms

$$\begin{aligned} \|z\|_{\mathsf{X}^r} &= \sum_{k\neq 0} \lambda_k^r (|\widehat{u}(k)|^2 + |\widehat{b}(k)|^2)^{1/2}, \\ \|z\|_{\mathsf{E}_T} &= \sup_{t\le T} \|z(t)\|_{\mathsf{X}^{-1}} + \mu \int_0^T \|z(t)\|_{\mathsf{X}^1}\, dt. \end{aligned} \tag{10.1}$$

Let $\mathbb{S}_H(t,s)$ be the exact constant-flux linear evolution. At each frequency its pair matrix, tensored with the spatial identity, is

$$-\lambda_k^2 \operatorname{diag}(\nu(t), \rho(t)) + i\kappa(H\cdot k) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{10.2}$$

Its Hermitian part is bounded above by $-\mu\lambda_k^2 I$; hence

$$\|\mathbb{S}_H(t,s;k)\| \le e^{-\mu\lambda_k^2 (t-s)}. \tag{10.3}$$

The time-dependent matrices need not commute. Both the decay bound and the fact that the spatial polarizations are only mixed by scalar pair coefficients hold for their exact time-ordered evolution.

Let $\mathfrak{B}$ denote the symmetric bilinear Duhamel operator for the quadratic MHD terms after this linear part has been removed. In particular $z = \mathbb{S}_H z(0) + \mathfrak{B}(z,z)$. It includes the Reynolds and Maxwell terms in momentum and both terms of induction. Set $\gamma(v) = \sqrt{\mu}\|v\|_{L^2(0,T;\mathsf{X}^0)}$. Fourier convolution and (10.3) give

$$\|\mathfrak{B}(v,w)\|_{\mathsf{E}_T} + \gamma(\mathfrak{B}(v,w)) \le C\mu^{-1}\gamma(v)\gamma(w). \tag{10.4}$$

The complete proof, including its labelled version used below, is in Appendix E.

Suppose the initial packet frequencies are

$$k_s = Qn_s\ell + \omega_s, \qquad 0 < |n_s| \le J_p, \quad |\omega_s| \le F_p, \quad \ell \in \mathbb{Z}^3 \setminus \{0\}. \tag{10.5}$$

For every complex solenoidal polarization a_s at k_s,

$$\ell \cdot a_s = -\frac{\omega_s \cdot a_s}{Qn_s}, \qquad a_s \cdot k_t = a_s \cdot \left(\omega_t - \frac{n_t}{n_s}\omega_s\right). \tag{10.6}$$

Thus $|a_s \cdot k_t| \le F_p(1 + J_p)|a_s|$, with no factor Q. This identity applies to every velocity and magnetic polarization, and (10.2) preserves it exactly. It therefore controls the full first quadratic interaction, including the velocity produced by Alfvén mixing; no Maxwell-only approximation has been made.

Lemma 10.1 (Complete collinear readout)**.** *Let an unknown real solenoidal joint background $z_0 = (u_0, b_0)$ have zero mean, Fourier band F_0, and supplied fixed-order smooth envelope. Let $|H| \le M$, and let an initially magnetic real packet $p_Q = (0, p_Q^B)$ have the frequencies (10.5), including real conjugates, and*

$$\|p_Q\|_{\mathsf{X}^{-1}} \le \sigma \le c\mu. \tag{10.7}$$

All background, tag, offset, polarization and upper frequency-ratio bounds are supplied before Q. Denote them collectively by a rational envelope $P \geq 2$. Given $0 < \zeta \leq 1$ and a duration cap, there are fixed computable constants and a rational clearing time

$$T = \frac{C_2 b_Q^2}{\mu Q^2}, \qquad n_* = C_1 b_Q, \qquad 2^{b_Q} \geq 2 + \frac{PQ}{\mu\zeta}, \tag{10.8}$$

where b_Q is the least such positive integer, with the following property for sufficiently large polynomially bounded integer Q. The actual unforced coupled endpoint from $z_0 + p_Q$ is

$$z(T) = z_0 + (\mathcal{L}_\rho(p_Q), 0) + \mathcal{R}_Q, \qquad \|\mathcal{R}_Q\|_{H^1} \leq C\zeta + \frac{P_*(1+b_Q)^2}{Q}. \tag{10.9}$$

Here P_* is a fixed computable polynomial in the earlier envelopes, μ^{-1} and σ, independent of Q. At a nonzero low frequency m, the leading velocity coefficient is

$$\widehat{\mathcal{L}_\rho}(m) = i\kappa \sum_{\substack{s,t:\ n_s+n_t=0 \\ k_s+k_t=m}} \mathbb{P}_m[(\widehat{p}_s^B \cdot m)\widehat{p}_t^B] \int_0^T e^{-\kappa^2(|k_s|^2+|k_t|^2)\int_0^r \rho(v)\,dv}\,dr. \tag{10.10}$$

The sum is ordered. No projection or coefficient-dependent propagation in this formula is an applied control.

For arbitrary incoming joint L^2, H^1 errors e_0, e_1 with the same velocity and magnetic means, and the actual mean-zero $L_t^\infty L_x^2$ disturbance of size η, the additional endpoint error is bounded by

$$C\left[e_1 + G_Q(e_0 + \eta T) + \eta\sqrt{T/\mu}\right], \qquad G_Q \leq CT\{P_4 + P_{\mathrm{freq}}\sigma Q^5\}. \tag{10.11}$$

Here P_4 bounds the background H^4 norm, and P_{freq} is a polynomial in $J_p, |\ell|, F_p{+}1$ independent of Q above separation. The bracket is required to be in an explicitly inverse-polynomial continuation ball. The perturbed solution then continues strongly on the whole window, and the estimate holds for every admitted energy realization and its one-sided endpoint trace.

Proof. Choose $Q \geq 8n_*(F_0 + F_p + 1)$, make T lie below its prescribed cap, and impose

$$\frac{C(\sigma + \sqrt{\mu T}P_0)}{\mu} \leq \frac{1}{16}, \qquad P_0^2 T \leq c\sigma, \qquad P_1 T \leq 1, \tag{10.12}$$

where P_i bounds $\|z_0\|_{\mathsf{X}^i}$ for $i = 0, 1$. These requirements have polynomial integer solutions: b_Q grows only logarithmically, whereas T decreases as $Q^{-2}b_Q^2$. Integer bit-length bounds, or $\log x \leq C_a x^a$ at one fixed positive $a < 1/4$, turn every requirement into a fixed-power sufficient inequality.

Put $v = \mathbb{S}_H p_Q$ and $z_2 = \mathfrak{B}(v, v)$. Equation (10.6), input decay and $\|p_Q\|_{\mathsf{X}^0} \leq PQ\sigma$ give

$$\sup_{t \leq T} \|z_2(t)\|_{\mathsf{X}^0} \leq P\sigma^2/\mu, \qquad \gamma(z_2) \leq P\sigma^2\sqrt{T/\mu}. \tag{10.13}$$

For the pure-packet solution write $z = v + w$. Its equation is $w = z_2 + \mathfrak{B}(v, w) + \mathfrak{B}(w, v) + \mathfrak{B}(w, w)$. The critical contraction, also valid after multiplying the packet by a complex parameter of modulus at most two, gives

$$\gamma(w) \leq C\gamma(z_2), \qquad \|z - v - z_2\|_{\mathsf{E}_T} \leq P\sigma^3\sqrt{T}\,\mu^{-3/2}. \tag{10.14}$$

Appendix E retains the integer carrier labels in this estimate. Cauchy estimates in the complex amplitude parameter give a geometric degree majorant. A degree-n zero-carrier term has physical frequency at most nF_p, so conversion to H^1 costs $C(1 + nF_p)^2$; the n^2 factor sums against that majorant. Thus the entire zero-carrier pure-packet remainder is bounded by the second quantity in (10.14), times a fixed polynomial.

Background leaves have carrier label zero and frequency at most F_0. Colouring leaves and using (10.4) gives the complete mixed and nonlinear-background zero-carrier bound

$$P(F_0, F_p, J_p, |\ell|) \left[\sigma P_0 \sqrt{T/\mu} + P_0^2 T\right]. \tag{10.15}$$

It includes all Lorentz and induction interactions with both components of the unknown background. This is a fixed-band statement: its envelope is selected before Q.

For every nonzero-carrier output at degree at most n_*, separation puts its physical frequency above $Q/2$. At a time slice of an n-leaf tree the active frequencies sum to the output m, so their squared lengths sum to at least $|m|^2/n$. Factoring half the scalar heat from each exact propagator gives the gain $e^{-c\mu Q^2 T/n}$. The remaining propagators still satisfy (10.3) with $\mu/2$. The complete higher-degree tail has geometric critical majorant and physical support at most PnQ; its H^1 conversion costs at most $P(nQ)^2$. The choices (10.8) pay both this tail and the finite-degree high sum by $C\zeta$. This is clearing by the actual evolution, including high modes created during the window. It is not an application of a spectral projector.

It remains to identify the low quadratic term. Remove the constant-H off-diagonal coupling from (10.2) and call the resulting diagonal evolution $D_k(t,s)$. Duhamel gives, on the high packet band,

$$\|\mathbb{S}_H(t,0;k) - D_k(t,0)\| \le C\frac{M}{\mu Q} e^{-c\mu Q^2 t}. \tag{10.16}$$

Indeed the direct bound is $CM|k|te^{-\mu\kappa^2|k|^2 t}$; spending half its decay gives the displayed inequality. In the full quadratic source this pays the generated Reynolds term, the Maxwell change and the quadratic magnetic component by P/Q, using again (10.6). For the diagonal initial magnetic packet the only quadratic source is its Lorentz term; its ordered Fourier convolution is exactly (10.10). Replacing its low-output propagator by identity costs $P(F_p^2 + MF_p)T$, since $|m| \le 2F_p$. Finally

$$\|\mathbb{S}_H(T,0)z_0 - z_0\|_{H^1} \le CT(\|z_0\|_{H^3} + M\|z_0\|_{H^2}).$$

Combining these bounds with (10.14) and (10.15), and substituting $\sqrt{T} = \sqrt{C_2}b_Q/(\sqrt{\mu}Q)$, proves (10.9).

For completeness the actual nominal reference satisfies

$$\int_0^T \|\nabla z(t)\|_\infty \, dt \le CP_1 T + C\sigma/\mu \le C. \tag{10.17}$$

Fixed-order differentiated MHD energy, with the constant-H cancellation, therefore gives

$$\sup_{t\le T} \|z(t)\|_{H^m} \le C_m\{\|z_0\|_{H^m} + P\sigma Q^{m+1}\}.$$

Appendix E proves the full-space perturbation estimate (10.11) from these bounds. It permits arbitrary spatial modes in errors and disturbances. Its direct bootstrap constructs the actual disturbed strong solution, after which identical-data weak–strong uniqueness gives the all-realization assertion. No uniqueness exponential selects its disturbance tolerance. □

10.2. **Rational polarizations and distinct tags.** Every retained nonzero output mode can be produced with the same carrier direction. Choose

$$L_0 = 2N + 1, \qquad \ell = (1, L_0, L_0^2). \tag{10.18}$$

For $0 < |m|_\infty \le N$, $m \cdot \ell$ is a nonzero integer: its highest nonzero positional digit dominates the lower digits. Also m is not parallel to ℓ, since a nonzero integer first component would force a third component of modulus at least $L_0^2 > N$. Set $q_m = \mathbb{P}_{\ell^\perp} m$ and $a = q_m/|q_m|^2$. For a positive tag n and a rational $d \perp m$, put

$$\begin{aligned} k &= Qn\ell, \qquad l = m - k, \qquad \Delta = 2Qn(m\cdot\ell) - |m|^2, \\ \beta &= -\frac{Qn|m|^2(d\cdot\ell)}{\Delta}, \qquad b = d - \beta a + \frac{2\beta m}{|m|^2}. \end{aligned} \tag{10.19}$$

For $Q \geq 6N^2$ these rational vectors obey

(10.20)
$$a \cdot k = b \cdot l = 0, \qquad \mathbb{P}_m[(a \cdot m)b + (b \cdot m)a] = d, \qquad |a| \leq C(1+N)^2, \quad |b| \leq C(1+N)^6|d|.$$

Indeed $a \cdot m = 1$, $a \cdot \ell = 0$, $b \cdot m = \beta$ and $\mathbb{P}_m(b + \beta a) = d$. Substitution of β gives $l \cdot b = 0$. The integer cross product $m \times \ell \neq 0$ gives $|q_m|^{-1} \leq |\ell|$, while $|\Delta| \geq Qn$ gives the bounds for β, b. No target coefficient or flux component is inverted. Taking $d = \mathbb{P}_m e_j$, omitting zero directions, covers every transverse polarization, including coordinate-axis output modes.

Use one representative of each real conjugate output pair and assign a distinct positive tag to every retained mode/cofactor/component pair. The initial batch has a first coefficient at k, a second at l, and their real conjugates. The only quadratic zero-carrier combinations are $k + l = m$, its conjugate, and zero-mode self-pairs. The last vanish because every nonlinearity is a divergence. Different tags cannot cancel. Higher tag relations require no exclusion: the complete higher remainder in Lemma 10.1 already pays them.

The known factor

$$\frac{\kappa(|k|^2 + |l|^2)}{Q^2\sigma^2} \leq \frac{P(N, J_p, |\ell|)}{\sigma^2} \tag{10.21}$$

is bounded independently of Q above separation. Thus the tag and polarization envelopes required before carrier selection are available in advance. Rational approximations of the fixed κ factor use absolute divided error, with the known field envelopes paying it.

10.3. **The internal resistivity signal and exact real phases.** The leading coefficient has an unknown resistivity normalization. It is cancelled by a physical field signal, not by a measured coefficient. Before final loading, keep the register bank in the exact parallel sector

$$u = e_2 v(x_3), \qquad B = H + e_2 b(x_3). \tag{10.22}$$

All nonconstant nonlinearities vanish. At code frequency $n \neq 0$,

$$\frac{d}{dt}\begin{pmatrix} v_n \\ b_n \end{pmatrix} = \begin{pmatrix} -\kappa^2 n^2 \nu & i\kappa n H_3 \\ i\kappa n H_3 & -\kappa^2 n^2 \rho \end{pmatrix}\begin{pmatrix} v_n \\ b_n \end{pmatrix}.$$

From $v_n(0) = 0$, $b_n(0) = r$, contractivity and the integral equation give

$$b_n(t) = r - \kappa^2 n^2 r \int_0^t \rho(s)\, ds + \epsilon_n, \qquad |\epsilon_n| \leq \tfrac{1}{2}(\kappa^2 n^2 + \kappa|n|M)^2 t^2 |r|. \tag{10.23}$$

Identical copies at two known distinct code frequencies n, N_c therefore supply the internal signal

$$\frac{b_n(t) - b_{N_c}(t)}{\kappa^2(N_c^2 - n^2)t} = \bar{\rho}_t r + O(Pt)|r|, \qquad \bar{\rho}_t = t^{-1}\int_0^t \rho(s)\, ds. \tag{10.24}$$

The copies, subtraction and routing are the physical register operations; the known denominator is paid and never involves r. Viscosity and Alfvén coupling are present in (10.23)'s remainder. Both copies refer to the same actual extracted code coefficient r. Its historical error is therefore common to the two leading constants and is not amplified by t^{-1}. Fresh copying, preparation and routing errors are assigned a tolerance at most a fixed multiple of $t\delta_{\mathrm{sig}}$ divided by their known polynomial gains, so the normalization in (10.24) pays them. These are choices after the signal time, not a redefinition that deletes old field errors. Velocity may be reset here, while it is still disposable. No value of $\bar{\rho}_t$ is observed or inserted into a later force.

Write the preconditioned even target without its degree-zero part as $W = \sum_j h_j \mathcal{R}_j$, with odd cofactor fields. At mode m, use the component coefficients of $\mathcal{R}_j$ with directions $d = \mathbb{P}_m e_r$. Let c be one such complex coefficient. Write the actual internal signal as $r_j^\rho = \rho_{\mathrm{sig}} h_j + \epsilon_j$, where $\rho_{\mathrm{sig}} = \bar{\rho}_{t_{\mathrm{sig}}}$ is a proof-only common average and ϵ_j contains its allotted code and

physical errors. This is an additive decomposition; no signal is divided by h_j. With a known positive duplication normalizer A_0, take magnetic Fourier coefficients

$$\widehat{p}^{B}(k) = Q\sigma \frac{r_j^{\rho}}{A_0} a, \qquad \widehat{p}^{B}(l) = Q\sigma \left[-iA_0 \frac{\kappa(|k|^2 + |l|^2)}{Q^2\sigma^2}\, c \right] b. \tag{10.25}$$

The component ϵ_j is paid separately by the bounded product map; the following normalization concerns its ideal part. Formula (10.20) already includes both ordered terms. If $c = (C - iD)/2$ corresponds to the real cofactor $C\cos(\kappa m\cdot x) + D\sin(\kappa m\cdot x)$, its second real packet quadrature in (10.25) is exactly represented by

$$\begin{aligned} p^{B}_{k,l}(x) &= 2Q\sigma \frac{r_j^{\rho}}{A_0} a \cos(\kappa k\cdot x) \\ &\quad + \frac{A_0\kappa(|k|^2 + |l|^2)}{Q\sigma}\, b[C\sin(\kappa l\cdot x) - D\cos(\kappa l\cdot x)]. \end{aligned} \tag{10.26}$$

The conjugate coefficients are included. In particular a cosine target requires the sine second packet, with this sign and normalization.

Put $a_m = \kappa^2(|k|^2 + |l|^2)$ and, only in the following formula, write $\rho(s)$ for the actual coefficient at readout start plus s. Set $R(s) = \int_0^s \rho(v)\, dv$. The leading normalized multiplier is governed by the exact identity

$$a_m \rho_{\mathrm{sig}} \int_0^T e^{-a_m R(s)}\, ds = 1 - e^{-a_m R(T)} + a_m \int_0^T [\rho_{\mathrm{sig}} - \rho(s)] e^{-a_m R(s)}\, ds. \tag{10.27}$$

Consequently its error from one is bounded by

$$e^{-a_m \mu T} + \frac{\delta_{\mathrm{sig}} + \Lambda\Delta t}{\mu}, \tag{10.28}$$

where δ_{sig} is the error in producing the internal coefficient and Δt is the entire prescribed elapsed time from signal generation to readout end. Generate the signal after the variable-degree arithmetic and compress the following preparation into this short time budget. All times are chosen from bounds, without a coefficient query.

Take A_0 large enough to bound the sum of the duplicated first-packet critical norms, for example a fixed multiple of the number of pairs times the uniform bound on $|a|$. Using (10.21) and the bound on b, there is a computed carrier-independent polynomial P_{tag} such that

$$\sum_{j,m,r} |c_{j,m,r}| \le \frac{\sigma^2}{C P_{\mathrm{tag}}} \tag{10.29}$$

makes the entire real packet list have critical norm at most σ. The first-signal norm uses $|r_j^{\rho}| \le 2$; the compensated second norm is paid by (10.29). This is a sum estimate over the whole batch, not a per-packet smallness assertion. The preconditioner may enforce this inequality before Q is chosen.

10.4. **Historical coefficients and fresh physical errors.** Use the actual computed register coefficients as the packet reference. Errors already present in those coefficients change the leading output through the finite product map in (10.25)–(10.27). On the bounded logical ball that map has a carrier-independent polynomial Lipschitz bound: the $Q^2\sigma^2$ in the Lorentz source cancels its known normalization, and ρ_{sig}/ρ is bounded by supplied positive diffusivity bounds. For example the error in a product satisfies $|\delta h\, c + h\, \delta c + \delta h\, \delta c|$, summed over the coefficient mass. Logical errors in the desired magnetic output similarly remain finite-band coefficient errors. They are never reclassified as an $O(Q^2)$ incoming-field error.

Fresh loader, smoothing, routing, impulse-rounding and disturbance errors are different. After Q is fixed they receive the genuinely smaller field tolerances in (10.11). Its gain is polynomial in Q, so these allowances remain inverse-polynomial. Together with (10.28) and Lemma 10.1, this gives one endpoint $(W, H + Z)$ with the allocated error, before amplification.

Neither repeated loading after unknown velocity nor an unknown register value as a mechanical force coefficient is required.

The supplied exact rational checking script tests (10.6), (10.20) and the real-phase normalization on finite rational examples, including zero signals and coordinate-axis outputs. It is an algebraic check; the preceding proofs, not those finite examples, establish the PDE and all-frequency estimates.

11. The nonlinear compiler and its resources

The preceding operations act on one evolving field. We now fix all choices from known data and show why their dependence is acyclic. A polynomial envelope below is generated from the finite estimates in the proof, with nonnegative rational coefficients. It is never an unknown physical norm.

11.1. **The historical coefficient error and the fresh field error.** The distinction between these two errors is essential. Let r denote the code coefficients actually extracted in the proof after the last restore, and let $r^\star(h)$ be their intended odd polynomial values. The register theorem gives a coefficient estimate

$$|r - r^\star(h)|_{\mathrm{code}} \le \delta_a, \qquad |r|_{\mathrm{code}} \le P(K, N, D). \tag{11.1}$$

The controller evaluates neither vector. The final decoder is analyzed relative to the packet list built from r, not from $r^\star$. Its fresh physical discrepancy is separately made small after the carrier is fixed. Its first router is the uniform physical restore on the whole incoming H^1 ball. The pinned finite-rank comparison annihilates the old off-code remainder, whose selected coefficients vanish; scalar heat before extraction does not change that fact. Choose this router's own fresh physical error after Q, then use its actual routed coefficients as the packet reference. Known attenuation contributes a separately paid coefficient error. Thus an old off-code allowance need not have anticipated the final carrier.

The leading readout coefficient is a known bilinear map of the two coded factors. Its normalized packet polarizations and tags have envelopes independent of Q, as proved in Section 10. On the known coefficient ball its Lipschitz constant is at most P_t, a computable polynomial independent of Q. Consequently

$$|\mathcal{V}(r) - \mathcal{V}(r^\star)|_{\mathrm{a}} + |\mathcal{Z}(r) - \mathcal{Z}(r^\star)|_{\mathrm{a}} \le P_t \delta_a. \tag{11.2}$$

This follows also directly from $ab - a'b' = a(b - b') + (a - a')b'$ and the paid coefficient bounds. The preconditioner has coefficient gain at most a fixed constant on the output band, uniformly in g, so it introduces no circular dependence in P_t.

In contrast, the map from coefficients to the high-frequency physical packet can have a norm growing with Q. A fresh error in preparing that packet must pay that norm and the rough-error gain of the readout lemma. Charging (11.1) as such a fresh field error would impose an unnecessary and circular carrier-dependent accuracy on the entire earlier arithmetic.

11.2. **A concrete order of compilation.** First canonicalize and truncate as in Lemma 3.1. Set a Euclidean output radius $N_b = 2N$, so every retained $|k|_\infty \le N$ lies in $|k| \le N_b$. Form U_0, Z_N, R_1, R_2, R_3 and the Horner schedule. Fix the logical code frequencies and all eleven register banks. Their number and largest frequency are polynomial in the retained coefficient count. This code is fixed before the final packet carrier.

Choose the known packet direction, tag catalogue and polarization envelopes from N_b and the code list. Their bounds in Section 10 are uniform for all sufficiently large Q; use those upper bounds, rather than evaluating them at a provisional carrier. Choose a positive dyadic critical budget $\sigma \le c\mu$, also small enough for the stated background and continuation constants. Its reciprocal is polynomial in R_*. Fix P_t large enough for the entire catalogue and all coefficient normalizations.

Construct the dyadic symbol of Section 9. Choose a dyadic integer $g \geq 2$ so that

$$\frac{CK}{g} \leq \frac{\sigma^2}{16P_t}. \tag{11.3}$$

The band contraction gives $W = C_{N_b}(U_N - U_0)$ satisfying the required weak packet budget. Equivalently precondition the three known cofactor coefficient lists by C_{N_b}; this multiplier does not mix spatial frequencies or parameter monomials. Evaluate its finite 3×3 matrices to an absolute rational error χ as in Appendix F. Use a rational Leray projection and real conjugate pairs, preserving divergence and reality exactly. Allocate χ below the divided final coefficient allowance.

Let $G_a = Cg^\beta$ be the terminal whole-space gain of the physical amplifier. Intermediate metric bounds depending on N are paid by its clocks and input radii; they are not charged once per letter or added to this terminal gain. Its smooth reference envelope depends only on K, N, g and the code bound. It is independent of the later carrier Q. First fix the final known-U_0 increment, with its reference envelope for $(A_{N_b}W(r), H + Z_N(r))$, endpoint allowance $\epsilon/256$ and admissible input radius r_f. Then fix the amplifier with endpoint allowance $\delta_{\mathrm{amp}} \leq \min(\epsilon/256, r_f/16)$ and let r_a be its admissible input radius. These are inverse-polynomial radii computed before Q; the reference fields are finite-band functions of the bounded code values, never of a high-frequency loader error. Choose the arithmetic and signal coefficient allowances so that

$$P_t\delta_a \leq \frac{\epsilon}{2^{12}G_a}, \qquad \chi K \leq \frac{\epsilon}{2^{12}G_aP_t}. \tag{11.4}$$

With $S_a \leq C(D+1)^3$, choose the square-multiplier error to satisfy both parts of Proposition 3.3:

$$\delta_h \leq \min\left\{\frac{1}{4C(1+S_a)}, \frac{\delta_a}{C'(1+K)(1+S_a)}\right\}.$$

The first cap bounds its exponential; the second pays the additive $CKS_a\delta_h$ term. Choose source and local errors

$$\delta_s \leq \min\left\{\frac{\delta_a}{C(1+K)}, \frac{\delta_h}{C'}\right\}, \qquad \delta_{\mathrm{loc}} \leq \frac{\delta_a}{C(1+S_a)}.$$

The h_i source allowance includes its effect on h_i^2: on the paid source ball that square has a fixed Lipschitz constant. Divide the permitted fresh drift of each permanent source once more by $C(1+S_a)$ so its accumulated error remains below δ_s. The register lemma realizes each assignment with these absolute errors and a common polynomial state envelope. Its actual smoothing and restore occur before the next coefficient reference is taken. All historical physical errors therefore remain in the state and enter the proved coefficient recurrence.

Now fix Q. Let ζ be a rational clearing allowance below

$$\min\{\epsilon/(2^{12}G_a),\ r_a/16,\ r_f/(64G_a)\}.$$

The readout estimates have the form

$$E_{\mathrm{read}} \leq C\zeta + P\frac{(1+\log Q)^b}{Q}, \tag{11.5}$$

including the finite-band unknown magnetic background. All coefficients of P and the fixed integer b are known at this point and independent of Q. Require this bound, the propagation drift, and all signal-normalization errors to be below their divided ϵ/G_a allowances. Require also the tag separations and the rational readout duration to fit its reserved clock. Require the total fresh readout error also to be at most $r_a/4$ and $r_f/(16G_a)$. Thus the actual output enters the amplifier's continuation ball and then the final increment's ball. All these carrier constraints use radii already fixed independently of Q. These inequalities have a polynomial integer solution: for fixed b, $(1+\log Q)^b \leq C_b\sqrt{Q}$, with computable C_b, so the square of the remaining polynomial envelope suffices after a fixed enlargement. An ordinary doubling

search with this explicit cap terminates without testing a PDE. The logarithmic truncation depth is computed from integer bit lengths.

Generate the internal resistivity signal only after arithmetic. Its short unforced generation time and the subsequent fixed-depth decoder time are chosen so that their total elapsed interval d_{sig} obeys

$$P_t \Lambda d_{\mathrm{sig}}/\mu \le \epsilon/(2^{12} G_a). \tag{11.6}$$

The signal's two known frequencies and difference normalization have already been fixed. Copies and routes are actual register operations; they never read the signal's value. After Q is fixed, assign the smaller fresh preparation, matrix-rounding and field tolerances required by the physical decoder and the whole-space readout perturbation bound. Their reciprocals are fixed polynomials in Q and the earlier envelopes.

The final decoder prepares $H + Z_N(r)$ and the entire collinear packet list with velocity close to zero. There is one unforced readout and clearing window. Apply the amplifier to its small velocity output while retaining the already written magnetic field. Finally apply the known joint-memory increment U_0. This last operation uses known Fourier coefficients and envelopes, not unknown register values. No packet is reloaded after the unknown velocity has been written.

11.3. **The endpoint error ledger.** For clarity allocate the total joint H^1 error to five groups, each at most $\epsilon/16$ after all later gains:

(1) historical coefficient, permanent-source and preconditioner rounding errors, using (11.2) and G_a;

(2) the fresh final-loader and internal-signal errors, using the readout's arbitrary-mode gain and then G_a;

(3) the complete readout remainder, physical high-mode clearing and background drift, using (11.5) and then G_a;

(4) the amplifier's own affine, diffusive, nonlinear, lowering, precision and disturbance errors;

(5) the final known increment and its disturbance allowance.

Choose the finitely many local divided budgets smaller if their displayed fixed sums demand it. The new operations have polynomial admissible input radii, so each fresh output allowance is also chosen within the next radius. The arithmetic recurrence, rather than a product of local gate condition numbers, handles its variable number of assignments. Thus the actual final state differs by less than $\epsilon/2$ from $(U_N(H/M), H + Z_N(H/M))$ in joint H^1. Adding (3.1) proves the required joint L^2 endpoint.

Every comparison has the same magnetic mean H. Gates add curls, induction pushforwards preserve the mean, all heat comparisons fix constants, and both disturbances have zero mean. The full inhomogeneous magnetic norms and mean terms in the estimates are therefore applicable throughout.

11.4. **Exact rational clock and unknown preparation.** Give all writing stages a total rational duration $T_w < \tau/4$, including register computations, signal preparation, loading, readout, amplification and final increment. Each service accepts a prescribed positive rational cap; for a variable list divide its group cap by its known call count. The final carrier is increased if needed for its unforced window to fit. The amplifier fixes its large known carrier heights before choosing the still shorter child increment clocks, as established in Section 9.

Reserve a rational preparation interval $T_p \le \tau/4$. Prepend the common eraser on the exact horizon $T_0 = \tau - T_p - T_w \ge \tau/2$ and then unforced preparation on T_p. Lemma D.2 gives a small joint H^1 handoff near $(0, H)$ at the prescribed time $\tau - T_w$, after choosing the eraser's joint L^2 tolerance and disturbance allowance inverse-polynomially small. Its regular intermediate time is used only in the proof; the force does not wait for a measured stopping event. The argument uses the actual non-autonomous energy inequality and the

strong energy trace of the eraser. All coefficients restricted to any later interval retain their ranges and Lipschitz bound. Translation of a rational time polynomial by its now known start time preserves rationality and degree. The final endpoint is exactly τ.

Take η to be a positive dyadic rational below every local disturbance allowance and the initial eraser allowance. The direct estimates, not a weak–strong uniqueness exponential, prove that this common allowance has polynomial reciprocal. Section 12 transfers the endpoint to all admitted realizations and their terminal traces.

11.5. **A terminating universal majorant.** We give the resource argument explicitly because a degree-dependent exponent would change the theorem. The retained input has at most $C(D+1)^3(N+1)^3$ keys, $N \le CR_*^2$, and $S_a \le CR_*^3$. The code, tags and polarization envelopes have fixed polynomial bounds. Equation (11.3) gives polynomial g, and the one auxiliary exponential is replaced by the fixed power Cg^β. The tolerances (11.4) are inverse-polynomial. The carrier argument above gives polynomial Q. All later clocks, matrix errors, preparation radii and disturbance allowances are inverse-polynomial in these quantities.

To produce universal constants algorithmically, first generate the fixed word, spatial certificates and constants in [1]. Use the full-matrix tail and annulus formulas in Section 9 to generate c_0, C_0, β. Fix once the finite Sobolev orders used in the variation, commutator and Fourier estimates. Product rules, finite derivative recursions, Fourier-shell sums and Young inequalities then generate positive-coefficient rational majorants for every service. Each constant depending only on these fixed orders is included as fixed precomputation. Exact rational upper enclosures suffice for π and other fixed constants.

Propagate degrees and coefficient sums through the acyclic construction: addition takes the maximum degree and adds sums; multiplication adds degrees and multiplies sums; a fixed power multiplies degree. A minimum of positive clocks is handled by summing reciprocal majorants. Replace logarithms by the explicit fixed-power bounds already used, and replace $e^{C_0 T_a}$ by Cg^β. Variable loops multiply their count into work, memory where retained, and output. They do not iterate a state-envelope polynomial:

- each arithmetic assignment returns to the same code band and live coefficient ball, with one uniform smoothing envelope;
- routing and decoding have fixed physical depth;
- repeated amplifier words use one accumulated metric and its whole-space gain certificate;
- there is one final loader, readout and amplifier.

This finite procedure outputs numerical and binary majorants $Q_{\mathrm{num}}(R_*)$ and $Q_{\mathrm{bit}}(L+R_*)$. Choose an integer p dominating their degrees and A dominating their coefficient sums and all fixed costs. They are effectively computable and independent of D. In particular there is no term P^{S_a} hidden in this choice.

Exact canonicalization reads complete denominators and may retain their L-bit information. Later physical coefficient choices use absolute rounding and omission thresholds. For a tiny target coefficient one never divides by its magnitude. Rounding through a rational curl potential or Leray projection preserves spatial constraints exactly, with a fixed polynomial accuracy cost. Products and sums of polynomially many rational clocks have bit height at most their total operand height. Expanding the finitely described ordinary time polynomials has the paid degree and output cost; compressed pulse derivatives and all emitted coefficients are counted. These observations give all bounds in Theorem 2.2. Since $\tau \le 1$, the impulse is also bounded by the force peak after enlarging A.

11.6. **Recovery of the original linear input class.**

Corollary 11.1. *The nonlinear response theorem contains linear magnetic targets through*

$$U(x,h) = V(x), \qquad Z(x,h) = M(\Pi(x) - I)h.$$

For the Sobolev input promise of Theorem 2.3, a polynomial canonicalization, truncation and envelope conversion gives its original L^2 endpoint and universal numerical/binary polynomial bounds. The separate linear construction retains its stronger time-degree bound of four.

Proof. First truncate the old complete lists at Euclidean radius $N_\ell = \lceil CMK/\epsilon \rceil$, using Lemma 4.2. Then

$$\|V - V_{N_\ell}\|_2 + M\|\Pi - \Pi_{N_\ell}\|_2 \le \epsilon/8$$

for a fixed sufficiently large C. For a finite field supported in this band, Cauchy–Schwarz with the Fourier H^1 weights gives

$$|F|_{\mathcal{X}} \le C(1+N_\ell)^{3/2}\|F\|_{H^1}.$$

Real sine/cosine conversion and the finite number of matrix entries cost only a fixed dimensional factor. Thus the new coefficient promise holds with a known rational $K' \le C(1+M)(1+K)(1+N_\ell)^2$. Apply Theorem 2.2 with degree $D = 1$, this K', and a smaller endpoint allowance. Its magnetic mean is H because $\langle \Pi \rangle_{N_\ell} = I$; every coefficient is solenoidal. The truncation error completes the original endpoint estimate. All new numerical parameters are polynomial in the old ones. This is a conversion of promises, not an identification of the coefficient sum with a Sobolev norm. The degree-four statement follows from the distinct fixed-depth proof in Appendix D, not from the general degree bound of the nonlinear theorem. □

12. Controlled continuation and the scope of the response

12.1. **Physical strong solutions and all admitted weak realizations.** All physical estimates above concern the actual disturbed equations, not only an ideal reference. We make the transfer argument explicit. For a fixed finite output force and fixed actual disturbance, Fourier Galerkin approximation gives a local strong solution from H^1 data. Pairing the two equations with $(I - \Delta)(u, B)$, using (A.1) and Young's inequality, gives a local bound

$$E' + c\mu\|(u,B)\|_{\mathcal{H}^2}^2 \le C\mu^{-3}(1+E)^3 + C\mu^{-1}\|(f+h_u, h_B)\|_{\mathcal{H}^0}^2, \qquad E = \|(u,B)\|_{\mathcal{H}^1}^2.$$

It is uniform in the Galerkin cutoff on a positive interval. Compactness and the equations give $L^\infty H^1 \cap L^2 H^2$, time derivative in L^2L^2, and strong H^1 continuity. The same inequality restarts the solution whenever its H^1 norm is bounded. Spatially constant positive time-dependent coefficients introduce no differentiated temporal term.

The special reset, hold, packet and global-heat estimates prove that this local solution continues through each designed segment. The large transient velocity carriers are finite and explicitly bounded; a generic lifespan estimate based on their peak is not imposed on the entire designed segment. Their cancellation coordinates supply the needed continuation estimates. Before writing starts, Lemma D.2 applies separately to each admitted weak realization and puts it in the prescribed small H^1 region at the same fixed time.

Let (U, B) be a controlled strong solution from one such writing start state, and let another admitted realization have the same state, force, coefficients and disturbance. Coupled relative energy, with $e = (u - U, B - \mathsf{B})$, gives

$$\frac{d}{dt}\|e\|_2^2 + c\mu\|\nabla e\|_2^2 \le C\mu^{-3}(1 + \|(U,\mathsf{B})\|_{\mathcal{H}^1}^4)\|e\|_2^2. \tag{12.1}$$

To see the exponent, integrate the transport terms by parts; the leading velocity–magnetic exchanges cancel. A remaining term is bounded by $C\|\nabla(U,\mathsf{B})\|_2\|e\|_4^2$. Interpolation $\|e\|_4 \le C\|e\|_2^{1/4}\|e\|_{H^1}^{3/4}$ and Young's inequality give (12.1). Testing the weak equations by time-smoothed strong fields and using the total-energy inequality justifies the same integrated estimate for a weak realization. The coefficient is integrable on the finite controlled interval. Its exponential multiplies zero initial discrepancy, so every admitted realization agrees with this strong solution. None of the disturbance allowances is extracted from that exponential: they were obtained from the direct polynomial estimates. Weak continuity and the one-sided energy convention identify the terminal trace with its strong trace.

Completion of Theorem 2.2. Lemma 3.1, Lemma 3.2 and Proposition 3.3 supply the finite coefficient problem and stable schedule. The physical register, restore and decoder estimates in Section 8 realize that schedule while velocity is disposable. The amplifier of Section 9 defines its known preconditioned target and gives a polynomial gain on arbitrary mean-zero errors. The one-batch readout of Section 10 writes that target and physically clears all remaining packet output on the already prepared magnetic background. The final known increment retains both unknown fields.

Section 11 fixes the code, weak budget, gain, historical errors and only then the final carrier and fresh field tolerances. Its exact rational clock, common preparation, error sum and finite majorant algorithm prove the endpoint and all resource bounds. Every operation preserves the unknown magnetic mean, and every force coefficient uses only known input and envelopes. The direct estimates give controlled strong continuation with the actual admitted disturbances. The preceding identical-data relative-energy argument covers every admitted weak realization and its one-sided terminal trace. No coefficient history or flux value is queried at any step. □

12.2. **Consequences and invariant boundaries.** The theorem implements one finite-time polynomial response map of an unknown conserved input. It permits nonlinear monomials, mixed flux components, spatially singular linear responses, and an even flux-dependent velocity in the same experiment. The compiler degree is variable and the resource exponent is universal; neither assertion follows from a fixed list of low-degree examples. The exact algebra in Example 3.4 and the singular linear example in Section 4 illustrate these distinct features.

Magnetic sign reversal sends (u, B) to $(u, -B)$ when the magnetic disturbance is reversed. Its invariants motivate the response class, but are not by themselves a proof that every such response is controllable. The construction earns that sufficiency under its stated coefficient promise. In particular, $Z(x, 0) = 0$ and the requested magnetic output at $H = 0$ is zero. This does not say the initial magnetic field is zero whenever its mean is zero. A nonzero mean-zero seed remains a possible initial field. If instead $B_0 = 0$ and $h_B = 0$, the induction equation preserves $B = 0$; there can be no uniform independently specified nonzero magnetic target for a class containing that preparation. Mechanical forcing never changes the conserved magnetic mean.

The seed-preparation paper [2] therefore remains distinct. Its Theorem 10.2 is conditional on Hypothesis D.1, including a magnetic-source channel and specified recursive prefix implementations. The present mechanical endpoint neither creates that channel nor proves those prefix clauses. No unconditional version of that theorem is claimed here.

12.3. **Model and resource limitations.** Both diffusivities are positive, spatially constant and independently Lipschitz in time, with supplied numerical lower bounds. The proof does not cover zero resistivity, arbitrary temporal switching or spatially varying transport coefficients. It retains the joint inhomogeneous H^1 preparation bound, zero velocity mean, complete rational finite target lists, globally supported periodic forcing, finite horizon and absolute additive $L_t^\infty L_x^2$ disturbance model. Fixed relative actuator errors, stochastic distributed fault models and timing errors are not included merely because an absolute force disturbance is tolerated.

The resource theorem counts large transient velocities, high carrier frequencies, small clocks, target preprocessing, precision and the enormous fixed shear-family construction. Its constants and powers can be extremely large. A polynomial in inverse diffusivity and absolute accuracy is not an affordability statement, and no comparison with electronic or fault-tolerant quantum computation follows from it. The controlled continuation argument supplies no arbitrary-data global regularity result for unforced three-dimensional MHD or Navier–Stokes.

12.4. **Further questions.** The response operation is compiled for one prescribed endpoint. A fixed native device admitting indefinitely reusable unknown updates, realistic relative

errors, localized physical actuators or autonomous fault-tolerant processing would require additional results. The present proof also does not classify the smallest useful response representation beyond finite polynomials with a paid coefficient bound. These are separate problems; their resolution cannot be inferred by repeating the endpoint theorem without paying a new preparation, smooth envelope and error budget.

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Appendix A. Effective estimates and rational arithmetic

We collect the elementary estimates used to make the polynomial convention effective. All space dimensions and differentiation orders in this appendix are fixed; none grows with a target catalogue.

A.1. **Products, interpolation and scalar smoothing.** On the unit three-torus, with inhomogeneous norms,

$$\begin{aligned} \|v\|_6 &\le C\|v\|_{H^1}, \qquad \|z\|_\infty + \|\nabla z\|_3 \le C\|z\|_{H^1}^{1/2}\|z\|_{H^2}^{1/2},\\ \|\mathcal{B}(v,z)\|_2 &\le C\|v\|_{H^1}\|z\|_{H^1}^{1/2}\|z\|_{H^2}^{1/2},\\ \|\mathcal{C}_v z\|_2 &\le C\|v\|_{H^1}\|z\|_{H^1}^{1/2}\|z\|_{H^2}^{1/2},\\ \|\mathcal{B}(v,z)\|_{H^{-3/4}} &\le C\|v\|_{H^1}\|z\|_{H^1}. \end{aligned} \tag{A.1}$$

Here and below C may be chosen a fixed computable dimensional constant. One elementary proof of the first inequality applies the periodic $W^{1,1} \to L^{3/2}$ inequality to $|v|^4$ and uses Hölder; the former inequality follows by applying the fundamental theorem of calculus along the three coordinate lines and then Hölder. This fixes a finite constant. For the supremum estimate, split the Fourier sum at radius $J \ge 1$. The low and high parts are bounded by $CJ^{1/2}\|z\|_{H^1}$ and $CJ^{-1/2}\|z\|_{H^2}$. Indeed the relevant shell sums are bounded by CJ and C/J. Take J comparable to $\|z\|_{H^2}/\|z\|_{H^1}$. Interpolation of ∇z between L^2 and L^6 gives its L^3 bound. Use L^6L^3 for $v \cdot \nabla z$ and $L^\infty L^2$ for $z \cdot \nabla v$ to obtain the next two inequalities. Finally $v \cdot \nabla z \in L^{3/2} \subset H^{-1/2} \subset H^{-3/4}$; duality of $H^{1/2} \subset L^3$ gives the last inclusion estimate. The Leray multiplier is bounded on every H^s with norm at most one. Smooth-reference products, for example,

$$\|\mathcal{C}_v\beta\|_2 \le C\|\beta\|_{W^{1,\infty}}\|v\|_{H^1}, \tag{A.2}$$

follow by the product rule. Fixed higher-order versions follow by finitely many Leibniz expansions and these estimates.

If $\mu \le \nu(t) \le 1$, the non-autonomous scalar Stokes propagator is exactly

$$S_\nu(t,s) = \exp\left(\Delta \int_s^t \nu(r)\,dr\right).$$

Its Fourier symbol and $\sup_{x>0} x^\theta e^{-x} < \infty$ for each fixed $\theta \ge 0$ give, for fixed $b \ge a$ and $0 < t - s \le 1$,

$$\|S_\nu(t,s)\|_{H^a \to H^b} \le C_{a,b}[\mu(t-s)]^{-(b-a)/2}. \tag{A.3}$$

The constant mode is included, since the right side is at least a fixed positive constant. In particular the $H^{-3/4} \to H^1$ kernel has integrable singularity $(t-s)^{-7/8}$. All constants can be enlarged to rationals by fixed integer bounds for these scalar maxima. No value of the actual coefficient profile is computed.

A.2. **Geometric and fixed-order constants.** A fixed smooth finite Fourier field has the rational derivative envelope (4.4). For a product its envelopes are bounded by finite convolution and Leibniz sums. For a spatial flow, the partition recurrence in Lemma 6.2 computes fixed-order derivative bounds after the rescaled coefficient integral has been made at most one. For pushforwards, inverse maps and their scalar principal metrics, use finite chain rules, the matrix inverse derivative identity, and the explicit positive inverse-ellipticity bounds. The energy calculation (5.2) absorbs each mixed product by $ab \le \theta a^2 + b^2/(4\theta)$ with a supplied rational θ. Thus its constants require only a finite list of rational additions, multiplications and fixed powers. The fixed Fourier multiplier commutators needed for Euler conjugations are imported at their proved signed orders from [1, Lemma 8.2]; their constants are included in the same fixed precomputation.

Choose the spatial order once after the fixed lowering depth is known: start from the two derivatives needed for joint energy, add the finite orders needed for the fourth Taylor jet, its remainder and the flow comparison, and add the derivative losses in each of the fixed lowering levels. This is a terminating integer calculation. The resulting order is universal. It is not necessary to differentiate once per temporal cell or once per target mode, and no such growing order is used here.

A.3. **Finite-bit implementation.** Canonicalization uses integer wavevector sign normalization, sorting and exact rational addition. The bit height of a common denominator is at most the sum of the input denominator heights. Polynomially many such arithmetic operations therefore have polynomial bit cost in the complete input length. Comparison with the truncation cutoff is a binary integer comparison.

For elementary real constants one may use Machin's identity

$$\pi = 16\arctan(1/5) - 4\arctan(1/239)$$

with alternating-series remainders. It computes b correct bits in polynomial bit work. Rational interval bisection computes a square root to an absolute tolerance in polynomial work in the operand length and requested digits, also when the interval contains zero. Sine and cosine at arguments bounded by a known numerical polynomial are computed by their Taylor series with a polynomial number of terms and divided guard digits. The finite phase quadratures in the imported velocity synthesis have the proved polynomial counts and nonzero conditioning bounds of [1, Sections 8.3–8.4]; those costs are retained.

Absolute approximation is important. A coefficient multiplying a field with known norm at most P needs error at most $\vartheta/(CP)$ to pay an assigned force error ϑ. Divide this budget by the known number of coefficients. No inverse of a tiny target coefficient occurs. Means and solenoidality of approximated target directions can be preserved by rational Fourier projection $\mathbb{P}_k$; its denominators are integer $|k|^2$ and its norm is at most one. All original input bits are read even when their frequencies are subsequently discarded.

Every emitted clock is chosen by an explicit positive rational formula, ceiling or finite minimum. A local polynomial of degree at most two, or at most four in the imported eraser, remains of that degree under $s = (t - t_0)/d$. Its expanded rational bit heights are bounded by a fixed multiple of the original heights and those of d, t_0. These facts justify the numerical/binary separation and the terminating majorant calculation in Section D.4.

The compiler is a total finite algorithm on the rational input domain. It uses the stated norm bounds as promises; it does not decide a norm inequality by an exact-real oracle. Outside those promises the same finite instructions still terminate, but the response conclusion is not asserted.

Appendix B. An actual coupled filter for finite packets

We prove the microscopic filter needed in the whole-field gate. It is different from the arbitrary-duration heat service: the following short time is tied to the packet scale. Define

$$\|v\|_{X^r} = \sum_{k\in\mathbb{Z}^3} (1+|k|)^r |\widehat{v}(k)|,$$

and use the sum of the two component norms for a velocity–magnetic pair. The equivalence with weighted pair Euclidean norms costs only an absolute factor.

Lemma B.1 (Coupled packet filter)**.** *Let $|H| \le M$ and $\mu \le \nu(t), \rho(t) \le 1$. Suppose $Q_{\mathrm{g}} = q^4 \ge 32(F_0+1)$ and*

$$b_0 = b_L + b_H = \sum_{|n|\le 4} A_n(x) e^{i\kappa n Q_{\mathrm{g}} x_3}, \qquad \operatorname{supp} \widehat{A}_n \subset \{|k| \le F_0\}, \tag{B.1}$$

is real, solenoidal and mean zero. Here $b_L = A_0$ and b_H contains the nonzero fast harmonics. Assume, for $r = 0, 1, 2$,

$$\|b_L\|_{X^r} \le E, \qquad \|b_0\|_{X^r} \le E q^{2+4r}, \tag{B.2}$$

where $E \ge 2$ is independent of q. For q above a computable polynomial threshold in E, F_0, M, μ^{-1}, the actual unforced MHD evolution from $(0, H+b_0)$ exists through $T = \mu^{-1} q^{-7}$ and satisfies

$$\|u(T)\|_{H^1} + \|B(T) - (H+b_L)\|_{H^1} \le P\{q^{-4} + \mu^{-1} q^{-7} + q^6 e^{-cq}\}. \tag{B.3}$$

The constants are uniform over the actual coefficient histories; no temporal regularity is required on this interval beyond measurability and the bounds. For initial joint errors e_0 in L^2 and e_1 in H^1, and disturbances of joint L^2 essential supremum η, the additional endpoint error is

$$P\{e_1 + q^3 e_0 + \eta\sqrt{T}\} \tag{B.4}$$

on inverse-polynomial input radii. It holds for all admitted energy weak realizations, with the one-sided total-energy endpoint convention.

Proof. Write $B = H + b$ and retain the Alfvén coupling in the linear part. At frequency k, its time-dependent pair matrix is

$$A_k(t) = -\kappa^2 |k|^2 \begin{pmatrix} \nu(t) & 0 \\ 0 & \rho(t) \end{pmatrix} + i\kappa (H\cdot k) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{B.5}$$

The second matrix is skew-Hermitian after its imaginary coefficient. Differentiating pair energy, rather than exponentiating a sum of noncommuting matrices, gives for its time-ordered propagator

$$\|E_k(t,s)\| \le e^{-\kappa^2 \mu |k|^2 (t-s)}, \qquad |k| \|E_k(t,s)\| \le C[\mu(t-s)]^{-1/2}. \tag{B.6}$$

This is uniform for arbitrary real H in its supplied ball.

All remaining nonlinearities are quadratic divergences. The Leray multiplier has norm at most one. Fourier convolution, the second estimate in (B.6) and $|k| \le |\ell| + |k-\ell|$ give the mild bilinear estimate

$$\|\mathrm{Bil}(Z,W)\|_{C_t X^r} \le C_r \mu^{-1/2} \sqrt{T} (\|Z\|_{C_t X^0} \|W\|_{C_t X^r} + \|Z\|_{C_t X^r} \|W\|_{C_t X^0}), \quad r = 0,1,2. \tag{B.7}$$

For $R_0 = Eq^2$, its contraction parameter is

$$\delta_0 = C\mu^{-1/2}\sqrt{T}\, R_0 \le P q^{-3/2}.$$

Choose q to make this at most $1/64$. Picard iteration gives the nominal solution with $\|(u,b)\|_{C_t X^r} \le P q^{2+4r}$.

Expand the mild solution into homogeneous binary trees in the initial packet. A degree-n term has n leaves and $n-1$ quadratic vertices. Catalan counting and (B.7) bound its X^1 norm by

$$CnR_1(C\delta_0)^{n-1}, \qquad R_1 = Eq^6.$$

The sum of degrees $n \geq 9$ is at most Pq^{-6}. Hence only a fixed collection of trees with at most eight leaves remains to be estimated.

Every edge in one of these trees has frequency $jQ_{\mathrm{g}}e_3 + \xi$, with $|\xi| \leq 8F_0$. A nonzero carrier therefore has modulus at least $Q_{\mathrm{g}}/2$. A nonroot derivative has size at most C_nQ_{g}; if output carrier is zero, its derivative and final H^1 weight have size at most $C_n(F_0+1)$ each.

Consider first a tree with nonzero output carrier. At each time slice the sum of its active edge carrier indices is the fixed nonzero output index. At least one active edge is therefore high. The sum of all high-edge lengths is at least T. Multiplying the contraction factors in (B.6) gives $e^{-c\mu Q_{\mathrm{g}}^2 T} = e^{-cq}$. All other derivative, amplitude and time-simplex factors are polynomial; more explicitly their largest scale is

$$(Eq^2)^n Q_{\mathrm{g}}^n T^{n-1} \leq P_n q^{7-n} \leq P_n q^6, \qquad 1 \leq n \leq 8.$$

Thus the entire high output, including high modes generated at intermediate times, is at most Pq^6e^{-cq} in H^1.

If the output carrier is zero but some leaf is high, there are at least two high leaves. Select one such leaf, and integrate its parent time first. Its heat factor supplies

$$\int_0^T e^{-c\mu Q_{\mathrm{g}}^2 s}\, ds \leq C(\mu Q_{\mathrm{g}}^2)^{-1}.$$

The other $n-2$ time variables may be bounded on $[0,T]$; enlarging the ordered simplex only enlarges the integral. The low root derivative and final H^1 weight are slow, while the other $n-2$ derivatives cost C_nQ_{g}. The contribution is at most

$$P_nR_0^nQ_{\mathrm{g}}^{n-2}(\mu Q_{\mathrm{g}}^2)^{-1}T^{n-2} \leq Pq^{-4}[Pq^{-1}]^{n-2}, \qquad 2 \leq n \leq 8. \tag{B.8}$$

This is a single selected heat integration, not an independence assumption. Increasing q makes the bracket bounded. Trees with only low leaves have uniform slow frequency bounds and contribute at most PT beyond the linear term. On their finitely bounded frequencies, integrating (B.5) gives PT difference from the identity. The mean remains exact because every quadratic term is a divergence. Summing the low, high and tail estimates proves (B.3).

We next pay arbitrary rough errors and disturbances. The nominal solution has computable bounds

$$L = \sup_t \|\nabla(u^r, B^r)\|_\infty \leq Pq^6, \qquad D_2 = \sup_t \|\nabla^2(u^r, B^r)\|_\infty \leq Pq^{10}. \tag{B.9}$$

These follow from its X^1, X^2 bounds. Coupled relative energy cancels the constant-H cross terms and the leading transport terms. Since $LT \leq P\mu^{-1}q^{-1}$, its exponential is bounded by a constant after the polynomial threshold is increased. For squared L^2 error E_0,

$$\sup_{t\leq T} \sqrt{E_0(t)} \leq P(e_0 + \eta T). \tag{B.10}$$

Here is the differentiated estimate needed to avoid an exponential in D_2. Subtract the two coupled systems and differentiate once. Terms where the derivative hits the error pair have the usual velocity–magnetic leading cancellation. A single reference derivative costs CLE_1. Two reference derivatives are paired with the undifferentiated error and therefore cost $CD_2\sqrt{E_0E_1}$. The error self interaction is bounded by interpolation and Young by $C\mu^{-3}E_1^3$, with half the joint H^2 dissipation absorbed. The disturbance costs $C\mu^{-1}\eta^2$. Thus

$$E_1' + c\mu\|\mathrm{error}\|_{H^2}^2 \leq C(1+L)E_1 + CD_2\sqrt{E_0E_1} + C\mu^{-3}E_1^3 + C\mu^{-1}\eta^2. \tag{B.11}$$

In the application the compared magnetic means agree. The displayed additional lower-order CE_1 also pays the missing Sobolev mass if a general initial discrepancy has a nonzero mean. Young's inequality at time scale T gives

$$CD_2\sqrt{E_0E_1} \le E_1/T + CTD_2^2E_0.$$

Apply (B.10), integrate once over the whole interval, and close a fixed $E_1 \le 1$ bootstrap. The exponential now contains only LT, an absolute constant from $1/T$, and $C\mu^{-3}T$, all bounded by the polynomial threshold. Taking square roots yields

$$\sup_{t\le T}\sqrt{E_1(t)} \le P\{e_1 + TD_2e_0 + \eta\sqrt{T}\} \le P\{e_1 + q^3e_0 + \eta\sqrt{T}\}.$$

The term $TD_2\eta T$ is included in the last allowance, since $q^3\sqrt{T}$ is polynomially bounded (and can be made at most one after including the supplied powers of μ^{-1}). This proves (B.4).

For each fixed output force and disturbance a local strong solution is constructed from its H^1 data. The strict bootstrap prevents loss of H^1 before T, so it continues throughout the interval. Identical-data weak–strong uniqueness identifies every admitted energy weak realization with it. This use has zero initial difference; the possibly large uniqueness exponent is not used to select η. The final strong trace agrees with the stated one-sided endpoint convention. □

For the jet in Lemma 6.3, take $F_0 = F + 4N$. Its remainder has H^1 norm Pq^{-2} and L^2 norm Pq^{-6}. In (B.4) it therefore costs only $P(q^{-2} + q^3q^{-6})$. This is why the exact image need not have finite Fourier support. The construction uses a finite jet only for the nominal packet estimate, and pays the complete exact image as a rough perturbation.

Appendix C. Exact imports and their applicability

The load-bearing external control foundation is [1], at its released version DOI. We use the following statements and constructions.

(1) Definition 2.1 and Theorem 2.2 give the admitted weak convention and the common compiler for (V, H). Our eraser target is zero, its incoming joint radius is M, and its horizon T_0 in (D.3) is positive rational and at most one. The restricted actual profiles retain their supplied ranges and Lipschitz bound; the final common disturbance allowance is smaller than that returned by the theorem. These verify all hypotheses.

(2) Lemma 4.1 gives exact induction return; Proposition 4.5 and Corollary 4.6 give the reset endpoint expansion and stable forward products; Lemma 4.7 gives the continuous actual-viscosity corrector. Proposition 4.8 and the coupled calculation in Section 5 give the raw joint remainder bound. Lemma 5.1 supplies the contraction and short recovery tail. The fields in these reset words belong to one fixed finite catalogue, and their impulses satisfy the small caps used in their proofs. Our new smooth-memory argument retains these carriers and proves its extra terms in Section 5; it does not import nonconstant memory as an unstated conclusion of the old theorem.

(3) Lemmas 8.1 and 8.2 give signed Sobolev propagator and Gårding bounds for the fixed fundamental-shear conjugations. The velocity writer uses only the fixed products allowed there. Its generic quadratic directions instead have the explicitly paid small impulses. Sections 8.3–8.5 and Proposition 8.3 provide the finite phase synthesis, depth-two certificates and actual closed leaves. Their target lists here are finite rational lists with known envelopes, so the same arithmetic construction applies. The additional magnetic diffusion and Lorentz terms for smooth memory are estimated in Lemma 5.4.

(4) Section 8.6 gives actual fixed-depth actuator lowering, the continuous correction, product generator, Euler defect and signed diffusion-form comparison. Every new lowering call fixes its inner amplitudes and geometry before refining its next grid. Its defects are placed below the same fixed coercive margins, and all original endpoints are retained. This is

the hypothesis needed to preserve the raw energy bound independently of reset accuracy. It is stronger than a qualitative saturation assertion.

These imports are proved mathematical operations; their uses involve no knowledge of the actual profiles or magnetic mean. The whole-field wavepacket geometry, smooth-memory estimates, general outer heat operation and fixed-depth response assembly are proved in the present article.

The nonlinear extension uses two further proof-level facts from the same released source. Lemma 4.2 bounds its continuous full matrix symbol on a fixed annulus. The proof of Proposition 4.4 matches the full low- and high-frequency leading matrices, not merely their symmetric parts, and bounds the full remainders by $2^{110}r^3$ and $2^{110}r^{-3}$. Lemma 9.1 combines those facts to obtain the stated norm bound. The dyadic sum, its all-mode inverse gain, noncommuting-scale comparison, and accumulated-metric extension of joint-memory known increments are proved in Section 9 and Appendix F. They are not assertions of the earlier endpoint Theorem 2.2 or of its stable forward repetition corollary.

Similarly, variable-depth physical register arithmetic, directional filtering and restoration, the smooth-family gate extension and the one-batch inverse-carrier readout are proved here. The imported constant-viscosity Navier–Stokes control literature does not discharge their unknown-history, coupled-MHD or resource hypotheses.

For comparison, [5, Theorem 2.5] replaces forcing directions along a given strong Navier–Stokes trajectory. Its Theorem 4.1 and Proposition 4.3 already supply the fundamental twelve-mode saturating space. These are predecessors of the algebraic control method, rather than unknown-preparation MHD or polynomial-resource imports. The return/linearization construction in [6, Proposition 1.2] uses actual-viscosity compensation in its force; it cannot be applied as a coefficient-blind physical lift here. The saturation and approximation methods of [3, 4] are likewise credited as precedents.

Galan's [7, Theorem 2.1] has controls in both equations for prescribed data. The numbered comparison to Rissel refers specifically to [8, Theorem 1.2]'s arXiv version 2: smooth known two-dimensional ideal data in a doubly connected domain, two controls and vanishing magnetic first-cohomology projections. Its Problem 1.4 for nonzero cohomology is a different two-dimensional ideal-MHD domain question and is not settled by the present periodic, positive-diffusivity theorem.

No abstract maximal-regularity theorem is used to replace a spatial coercivity or smoothing estimate. The scalar-principal calculation (5.2), the actual Euler commutator import above, and the physical energy arguments give the required fixed spaces and bounds. The Fourier estimates in Appendix B are proved for the actual coupled time-ordered propagator, rather than imported from a constant-viscosity Navier–Stokes theorem on a different domain.

Appendix D. The fixed-depth linear compiler and preparation

We now join the proved operations. All tolerances below are rational. A polynomial envelope is a polynomial with nonnegative rational coefficients, generated by the finite formulas in the preceding estimates. Its parameters are supplied numerical bounds, not norms of unknown solutions.

D.1. **Replacing each outer selector.**

Lemma D.1 (Selective positive filtering)**.** *At each of the three selections in Proposition 4.3, positive mechanical filtering can replace the idealized Fourier cutoff, uniformly over the unknown profiles and $|H| \le M$. Given an output tolerance ξ and duration cap S, its carrier frequency, exact positive duration and force have polynomial bounds in the lower-frequency reference data, $\xi^{-1}, S^{-1}, \mu^{-1}, \Lambda$. Its physical input is the full smooth field in (4.8), plus a sufficiently small arbitrary mean-zero H^1 discrepancy and a supplied velocity bound. Its output velocity is at most ξ after a further reduction of the budgets if needed.*

Proof. Choose its duration $0 < T \le S$ from the retained reference first, so that

$$T \sup_{|H|\le M} \|\Delta(L_j H)\|_{H^1} \le \xi/16. \tag{D.1}$$

The supremum has a rational envelope from the known matrix. Then choose the integer Q_j to satisfy (4.6) and

$$C_j(1+Q_j)^2 \exp(-\kappa^2 \mu T Q_j^2/8) \le \xi/16. \tag{D.2}$$

This requires only a fixed polynomial choice. Indeed, $e^{-x} \le 24x^{-4}$, and for $Q \ge 1$ the left side is at most $CC_j(\mu T)^{-4}Q^{-6}$. A sufficiently large integer power of $2 + C_j + \xi^{-1} + (\mu T)^{-1}$ and the separation bound suffices. No logarithm or spectral optimization is required.

Only now compute a full smooth envelope for $L_j + W_j(Q_j)$ and invoke Lemma 7.1 on the already selected time T. For every effective heat time $D \in [\mu T/2, T]$, scalar heat contraction and the fundamental theorem of calculus give

$$\|e^{D\Delta} L_j H - L_j H\|_{H^1} \le T\|\Delta L_j H\|_{H^1}, \qquad \|e^{D\Delta} W_j H\|_{H^1} \le e^{-\kappa^2 \mu T Q_j^2/8} \|W_j H\|_{H^1}.$$

The heat lemma pays physical errors and incoming discrepancies with its remaining budget. Its parameter ζ may be made smaller than ξ to ensure the required velocity bound. All these reductions are polynomial. The large full envelope changes the number of short reset-and-hold cells, not the previously chosen T or the two inequalities above. □

D.2. **An acyclic order of choices.** There are nine writing operations: five gates, three outer filters and the final velocity writer. Give each a duration cap $\tau/32$. We specify them in reverse order, while their eventual execution is chronological.

(1) Compile the smooth-memory velocity writer of Lemma 5.4 for target V_N, memory $\Pi_N H$, and a small fixed fraction of ϵ as its error budget. Record its admissible joint input radius r_w and choose the third filter's output budget below $r_w/8$.

(2) For filter three the retained field is $\Pi_N H$, independent of Q_3. Choose T_3, then Q_3, then its full heat parameters as in Lemma D.1. Its nominal input is the explicit fifth-gate image of $\Sigma_2 H$. This has a known smooth envelope depending on Q_3; no microscopic frequency of that gate is needed to bound its endpoint.

(3) Compile gate five from $\Sigma_2 H$, with scalar (4.15), to meet the third filter's input radius. Its finite implementation has an explicit gain G_5, additive error allocation and admissible input radius. Pass the smaller of that radius and its divided output budget backward.

(4) Filter two retains $\Sigma_2 H$, whose pilot frequency Q_3 is now fixed. Choose T_2, Q_2 and its heat parameters, then compile gate four.

(5) Filter one retains $\Sigma_1 H$, with Q_2 now fixed. Choose T_1, Q_1 and its heat parameters, then compile gate three.

(6) The two encoding gates now have fully specified coefficients and references. Compile them in reverse order and obtain the required initial joint H^1 radius β_0 about $(0, H)$.

For any subroutine with estimate $e_{\text{out}} \le G e_{\text{in}} + b$, assign b a fixed fraction of its output budget, construct the subroutine and its polynomial G, and only then choose the incoming budget. Intersect it with the proved admissible input radius. For filters use the corresponding radius statement of Lemma 7.1; its global proof also gives a gain at most two after the stated safety reductions. All intersections are finite minima of positive rationals.

Each retained reference is independent of its discarded outer carrier. No later reference contains an earlier discarded microscopic packet frequency. Such remnants belong to the rough discrepancy, whose higher derivatives are never requested. Thus these instructions define a directed acyclic algorithm. The fixed number of gate/filter substitutions is the reason that composing their polynomial bounds remains polynomial.

D.3. **Prescribed preparation and exact terminal time.** Let D be the exact sum of the nine compiled durations. Then $D \le 9\tau/32$. Set

$$a = \tau/8, \qquad T_0 = \tau - D - a > \tau/2. \tag{D.3}$$

Use [1, Theorem 2.2] with zero target on $[0, T_0]$, followed by zero chosen force for the prescribed interval of length a. The eraser's accuracy is chosen after β_0 is known. We also retain the controlled strong continuation and identical-data weak–strong argument in [1, Section 10]: every admitted eraser realization agrees with its controlled strong solution and has a strong H^1 left trace at T_0. Passing to T_0 through the good start times in its energy inequality therefore permits the deterministic restart used below. An L^2 endpoint statement alone would not justify that restart.

Lemma D.2 (A fixed preparation time)**.** *The eraser accuracy and a positive disturbance allowance can be chosen inverse-polynomially in $M, \mu^{-1}, a^{-1}, \beta_0^{-1}$ so that every admitted realization satisfies*

$$\|(u, B - H)(T_0 + a)\|_{\mathcal{H}^1} \le \beta_0. \tag{D.4}$$

The force never observes or selects a regular time.

Proof. Write $b = B - H$. The constant-field couplings in both equations cancel in the joint energy. If the eraser ends with L^2 error at most d_e, the unforced energy inequality and Poincaré give

$$\sup_{0 \le t \le a} \|(u, b)\|_2 \le d_e + \eta a, \qquad \int_0^a \|\nabla(u, b)\|_2^2 \, dt \le C\mu^{-1}(d_e + \eta a)^2.$$

This inequality starts at the deterministic time T_0: equivalently, subtract the strong eraser's energy identity through T_0 from the admitted total-energy inequality starting at zero, then remove the conserved mean contribution $|H|^2/2$. For each weak realization there is an energy-regular time in the first half of this interval with joint H^1 norm at most $C(d_e + \eta a)/\sqrt{\mu a}$. It is used only in the proof. For a mean-zero fluctuation pair, the same constant-H cancellation after one derivative, the product estimate in Appendix A, and Poincaré yield

$$(Z^2)' + c\mu Z^2 \le C\eta^2/\mu \quad \text{while } Z = \|(u, b)\|_{\mathcal{H}^1} \le c_0\mu. \tag{D.5}$$

Indeed the remaining quadratic energy is at most CZ times the joint H^2 dissipation, and is absorbed for this small Z. Reduce β_0 below $c_0\mu/8$, take $d_e \le c\beta_0\mu a$ and $\eta \le c\mu\beta_0$ with a fixed sufficiently small rational c. The regular-time bound and (D.5) then give continuation and (D.4) at the fixed end of the wait. Identical-data weak–strong uniqueness on the remaining wait justifies using its strong energy estimate for each original weak realization. No regular time is part of the emitted schedule. □

Execute the five gates and three filters from this preparation, then the final velocity writer. Their backward budgets may be chosen to give joint endpoint error at most $\epsilon/4$ about $(V_N, \Pi_N H)$; all scalar splittings and rational approximations receive fixed fractions of this budget. Equation (4.2) gives (2.11), with room to spare. The exact clock (D.3) ends at τ. Translation of a rational time polynomial to its physical time origin is rational and does not change its degree.

D.4. **Universal numerical and binary bounds.** All required smooth envelopes after truncation are polynomial in R by Lemma 4.2. Each gate and filter has a fixed polynomial contract, including its inverse admissible radius, inverse disturbance allowance, duration reciprocal, force bounds and work counts. The number of gate/filter substitutions is fixed. The inner reset catalogue has a fixed lowering depth, and variable Fourier velocity writing has at most two quadratic stages. Loops within a gate or heat operation use one proved state envelope: a loop multiplies cost by its count, rather than composing a state-bound polynomial that many times.

Here is a precise way to define universal constants without hiding an oracle. First generate the fixed reset and signed-square certificates of [1]. Choose a fixed spatial order sufficient for their finite lowering depth and the derivatives in Sections 5–7. The derivative recurrences, finite product rules, rational Fourier-shell bounds and Young inequalities in Appendix A generate nonnegative rational polynomial majorants for every new estimate. Record for each arithmetic node its degree and coefficient sum, and separately its operand-bit and operation bounds. Addition adds coefficient sums; multiplication adds degrees and multiplies the sums; a fixed power multiplies the degree. Replace a minimum of positive clocks by the sum of their reciprocal bounds. A loop multiplies work and output by its proved count. Apply these rules in the acyclic order above and through the fixed lowering and depth-two synthesis trees.

This is a terminating finite procedure, producing numerical and binary majorants $Q_{\mathrm{num}}(R)$ and $Q_{\mathrm{bit}}(L+R)$. Include the fixed precomputation's complete work, memory and description in their constants. Let p dominate their degrees and let an integer A dominate their coefficient sums and fixed costs. This proves the asserted universal bounds. It does not assert that an enormous fixed table has been executed.

For completeness, long denominators do not invalidate the ledger. Canonicalization reads them in full. Subsequent amplitude and root choices use absolute divided tolerances, not the inverse of a tiny target coefficient. Finite Fourier products, certified π enclosures and elementary quadrature have the paid bit bounds in Appendix A. Repeated cells reuse stored rational parameters. Summing polynomially many rational durations has bit height at most their total bit height, hence remains polynomial in $L+R$. Ordinary-time polynomial expansion has fixed degree. The only degree-four pulse is the imported initial eraser; all newly emitted reset, setter and lowering pieces have degree at most two, while holds emit zero. Thus degree at most four is proved for the whole force. Finally $\tau \le 1$ implies the impulse bound from the force-peak bound in (2.12).

Completion of Theorem 2.3. Lemma 4.2 and Proposition 4.3 provide the finite reference sequence. Lemmas 5.3 and 5.4, Proposition 6.1, and Lemmas 7.1 and D.1 provide its physical operations, with their stated incoming-state and mean hypotheses. Every reference and actual field has the same mean H: gates add curls, pushforwards preserve flux, and heat and the admitted disturbances preserve the zero mode. Thus those hypotheses hold at every call. Section D gives the acyclic backward budgets, fixed preparation and exact clock. Its error sum proves the endpoint and its ledger proves all numerical, binary and force-degree bounds. The preceding continuation and identical-data argument supplies the all-realization quantifier. All choices use rational data and envelopes; the actual profiles and H occur only in the proof. □

Appendix E. Critical estimates and complete physical readout clearing

This appendix proves the Fourier estimates used in Lemma 10.1. All Fourier norms are on the mean-zero pair $z = (u, B-H)$; the conserved constant H is included exactly in $\mathbb{S}_H$, not treated as a small perturbation.

E.1. **A bilinear estimate for the time-ordered propagator.** Let $\mathfrak{N}(z,w)$ be the symmetric bilinear polarization of the quadratic MHD nonlinearity. If $z=(u,b)$ and $w=(v,c)$, its momentum component is

$$-\tfrac{1}{2}\mathbb{P}[(u\cdot\nabla)v+(v\cdot\nabla)u]+\tfrac{1}{2}\mathbb{P}[(b\cdot\nabla)c+(c\cdot\nabla)b],$$

and its induction component is

$$\tfrac{1}{2}[(b\cdot\nabla)v+(c\cdot\nabla)u-(u\cdot\nabla)c-(v\cdot\nabla)b].$$

It is solenoidal and has zero mean. Its Duhamel operator is

$$\mathfrak{B}(v,w)(t)=\int_0^t \mathbb{S}_H(t,s)\mathfrak{N}(v,w)(s)\,ds.$$

Convolution and the divergence form give

$$|\widehat{\mathfrak{N}(v,w)}(k)| \le C\lambda_k \sum_{l+m=k} |\widehat{v}(l)||\widehat{w}(m)|.$$

Use (10.3) first with the critical inverse weight and then with the time-integrated positive weight. At each nonzero output,

$$\mu \int_s^T \lambda_k^2 e^{-\mu\lambda_k^2(t-s)}\, dt \le 1.$$

Fubini and the triangle inequality therefore prove

$$\|\mathfrak{B}(v,w)\|_{\mathsf{E}_T} \le C \int_0^T \|v\|_{\mathsf{X}^0}\|w\|_{\mathsf{X}^0}\, dt. \tag{E.1}$$

At each time, Fourier Cauchy–Schwarz gives $\|v\|_{\mathsf{X}^0}^2 \le \|v\|_{\mathsf{X}^{-1}}\|v\|_{\mathsf{X}^1}$. Thus $\gamma(v) \le \|v\|_{\mathsf{E}_T}/2$, after an irrelevant fixed normalization of the sum norm, and (E.1) implies (10.4). All assertions hold for complex fields. No commutation of the pair matrices has been used.

For $v = \mathbb{S}_H p_Q$, Minkowski in time and $\int_0^\infty e^{-2\mu\lambda_k^2 t}\, dt = (2\mu\lambda_k^2)^{-1}$ give $\gamma(v) \le C\sigma$. For $L = \mathbb{S}_H z_0$, one has $\gamma(L) \le \gamma_0 := \sqrt{\mu T} P_0$. The fixed point for $z = L + v + w$ is a contraction in the ball

$$\|w\|_{\mathsf{E}_T} \le C\mu^{-1}(\sigma + \gamma_0)^2 \tag{E.2}$$

under (10.12). The possibly large $\|L\|_{\mathsf{E}_T}$ is never required to be small; only its short-time norm γ_0 enters the bilinear map. In particular $\|w\|_{\mathsf{E}_T} \le C\sigma$, because $P_0^2 T \le c\sigma$ and $\sigma \le c\mu$. This constructs the actual nominal unforced solution and yields (10.17). Smooth approximation and the fixed spatial energy bounds identify it with the physical strong solution.

E.2. **Carrier labels and the improved higher remainder.** Retain the integer carrier label in addition to the physical frequency: an initial packet coefficient has (k, j) with $k = Qj\ell + \omega$; a background coefficient has label zero. Products add both entries, and the propagator acts at the physical k. Define labelled $\mathsf{X}^r, \mathsf{E}_T, \gamma$ by summing also over the label. Equation (E.1) is unchanged, because its proof uses only convolution, nonnegative sums and physical decay. At physical frequency zero every nonlinear coefficient vanishes individually as a divergence, so the critical inverse weight causes no ambiguity. Summing labels to recover the physical field is a contraction. These facts justify estimating a carrier sector without assuming that cancellations in the physical norm bound its separate label contributions.

For pure packets, (10.6) gives

$$\|\mathfrak{N}(v,v)(t)\|_{\mathsf{X}^0} \le PQ^2\sigma^2 e^{-c\mu Q^2 t}.$$

The output propagator is contractive on X^0. Integration proves (10.13), in the labelled norms as well. For $z = v + w$, the ordinary contraction first bounds $\gamma(w) \le C\sigma^2/\mu$. Its equation and (10.4) then give

$$\gamma(w) \le \gamma(z_2) + \frac{C}{\mu}[2\gamma(v) + \gamma(w)]\gamma(w) \le C\gamma(z_2).$$

The same estimate in E_T on the remaining three bilinear terms proves (10.14). This argument also covers large polynomial geometry factors in the sharper z_2 bound: the ordinary critical estimate still supplies the small contraction ball.

Multiply the initial packet by a complex parameter a with $|a| \le 2$, choosing the universal constant in $\sigma \le c\mu$ sufficiently small for that disk and for half-heat estimates below. The convergent Picard series is analytic there, including in the labelled Banach space. For the remainder $R(a) = z(a) - av - a^2 z_2$, Cauchy's coefficient formula gives

$$\|R_n\|_{\mathsf{E}_T} \le C2^{-n} P\sigma^3\sqrt{T}\,\mu^{-3/2}, \qquad n \ge 3. \tag{E.3}$$

A degree-n coefficient with total carrier zero has $|k| \le nF_p$. On this sector $\|f\|_{H^1} \le C(1+nF_p)^2\|f\|_{\mathsf{X}^{-1}}$. The sum of $n^2 2^{-n}$ is finite. Thus (E.3) gives the whole zero-carrier higher bound, not just a bound on one selected output or a fixed-order Taylor jet.

For backgrounds use two leaf colours. Put $S = \sigma + \gamma_0$ and $\theta = CS/\mu \le 1/4$, enlarging C to include Catalan counting. Every degree-n nonlinear term has labelled critical majorant $C^{n-1}\mu^{1-n}S^n$. The terms with at least one background leaf have the sharper coloured majorant

$$C^{n-1}\mu^{1-n}[(\sigma+\gamma_0)^n - \sigma^n], \qquad n \ge 2. \tag{E.4}$$

Their zero-carrier support is at most $n\max(F_0, F_p)$. Summing the corresponding n^2 factors against (E.4) gives (10.15): the first two-colour term is $C\mu^{-1}(\sigma\gamma_0 + \gamma_0^2)$, and the rest is a convergent geometric derivative series. This includes pure-background nonlinear terms as well as every mixed Reynolds, Maxwell and induction tree. The linear background itself is retained separately.

E.3. **Clearing of the complete remaining physical field.** In a tree with n leaves, every time slice has at most n active edge frequencies whose sum is the root frequency m. Hence $\sum_{\text{active}} |k_e|^2 \ge |m|^2/n$. The exact propagator factors as

$$\mathbb{S}_H(t,s;k) = e^{-\mu\lambda_k^2(t-s)/2}\widetilde{\mathbb{S}}_H(t,s;k),$$

where $\widetilde{\mathbb{S}}_H$ has diffusivities $\nu - \mu/2, \rho - \mu/2$ and the same skew-Hermitian coupling. It is contractive with the remaining positive diffusivity. On a whole tree the extracted scalar factors multiply to at most $e^{-c\mu|m|^2T/n}$. This extraction is valid for the time-ordered simplex, not an assumption of independent interaction times.

All degree-n outputs have support at most PnQ after separation. For degrees $n \ge 2$ the critical majorant is bounded by $C\mu\theta^n$, so their entire tail past $L \ge 2$ obeys

$$\sum_{n>L} \|z_n(T)\|_{H^1} \le CPQ^2\mu(L+1)^2\theta^L. \tag{E.5}$$

The degree-one packet has critical norm at most $C\sigma$ and all its frequencies are high. The degree-one background has no frequencies above $Q/2$. For $2 \le n \le L$, nonzero carrier implies $|m| \ge Q - n\max(F_0, F_p) \ge Q/2$. Their full high sum is bounded by

$$CPQ^2\left[\sigma e^{-c\mu Q^2 T} + \mu\sum_{n=2}^{L} n^2\theta^n e^{-c\mu Q^2T/n}\right]. \tag{E.6}$$

Choose $L = n_* = C_1 b_Q$ first. A fixed C_1 makes (E.5) at most ζ; the factor L^2 is harmless against its geometric decay. Choose the fixed C_2 in (10.8) so $c\mu Q^2T/L$ dominates the logarithm of the polynomial prefactor in (E.6) and ζ^{-1}. Its right side is then at most $C\zeta$. These choices are explicit integer comparisons and fixed bounds, not tree enumeration or evaluation of the physical solution. All outputs are accounted for: zero carrier at degree at most L is in its stated slow band, nonzero carrier is physically high, and all degrees above L have been estimated in the whole space.

E.4. **Arbitrary spatial perturbations and the endpoint trace.** The nominal smooth solution satisfies (10.17). The standard differentiated coupled energy, including constant-H cancellation, therefore gives for fixed $m \ge 4$

$$\sup_{t\le T}\|z(t)\|_{H^m} \le C_m[\|z_0\|_{H^m} + P\sigma Q^{m+1}]. \tag{E.7}$$

One way to see the constant is to retain the tame commutator coefficient $C_m(\|\nabla u\|_\infty + \|\nabla b\|_\infty)$, whose time integral is bounded by a fixed constant. Spatially constant time-dependent diffusivities produce favorable dissipation and no differentiated coefficient. In particular

$$D_2 := \sup_{t\le T}\|D^2 z(t)\|_\infty \le C(P_4 + P_{\text{freq}}\sigma Q^5),$$

where P_{freq} depends polynomially on the tag/direction/offset envelopes and is independent of Q above separation.

Let the actual initial discrepancy have norms e_0, e_1. Relative joint L^2 energy against the nominal reference gives

$$\sup_{t\le T} E_0(t) \le C(e_0 + \eta T), \tag{E.8}$$

because its coefficient is the integrated gradient already bounded. This comparison uses the full coupled cancellation and holds for energy weak realizations as well. It imposes no packet geometry on their errors or disturbances.

For a local actual strong perturbation let $Y = E_1^2$ be squared joint H^1 error and let E_2 be its joint H^2 norm. Differentiating the difference equations and integrating by parts gives

$$Y' + c\mu E_2^2 \le CG_1(t)Y + CD_2E_0\sqrt{Y} + C\mu^{-3}Y^3 + C\mu^{-1}\eta^2, \tag{E.9}$$

where $G_1 = \|\nabla z\|_\infty$ has bounded integral. First reference derivatives multiply Y; a second reference derivative always has the undifferentiated error and hence the already controlled E_0. The leading velocity–magnetic differentiated terms cancel. The remaining self-error terms are bounded by the usual H^1/H^2 product and Young inequality, giving $C\mu^{-3}Y^3$.

Use $CD_2E_0\sqrt{Y} \le Y/T + CTD_2^2E_0^2$ and (E.8). Integrating with a bounded Gronwall factor, and closing the cubic term by first exit, gives

$$\sup_{t\le T} E_1(t) \le C[e_1 + TD_2(e_0 + \eta T) + \eta\sqrt{T/\mu}].$$

For example require the bracket to be at most a fixed small multiple of $\min(1, (\mu^3/T)^{1/4})$. All inverse allowances are polynomial in the displayed parameters. This proves (10.11) with $G_Q = TD_2$ and provides actual strong continuation on $[0, T]$. Compare every admitted weak realization to that disturbed strong solution with identical data, profiles, force and disturbance. Weak–strong uniqueness then identifies them; its exponential acts on zero discrepancy and has not selected an error allowance. Strong continuity and the stated weak-left energy convention identify the terminal field. Thus arbitrary-mode disturbances, including modes transverse to the nominal packet geometry, are covered.

E.5. **Scope of the background statement.** Lemma 10.1 uses a finite-band unknown background. This is precisely the reference produced by the final decoder: its unknown computed coefficients multiply the retained output Fourier fields. Its fresh physical discrepancy is paid separately by (10.11). No assertion that an arbitrarily rough disturbed physical state equals that finite-band reference is made. The proof does not apply a truncation or a projector to delete errors. In particular, the inverse-carrier statement here does not follow by inserting a growing background cutoff into (10.15); such an insertion must retain the resulting cutoff factor or prove an additional weighted estimate.

Appendix F. Multiplier estimates and finite amplification arithmetic

This appendix supplies the fixed-metric commutator estimate used in Section 9. It is a spatial estimate for explicit known multipliers, not an abstract maximal-regularity import. Word comparisons are on the solenoidal spaces of the model. Full-vector extensions are used only to compute principal parts. The actual-prefix inverse comparisons below are applied to the combined operators, which preserve solenoidality and the required means.

F.1. **Symbol differences without real-frequency regularity at resonances.** Extend the transverse multipliers by identity on the longitudinal direction and at zero. Let $\mathcal{A}(k)$ be any nominal amplification prefix made from the factors $e^{-s^4 D_j(k)}$ with total auxiliary time at most $T_a + s^4$. Then

$$\|\mathcal{A}(k+v) - \mathcal{A}(k)\| \le P(N,g)\frac{1+|v|}{1+|k|}, \qquad k, v \in \mathbb{Z}^3. \tag{F.1}$$

The same bound holds for $\mathcal{A}^{-1}$.

To prove this, first work where $|\xi| \geq C\lambda_{J_N}$, with a fixed sufficiently large C. The rational symbol formula and its moment cancellations give, for every fixed r,

$$|\partial_\xi^r S(\xi/\lambda_j)| \leq C_r \lambda_j^2 |\xi|^{-2-r}.$$

One can differentiate the high-frequency expansion directly: all shifted denominators are separated from zero in this region and the finite matched coefficients cancel the earlier powers. As $\sum_j \lambda_j^2 \leq C\lambda_{J_N}^2$, the sum of the differentiated symbols has bounds $C_r|\xi|^{-r}$ on this region. For matrix exponentials use

$$\partial_\ell e^{tB(\xi)} = \int_0^t e^{(t-r)B(\xi)} (\partial_\ell B(\xi)) e^{rB(\xi)}\, dr.$$

In a product of factors the exponential norm bounds use the sum of their time lengths and the pointwise sum of symbol norms, so the result is polynomial in g and T_a, not exponential in the number of factors. This proves a first derivative bound $P(g, T_a)/|\xi|$ for every prefix and its inverse.

If $|k| > 2C\lambda_{J_N}$ and $|v| < |k|/2$, integrate this derivative along the segment from k to $k+v$. If $|v| \geq |k|/2$, the uniform norm bound proves (F.1) directly. For the remaining $|k| \leq 2C\lambda_{J_N}$, use the norm bound again and multiply its constant by $1 + 2C\lambda_{J_N}$. Since $\lambda_{J_N} \leq 2N_0 N$, this is a polynomial. This last argument treats every finite shifted-zero lattice argument directly. It assumes no unjustified smoothness through a real resonance. Higher fixed differences, when needed, follow by the same split and repeated Duhamel differentiation.

F.2. **The two commutator scales and genuine coercivity.** For every needed fixed integer s,

$$\|[\phi, \mathcal{A}]\|_{H^{s-1} \to H^s} + \|[\phi, \mathcal{A}^{-1}]\|_{H^{s-1} \to H^s} \leq P_s(N, g) \|\phi\|_{H^{|s|+6}}. \tag{F.2}$$

Indeed its Fourier matrix from k to $k+v$ is $\widehat{\phi}(v)[\mathcal{A}(k) - \mathcal{A}(k+v)]$. The ratio of Sobolev weights costs at most $C_s(1+|v|)^{|s|}(1+|k|)$. Equation (F.1) cancels the last factor. Young's convolution inequality and Cauchy–Schwarz for the fixed Fourier-shell sum prove (F.2). In particular both $H^{-1} \to L^2$ and $L^2 \to H^1$ are available.

Let $L = \sum a_{ij}(x)\partial_i\partial_j + \sum b_i(x)\partial_i + c(x)$ have scalar principal coefficients with supplied smooth bounds. Derivatives commute with $\mathcal{A}$, and

$$[L, \mathcal{A}] = \sum [a_{ij}, \mathcal{A}]\partial_i\partial_j + \sum [b_i, \mathcal{A}]\partial_i + [c, \mathcal{A}].$$

The first sum maps H^1 to L^2 by $\partial_i\partial_j : H^1 \to H^{-1}$ and the $s = 0$ case of (F.2); the other terms have the same or lower order. Therefore

$$\mathcal{A}^{-1} L \mathcal{A} = L + R_1, \qquad \|R_1\|_{H^1 \to L^2} \leq P. \tag{F.3}$$

If L is uniformly elliptic, integration against Jz and Young's inequality give a polynomial H^1/H^2 Gårding bound. Common's fixed fundamental-shear products already have this scalar principal part and first-order remainder. Composing its remainder with $\mathcal{A}^{\pm 1}$ preserves the $H^1 \to L^2$ bound. For small generic quadratic excursions use their small-impulse commutator estimate instead. Thus this argument covers exactly the partial maps used in the known-increment construction.

For the actual accumulated prefix $\mathcal{Q} = \mathcal{A} + E$, suppose $\|E\|_{H^r \to H^r} \leq \omega$ for $r = 0, 2$ and sufficiently small ω. Since $\|\mathcal{A}^{-1}\| \leq 1$, the Neumann identity gives $\|\mathcal{Q}^{-1}\| \leq 2$ on those orders and $\|\mathcal{Q}^{-1} - \mathcal{A}^{-1}\| \leq 2\omega$. Expanding the difference of the two conjugations yields

$$\mathcal{Q}^{-1} L \mathcal{Q} - \mathcal{A}^{-1} L \mathcal{A} = (\mathcal{Q}^{-1} - \mathcal{A}^{-1}) L \mathcal{Q} + \mathcal{A}^{-1} L E.$$

It has $H^2 \to L^2$ norm at most $P(N, g)\omega$. Pairing with Jz therefore gives a diffusion-form error at most $P\omega\|z\|_{H^2}^2$, absorbable by a polynomial reduction of the word tolerance. The signed-scale variants follow by the same formula and formal adjoints. This proves (9.17). It is stronger than bounded invertibility of the coordinate maps.

F.3. **Insertions inside a word.** At a letter boundary the accumulated prefix is $P_\circ(\mathcal{A}+E)$, where $P_\circ$ is a partial scale word. Small impulse does not imply that $P_\circ - I$ has small operator norm. We instead estimate its action on the diffusion form.

For a smooth solenoidal direction ξ, use the full-vector extension from the proof of [1, Lemma 8.2],

$$L_\xi^{\mathrm{ext}} = \xi\cdot\nabla + R_\xi, \qquad R_\xi z = \mathbb{P}[(z\cdot\nabla)\xi] - \nabla\Delta^{-1}\partial_j[(\partial_i\xi_j)z_i].$$

The inverse Laplacian is zero on constants. Each Riesz factor has bounded lattice symbol, so multiplication estimates bound R_ξ on L^2 and H^2 by fixed spatial norms of ξ. For a scalar second-order differential operator L_0, the third derivatives in $[L_0, \xi\cdot\nabla]$ cancel. The remaining differential operator is of order at most two. Also both L_0R_ξ and $R_\xi L_0$ map H^2 to L^2. Thus

$$\|[L_0, L_\xi^{\mathrm{ext}}]\|_{H^2\to L^2} \le P(\|L_0\|_{C^m}, \|\xi\|_{C^m})$$

at one fixed m, with $\|L_0\|_{C^m}$ denoting the sum of its coefficient norms. Duhamel's commutator formula, with the small-impulse L^2, H^2 propagator bounds, gives

$$\|Q_\xi(a)^{-1}L_0Q_\xi(a) - L_0\|_{H^2\to L^2} \le P|a|.$$

There are a fixed number of letters in a partial scale word. Their impulses are s/λ_j times fixed constants and their spatial norms are polynomial in $\lambda_j \le 2N_0N$. Telescoping these conjugations therefore costs at most $P(N)s$.

If the leaf diffusion is $L = L_0 + R_1$, boundedness of $P_\circ^{\pm1}$ on H^1, L^2 keeps $P_\circ^{-1}R_1P_\circ$ of order at most one, in the precise sense $H^1 \to L^2$. Conjugating the scalar part by $\mathcal{A}$ uses (F.3); conjugating the small $H^2 \to L^2$ defect costs at most its polynomial multiplier bounds. The actual E error is paid by the preceding expansion. Hence the full additional diffusion-form defect is $P(N,g)(s+\omega)$, below a fixed ellipticity fraction by polynomial caps. Leaf geometry is bounded before the seed amplitudes are selected: closed phase-conjugation magnitudes depend only on spatial frequencies, and generic quadratic phases have a prescribed small cap. The same calculation applies to the small generic prefixes inside the at most two polarization stages; their common phase cap is fixed from the same frequency and ellipticity envelope. This ensures that these caps on s precede the carrier speed and its child clocks.

F.4. **Known three-dimensional matrix computations.** Every retained $D_N(k)$ is computed from a fixed rational rotation list and known integer frequencies. The explicit symbol has removable shifted-zero arguments, which are handled by the formula of [1, Lemma 4.2]. Approximate the sole fixed irrational factor κ^4 using a rational enclosure of π. For example the geometric arctangent series in $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ supplies an enclosure with b binary digits in $O(b)$ rational terms. All denominators and matrix arguments concern known data.

Put $B = T_aD_N(k)$ and let $b \ge \|B\|$ be its known rational bound. Here $b \le C_0T_a = O(\log g)$, with fixed constants. The Taylor polynomial has error at most

$$e^b\,\frac{b^{m+1}}{(m+1)!}.$$

For an absolute error χ, a fixed sufficiently large multiple of $1 + b + \log_2(\chi^{-1})$ terms suffices, by the elementary factorial bound $m! \ge (m/e)^m$. Errors in the matrix argument are paid by $\|e^{B+E} - e^B\| \le e^{b+\|E\|}\|E\|$. Fixed-point rational arithmetic with divided guard digits thus has polynomial work and bit lengths in the known input lengths, $\log g$ and $\log(\chi^{-1})$. Scaling and squaring is an equivalent option, but no eigenvector, spectral or norm oracle is needed.

Apply the resulting matrix to the known finite coefficient list and then apply the rational projector $\mathbb{P}_k$. Work with conjugate frequency pairs to preserve reality, and retain the zero mode exactly. Only transverse vectors are contracted by g^{-1}; no contraction claim is made on the artificially extended longitudinal direction. For a transverse coefficient vector v, $|C_N(k)v|_1 \le \sqrt{3}g^{-1}|v|_1$. Allocate the total rounding error below the requested final budget

divided by Cg^β and the known coefficient count. Neither H nor an actual coefficient history enters this computation.

F.5. **Terminating polynomial ledger.** All required spatial orders are fixed before the input is read. The rotation list, its denominator clearer and word lengths are fixed precomputation. On a variable input, $2^{J_N} \le 2N$; dyadic scaling, the fifth-order derivative recursions, and the signed commutator bounds therefore give polynomials in N. The only exponential in variable auxiliary time is bounded by Cg^β. The chosen s^{-1}, repetition count, total absolute impulse, carrier speed and every divided tolerance are then polynomials in the displayed numerical parameters. Carrier speeds are fixed before choosing their child clocks.

Known increments have fixed spatial synthesis depth and polynomial counts. Their prefixes are bounded by sums of known increments, and their errors are allotted in one fixed accumulated metric. A loop multiplies cost by its count; it does not repeatedly compose a state-envelope polynomial. Source Fourier coefficients and their complete binary lengths are read in full. Tiny coefficients use absolute omission/rounding allowances, without division by their magnitudes. Rational root enclosures, ordinary-time polynomial expansion, and summing polynomially many rational clock lengths have the bounds in Appendix A.

Represent every numerical estimate by a positive-coefficient polynomial. At a product add degrees and multiply coefficient sums; at a fixed power multiply the degree; at a positive minimum sum reciprocal bounds; at a loop multiply the operation/output count by the proved loop bound. Perform these operations through the fixed construction tree above. Include the entire finite precomputation. This terminating computation gives the universal polynomial and the reciprocal disturbance radius in Lemma 9.5. No exponent depends on the nonlinear degree or a target catalogue.

Appendix G. Variations, directional filters and register estimates

This appendix proves the programme-specific estimates used in Section 8. All derivative orders are fixed once and for all, beyond the orders needed by the fixed lowering depth. Constants below are computable from finite spatial product rules and the structural estimates already proved in the linear foundation.

G.1. **Homogeneous and quadratic variations of the actual reset.** We use the actual coordinates of Lemma 5.2, including its continuous actual-viscosity corrector and exact product lowering geometry. The changes of variables, carrier and correctors are independent of the initial state being varied. On a word the transformed state $X = (v, b)$ has the form

$$X_t = \mathcal{A}(t)X + \mathcal{L}(t)X + \mathcal{Q}_t(X, X) + g(t). \tag{G.1}$$

The block-diagonal diffusion has common H^1/H^2 coercivity, and all remaining coefficients have polynomial bounds. The large carrier and the time derivative of its coefficient-dependent corrector have cancelled. The reference remainder has the common raw H^1 bound P independent of the requested endpoint accuracy and compression. At a word boundary, velocity coordinates jump by $W(s)$ and magnetic coordinates are unchanged.

For an initial direction Y_0, the actual homogeneous variation solves

$$Y_t = \mathcal{A}Y + \mathcal{L}Y + \mathcal{Q}_t(X, Y) + \mathcal{Q}_t(Y, X), \qquad Y(0) = Y_0. \tag{G.2}$$

This equation contains no differentiated additive residual. The elementary three-dimensional products give

$$\|\mathcal{Q}_t(X, Y) + \mathcal{Q}_t(Y, X)\|_2 \le P\|X\|_{H^1}\|Y\|_{H^1}^{1/2}\|Y\|_{H^2}^{1/2}.$$

For terms with a derivative on X, put Y in L^∞ and ∇X in L^2; for derivatives on Y use L^6L^3. The fixed Euler/induction coordinates and lowering defects multiply the bound by a known polynomial. Pairing with the order-one energy and absorbing its H^2 factor gives

$$(\|Y\|_{H^1}^2)' + P^{-1}\|Y\|_{H^2}^2 \le P\|Y\|_{H^1}^2. \tag{G.3}$$

At complete words the energy multiplier is at most $(1+Cs^5)^2$. The same weighting as in the raw reset proof therefore gives

$$\sup_{t\le d}\|Y\|_{H^1}^2 + P^{-1}\int_0^d \|Y\|_{H^2}^2 \le e^{Pd+Cns^5}\|Y_0\|_{H^1}^2. \tag{G.4}$$

The word choice permits both its contraction and an independently prescribed small budget for ns^5. Taking $Y_{u,0}=0$ proves the normalized magnetic bound $\|C\|\le 1+\delta$ when $Pd+Cns^5$ is sufficiently small.

For two directions $Y_{1,0}, Y_{2,0}$ the mixed second variation has zero initial state and equation

$$\begin{aligned} Z_t &= \mathcal{A}Z + \mathcal{L}Z + \mathcal{Q}_t(X,Z) + \mathcal{Q}_t(Z,X) \\ &\quad + \mathcal{Q}_t(Y_1,Y_2) + \mathcal{Q}_t(Y_2,Y_1). \end{aligned} \tag{G.5}$$

Its restarts are the same linear restarts. The squared L^2 norm of the last line is bounded by

$$P(\|Y_1\|_{H^1}^2\|Y_2\|_{H^1}\|Y_2\|_{H^2} + \|Y_2\|_{H^1}^2\|Y_1\|_{H^1}\|Y_1\|_{H^2}).$$

Cauchy–Schwarz in time and (G.4) bound its integral by $P\sqrt{d}\|Y_{1,0}\|_{H^1}^2\|Y_{2,0}\|_{H^1}^2$. The energy of (G.5), with half the dissipation absorbing its source, consequently gives

$$\|D^2F(X_0)[Y_{1,0},Y_{2,0}]\|_{H^1} \le Pd^{1/4}\|Y_{1,0}\|_{H^1}\|Y_{2,0}\|_{H^1}. \tag{G.6}$$

The estimate is uniform on a common continuation ball. One obtains the derivatives first in Fourier Galerkin systems. Difference quotients satisfy the same energies, and their remainders satisfy the equation above; strong convergence and the uniform bounds identify the Fréchet derivatives of the actual endpoint map. Taylor's integral formula then gives its quadratic remainder. This proves a variation theorem, rather than differentiating an inequality with an additive error.

Here is a safe strong estimate on smooth test vectors. The pure transformed magnetic diffusion has scalar principal part and bounded spatial coefficients. The spatial energy hierarchy and its H^1 adjoint give

$$\|U(t,s)\|_{H^{-3/4}\to H^1} \le P(t-s)^{-7/8} \tag{G.7}$$

on a polynomially short interval. The proof is the fixed-order hierarchy used in the common-form argument: the H^1 adjoint smooths to H^3 at rate $(t-s)^{-1}$, interpolation at $7/8$ gives $H^{11/4}$, and the H^1 pairing applies J before physical duality. No temporal derivative of the coefficients is used. The magnetic coordinate has no jump at complete words, so this is one evolution over the full reset.

For initial magnetic variation $v\in H^3$ and zero velocity variation, subtract the fixed vector v in its magnetic equation. Its source is $\mathcal{A}_B(t)v$ plus the linearized induction terms. The former has $H^{-3/4}$ norm at most $P\|v\|_{H^3}$; the latter has norm at most $P\|Y\|_{H^1}$, because the bilinear bracket maps $H^1\times H^1$ to $H^{-3/4}$. Equations (G.4)–(G.7) and Duhamel therefore give

$$\|(C-I)v\|_{H^1} \le Pd^{1/8}\|v\|_{H^3}. \tag{G.8}$$

This argument uses only a raw H^1 bound for the reference; it does not assume higher regularity of an arbitrary incoming error.

Stable forward Euler products applied to the velocity variation give

$$\|Y_{u,\mathrm{raw}}\|_2 \le \gamma_w\|Y_{u,0}\|_{H^1} + P\sqrt{d_r}\|Y_0\|_{H^1}.$$

The diffusion integral is paid by Cauchy–Schwarz and (G.4); the linearized quadratic sources give a positive power at least $d_r^{3/4}$. On the final unforced tail of length l, scalar Stokes smoothing and $H^{-3/4}$ products give

$$\|Y_{u,\mathrm{out}}\|_{H^1} \le \frac{C\|Y_{u,\mathrm{raw}}\|_2}{\sqrt{\mu l}} + Pl^{1/8}\|Y_0\|_{H^1}. \tag{G.9}$$

Choose l first from the desired derivative tolerance, γ_w next, and d_r last. The raw envelope does not depend on γ_w. The second-variation proof applies also to the tail; composition gives

(G.6) with $d = d_r + l$ after enlarging P. All inequalities involve fixed rational powers. The known word choice, followed by explicit minima for these clocks, gives polynomial costs.

The other off-diagonal derivative is also small. If the incoming magnetic variation is zero, its energy is forced by the velocity variation and the reference magnetic field. The product bound with the variation in L^∞ and the reference derivative in L^2 gives an integrated source at most $P\int\|Y_u\|_{H^1}\|Y_u\|_{H^2} \le P\sqrt{d}\|Y_{u,0}\|_{H^1}^2$. Hence

$$\|D_u F_B\|_{H^1\to H^1} \le Pd^{1/4}. \tag{G.10}$$

The continuous magnetic coordinate has no restart jump. This proof uses the raw reference H^1 bound, not a pointwise reference H^2 bound.

The same argument proves the transition statement. For a shear pulse solve $x' + \kappa^2\nu x = q'$ only in the proof. Since $x - q$ solves $(x-q)' + \kappa^2\nu(x-q) = -\kappa^2\nu q$, (8.5) follows by variation of constants. Normalize by $\operatorname{diag}(E_{\alpha(t)}, M_{\alpha(t)})$. The published fundamental-shear estimates and genuine Gårding lemma have polynomial constants in $1 + |a|$. Their full signed estimates are used, rather than bounded invertibility alone. Normalized variation energies are exactly the calculations above with enlarged polynomials. At the endpoint the extra phase is $\alpha(d_c) - a$; its norm is at most $e^{P|\alpha(d_c)-a|}$ and its $H^3 \to H^1$ difference is $P|\alpha(d_c) - a|$. Append a reset to remove the bounded velocity tail. In the chain rule its cross velocity-to-magnetic block is bounded by (G.10); the pulse's other derivative blocks have polynomial bounds in a. Choose the reset clock after those bounds, so the extra block product is below its divided allowance. For a fixed incoming prefix G, use the magnetic coordinate $M_{\alpha(t)}G$ throughout. Coercivity and product constants merely acquire the supplied polynomial geometric bounds of G; the initial and final material coordinates are the same. Thus this argument introduces no fixed norm-conversion loss in a sequence of transitions. This completes Lemma 8.1 for actual twelve-direction transitions.

G.2. **Pinned jump operators and two-sided smoothing.** For a nominal physical transition with fixed incoming velocity u, the magnetic endpoint obeys $F_B(u, 0) = 0$. After intended shear normalization, define

$$C(u,b) = \int_0^1 D_B F_B(u, tb)\, dt, \qquad F_B(u,b) = C(u,b)b. \tag{G.11}$$

Every radial reference lies in the same supplied ball. The preceding variation estimates give $\|C\| \le 1 + \delta$ and $\|(C - I)v\|_{H^1} \le \varepsilon\|v\|_{H^3}$ uniformly. These are operators pinned for a proof along the actual trajectory; the force does not evaluate them. The second derivative estimate also bounds their variation over the common ball. Actual disturbances contribute a direct fresh endpoint error, by the same energy estimate with an L^2 source; its allowance can be divided by the known number of jumps.

For the H^1 adjoint, expansion of a square gives

$$\begin{aligned}\|(C^* - I)v\|_{H^1}^2 &\le ((1+\delta)^2 - 1)\|v\|_{H^1}^2 + 2|\langle (C-I)v, v\rangle_{H^1}| \\ &\le C\delta\|v\|_{H^1}^2 + 2\varepsilon\|v\|_{H^1}\|v\|_{H^3}.\end{aligned} \tag{G.12}$$

The adjoint sign or pairing cannot be omitted. A preceding hold smooths an interior or terminal jump by the forward $H^1 \to H^3$ bound. An initial jump is smoothed by the following hold, using (G.12) and its backward H^1 adjoint.

On one macro with $PT_m \le c$ and $\sum_j \delta_j \le c$, all outside H^1 factors in a product telescope are uniformly bounded. If every hold lies in $[\ell_0, 2\ell_0]$, the total jump error is bounded by

$$PN_h\ell_0^{-3/2}\sqrt{\max_j(\delta_j + \varepsilon_j)}. \tag{G.13}$$

This conservative bound follows by applying one adjacent heat factor in each term. It requires no higher Sobolev bound for a long product of jumps. All jump tolerances are chosen after ℓ_0, N_h are fixed.

G.3. **Full vector averaging and actual coupled holds.** Let $\Phi_a(x) = x + as(x_j)e_i$ and $E = e_i \otimes e_j$. $D\Phi_a = I + as'E$, $E^2 = 0$, so the material-coordinate scalar metric is

$$g = I - as'(e_i \otimes e_j + e_j \otimes e_i) + a^2(s')^2 e_i \otimes e_i.$$

The product derivative in $\mathcal{M}_a^{-1}\Delta\mathcal{M}_a$ contributes $2as''E\partial_j - 2a^2s's''E\partial_i + as'''E$. Together these give (8.6). Opposite signs cancel its linear terms, while sine and cosine give $\langle (s')^2\rangle = \kappa^2/2$ and $\langle s's''\rangle = 0$. This proves the complete vector average (8.7), including its action on H.

Use equal holding lengths for all four phases of a cycle. During a hold the magnetic homogeneous generator is $A_r(t) = \rho(t)M_r^{-1}\Delta M_r$. Its scalar principal part, common form bounds and forward/backward smoothing constants are polynomial in $1 + |a|, \mu^{-1}$. Freeze ρ only for the proof at the physical cycle start. A forward solution and backward H^1 adjoint give the whole-macro freezing error

$$\|U - \widetilde{U}\|_{H^1 \to H^1} \le P\Lambda\ell, \tag{G.14}$$

where ℓ is the maximum physical cycle length. The integrated H^2 energies supply the two factors; the number of cycles does not multiply the bound.

For frozen cycles the average is the constant-coefficient operator $D_r = \rho_r(\Delta + \chi\partial_{ii})$. Put $K(t) = \int_{t_r}^t [\widetilde{A}(s) - D_r]ds$ in holding time. It vanishes at cycle endpoints and has both $H^3 \to H^1$ and adjoint $H^3 \to H^1$ norm at most $P\ell$. The latter follows from the H^1 adjoint identity $K^\dagger = J^{-1}K_{L^2}^* J$. The primitive has order two, so $H^3 \xrightarrow{J} H^1 \xrightarrow{K_{L^2}^*} H^{-1} \xrightarrow{J^{-1}} H^1$ is the actual derivative count.

In the forward/backward comparison integral split two layers of length $d \le T_m/4$. Their integrated/pointwise H^2 estimates cost $P\sqrt{d/T_m}$. On the interior integrate K_t by parts. There are no internal cycle endpoint terms. Endpoint and interior terms cost $P\ell/d$, using $H^1 \to H^3$ smoothing on both sides. The anisotropic heat comparison is contractive, has baseline ellipticity at least μ, and has polynomial upper coefficient bounds, so precisely these estimates apply to it. Choose $d = \sqrt{\ell T_m}$, or use uniform energies if the mesh is coarse. We obtain

$$\|\widetilde{U} - e^{D_m(\Delta + \chi\partial_{ii})}\|_{H^1 \to H^1} \le P(\ell/T_m)^{1/4}. \tag{G.15}$$

This is the common-form proof, with its new full-vector applicability verified explicitly. Isotropy was never needed.

We now include the coupled dynamics. At the start of every hold reset velocity below r. While the magnetic coordinate is in its common H^1 ball, scalar Stokes smoothing gives

$$\sup_{\text{hold}} \|u\|_{H^1} \le r + P\ell^{1/8} =: U_0. \tag{G.16}$$

This uses only the physical $H^{-3/4}$ product bound, and a short coupled bootstrap. In the magnetic frame the exact extra source is a bracket. Its squared L^2 norm is at most $PU_0^2\|c\|_{H^1}\|c\|_{H^2}$. The actual magnetic energy on all holds, weighted to remove the factors $(1 + \delta_j)^2$ at jumps, gives one common H^1 bound and one summed H^2 dissipation bound. Compare it with the homogeneous evolution using the same pinned operators (G.11). Their error has no new homogeneous jump loss. Its source energy gives

$$E_{\text{fb}} \le PT_m^{1/4}(r + P\ell^{1/8}) + \text{divided fresh jump and disturbance errors}. \tag{G.17}$$

Indeed $\int_{\text{holds}} \|c\|_{H^2} \le \sqrt{T_m}(\int_{\text{holds}} \|c\|_{H^2}^2)^{1/2}$. Taking the square root of the error energy proves the stated positive time powers. It is a single macro estimate, not a sum of local hold errors. No smoothing kernel across arbitrary jumps is presumed.

Choose U_0 small and the mesh from (G.13), (G.14), (G.15) and (G.17). Each inequality contains only fixed powers. Divide a long filter into polynomially many macros with $PT_m \le c$. Their enhanced heat comparisons are commuting contractions. Telescoping the actual macro errors therefore keeps the same incoming H^1 ball, up to the sum of divided absolute errors, at each macro. It does not exponentiate a raw Gårding bound over the whole filter.

For exact time, fix macro and cycle counts first and promise each hold length in $[\ell_0, 2\ell_0]$. Choose all jump tolerances with the worst lower length ℓ_0. Compile transitions and resets with total duration at most half the corresponding macro cap. Their durations are rational and known. The remaining time divided by its number of holds is the actual common hold length. This fills each macro and hence any prescribed total T. Coefficient freezing uses the maximum physical cycle length, so omitted transition time does not invalidate the Lipschitz estimate. The analysis quantity D_i is the sum of frozen resistivity times the holds and belongs to $[\mu T/2, T]$. All actual force coefficients remain rational; χ is a known comparison constant involving κ, never an observed coefficient. This completes Lemma 8.2.

G.4. **The whole-field gate on an arbitrary smooth family.** There is a small additional point when the gate in a restore follows positive heat: its reference need not have finite Fourier support.

Lemma G.1 (Smooth-family gate)**.** *The physical gate of Proposition 6.1 applies to an arbitrary unknown solenoidal smooth family β of mean H with a supplied fixed high Sobolev envelope, without requiring its coefficients or linearity in H. Its force uses only that envelope and the known finite Fourier scalar ϕ. Its rough-error and resource bounds remain fixed polynomials.*

Proof. Use the actual assigned flows to track the full β; their proof needs only its spatial envelope. In the passive calculation truncate β only in the proof at $|k| \le q$. Choose a fixed supplied order $m \ge 20$ and the larger fixed orders needed by the lift. Then

$$\|\beta - \beta_{\le q}\|_{H^1} \le Pq^{1-m}, \qquad \|\beta - \beta_{\le q}\|_2 \le Pq^{-m}.$$

The combined two-flow induction map has H^1 gain at most Pq^{12} and L^2 gain at most Pq^4, by the explicit lifted geometry. Thus its exact image tail contributes at most Pq^{13-m} in H^1 and Pq^{4-m} in L^2. The microscopic packet's additional q^3L^2 gain costs Pq^{7-m}, still smaller than Pq^{-2}.

The growing slow cutoff must not enter the packet error as a new uncontrolled polynomial. Use instead its weighted slow Fourier envelopes. For every fixed r below the supplied order, the modulating coefficients of the finite jet have slow X^r sums at most $P_r q^2$, while its zero carrier has slow X^r sum at most P_r, independent of the cutoff. In every low-output tree the two root weights obey

$$(1 + |\xi_1 + \cdots + \xi_n|)^2 \le C_n \sum_j (1 + |\xi_j|)^2.$$

Assign them to one leaf in its convolution estimate. This replaces a maximum slow frequency by its bounded slow X^2 envelope. The packet proof's low/high and low-only bounds therefore have constants independent of the cutoff q. More explicitly, only degrees $2 \le n \le 8$ require separate treatment; its unchanged contraction tail is Pq^{-6}. Writing $Q_{\mathrm{g}} = q^4$ and $T = \mu^{-1}q^{-7}$, a low-output tree with a high leaf is bounded by

$$P_n q^{2n} Q_{\mathrm{g}}^{n-2} (\mu Q_{\mathrm{g}}^2)^{-1} T^{n-2} \le Pq^{-4}(Pq^{-1})^{n-2}.$$

One selected high-leaf heat integration supplies the inverse μQ_{g}^2; every nonroot derivative costs at most $C_n Q_{\mathrm{g}}$. The two slow root weights are paid by the marked leaf's X^2 envelope above, with no factor from the cutoff. For trees whose leaves all have zero carrier, their coefficient mass and marked root moment are bounded independently of q. The same finite simplex estimate gives $PT(PQ_{\mathrm{g}}T)^{n-2}$, summable since $Q_{\mathrm{g}}T = \mu^{-1}q^{-3}$. Their linear drift is bounded by $T(\|A_0\|_{X^3} + M\|A_0\|_{X^2}) \le PT$. High edges remain separated because $q^4 \ge 32(q + 4N + 1)$; the complete high-output estimate and arbitrary-mode perturbation proof of Appendix B are unchanged. The Taylor remainder uses supremum derivatives and remains Pq^{-2} in H^1 and Pq^{-6} in L^2; truncation does not increase the needed derivative envelopes, by the supplied fixed higher Sobolev norm.

Finally $\|T_i(\phi)(\beta-\beta_{\le q})\|_{H^1} \le P\|\beta-\beta_{\le q}\|_{H^2} \le Pq^{2-m}$. These estimates prove the same gate accuracy on β after a fixed polynomial choice of q. No values of β or actual coefficients are evaluated. This proves the extension used in restoration and decoding; it is stronger than simply renaming ΣH in the earlier statement. □

G.5. **The short cubic code calculation and complete clearing.** Let $Q_c = q^4$, $A_c = q^6$, $t_c = q^{-11}$ and, for a code frequency n,

$$k_1 = Q_c e_1, \quad k_2 = Q_c e_2, \quad k_3 = -Q_c(e_1+e_2) + ne_3, \qquad b_1 = e_2,\ b_2 = e_3,\ b_3 = e_1 - e_2.$$

For a pair a, b and its remaining index c define

$$F_{ab} = \mathbb{P}_{k_a+k_b}[(b_a \cdot k_b)b_b + (b_b \cdot k_a)b_a], \quad D_{ab,c} = (b_c \cdot ne_3)F_{ab} - (F_{ab} \cdot ne_3)b_c.$$

Direct rational projection gives

$$\begin{aligned} D_{12,3} &= -Q_c n(e_1 - e_2), \\ D_{13,2} &= -Q_c n e_1 + \frac{2Q_c n^3}{Q_c^2 + n^2} e_2, \\ D_{23,1} &= \frac{Q_c n(Q_c^2 - n^2)}{Q_c^2 + n^2} e_2. \end{aligned}$$

Their sum is exactly $-2Q_c n(e_1 - e_2)$. With identity linear propagators the two time integrations therefore give the cubic magnetic coefficient

$$\kappa^2 A_c^3 t_c^2 Q_c n\, s_1 s_2 s_3 (e_1 - e_2) = \kappa^2 n s_1 s_2 s_3 (e_1 - e_2). \tag{G.18}$$

For the real packets (8.17), $g_n = r_n/(2i\kappa^2 n)$. The third packet's coefficient on the negative horizontal carrier is $iA_c g_n$, and the positive one is $-iA_c g_n$. Equation (G.18) for the two opposite triads contributes $h_i^2 r_n/2$ twice. Thus the real slow output is precisely $h_i^2 r(z)(e_1 - e_2)$. A cosine in that third horizontal factor would cancel rather than add.

We justify the finite-amplitude expansion and the idle bank. Include the constant-H Alfvén coupling and both actual time-dependent diffusivities in the exact Fourier propagator. Its norm is at most one, by pair energy; the matrices need not commute. A homogeneous degree-n Duhamel term has frequency at most $Cn(Q_c + F)$. If $R_0 \le PA_c$ bounds all high packet amplitudes and the bounded idle bank, Catalan recurrence with its $n/(n-1)$ time-integration factor gives

$$\|Z_n(t)\|_{X^r} \le C_r^n n^r R_0^n Q_c^{n-1} t^{n-1} \tag{G.19}$$

for each fixed r, once q dominates the code envelope. The series converges since $CQ_c R_0 t_c \le P/q < 1/4$. It constructs the actual smooth coupled solution, with polynomial fixed-order envelopes.

At a zero-horizontal-carrier root, its derivative and final H^1 weight cost at most $Cn(1+F)$, instead of Q_c. Consequently the slow degree-n term for $n \ge 2$ is at most

$$C^n n^2 P(F) R_0^n Q_c^{n-2} t_c^{n-1}. \tag{G.20}$$

The sum of degrees at least five is Pq^{-2}. Degree-two slow high-packet interactions vanish: equal-family polarizations annihilate every same-family derivative, including different third-bank code frequencies, and distinct families have nonzero horizontal carrier. At frozen identity propagators the magnetic field has odd packet degree, because velocity starts at zero. Thus its pure fourth-degree magnetic term vanishes even for a whole register with several offsets. With several code offsets, zero horizontal carrier does not force zero vertical frequency; odd magnetic parity supplies this cancellation.

On the fixed degree-three/four frequencies, replacing any linear propagator by identity costs $Pt_c(Q_c^2 + MQ_c) \le Pq^{-3}$, by its finite matrix integral equation. The cubic slow leading scale is one, and the fourth-degree slow majorant is Pq^{-1}, so their errors are Pq^{-3} and Pq^{-4} respectively. This proves the material-independent cubic to the common Pq^{-2} accuracy. Each coloured tree with a bounded idle leaf gains A_c^{-1} relative to a high leaf. Its degree-three/four

slow bounds are Pq^{-6}, Pq^{-7}; one-high/one-idle quadratic output is high, and all-idle evolution is Pt_c. These estimates rigorously retain the entire bounded idle code bank.

For arbitrary rough errors, the actual reference has $L = \sup\|\nabla(u,B)\|_\infty \le Pq^{10}$ and $D_2 = \sup\|\nabla^2(u,B)\|_\infty \le Pq^{14}$. Hence $Lt_c \le P/q$. Relative L^2 energy and differentiated H^1 energy give, on an inverse-polynomial continuation ball,

$$\sup_{t\le t_c}\|\text{error}\|_{H^1} \le P(e_1 + q^3 e_0 + \eta\sqrt{t_c}). \tag{G.21}$$

As in the earlier packet proof, second reference derivatives multiply L^2 error: use $CD_2\sqrt{E_0E_1} \le E_1/t_c + Ct_cD_2^2E_0$. The large D_2 is not put into an exponential. This also proves actual strong continuation and admits arbitrary-mode disturbances.

Every nonzero horizontal carrier has modulus at least Q_c, including all degrees of the convergent series. At the actual pulse endpoint use its smooth magnetic field as the proof reference in Lemma 7.1; its supplied envelope is the polynomial from (G.19). The actual velocity has a paid polynomial bound. Use the prescribed heat duration q^{-7}, so the whole high output is at most $Pq^{10}e^{-c\mu q}$ in H^1. The intended code has baseline heat bias at most $P(F)q^{-7}$, and the slow remainder contracts. The general heat lemma is essential: the large full envelope changes its cell count, not this prescribed duration. Final physical restoration leaves only the intended finite code plus allotted logical and off-code errors. This completes the physical multiplication proof.

G.6. **Genuine smoothing and the common boundary envelope.** Let the actual incoming joint H^1 state have norm at most P_c. The zero-disturbance reference from that same state has, on $s \le c\mu^3 P_c^{-4}$, the usual actual local bound $\sup\|(u^0,B^0)\|_{H^1} \le 2P_c$. Its magnetic Duhamel formula against scalar heat and the strict-subcritical bracket estimate give

$$\|B^0(s) - e^{\Delta\int_0^s \rho}B_{\text{in}}\|_{H^1} \le P_c^C\int_0^s (s-r)^{-7/8}dr \le P_c^C s^{1/8}.$$

An actual $L^\infty L^2$ disturbance contributes $P_c^C\eta\sqrt{s/\mu}$ by direct difference energy. It is not silently differentiated in space.

For completeness the zero-disturbance reference has a polynomial positive-time envelope at every fixed order. Fractional Leibniz and $H^1 \hookrightarrow L^6$ give, for fixed integer $m \ge 1$,

$$\|\text{div}(z\otimes z)\|_{H^{m-1}} \le C_m\|z\|_{H^1}\|z\|_{H^{m+1/2}}.$$

Interpolation and Young yield the spatial energy hierarchy

$$E_m' + c\mu E_{m+1} \le C_m\mu^{-3}P_c^4E_m, \qquad E_m = \|(u^0,B^0)\|_{H^m}^2.$$

Multiplying successively by t^{m-1} proves $\sup_{0<t\le s} t^{m-1}E_m(t) \le P_m(P_c,\mu^{-1})$ by induction from the integrated preceding order. The horizon is chosen below a fixed safety fraction for the finitely many required m. No coefficient time derivative occurs. Thus the reference at the prescribed positive time has the claimed computable smooth envelope.

After the actual restore, the coefficients used in the next proof are the actual extracted coefficients. The restore physically removes the off-code field, with any remaining extraction bias paid as fresh logical error. Set each permanent source's initial error below $e_s/4$ and its fresh per-assignment error below $e_s/[C(S_a+1)]$. Source drift, including unknown heat attenuation, then remains below e_s over the complete programme. Choose S_ae_s inverse-polynomially small. Multipliers slightly above one are charged by their source error times the common live mass, or by the bounded scalar exponential in Proposition 3.3.

Every complete assignment restores the same code band and the same P_c bound. Its temporary high smooth envelope is never fed forward as the next boundary envelope. Consequently all requested local accuracies, clocks and disturbance allowances are fixed polynomials in S_a, and $S_a \le C(D+1)^3$. This is the uniform-degree resource property needed in Lemma 8.6.

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